Opening Ceremony, Welcome to Country, Announcements & Housekeeping

9:00 - 9:30am Monday, 1st December, 2025
405.201 Haydn Williams LT
Chairs Lynne Giles

Keynote: Sofia Villar

9:30 - 10:30am Monday, 1st December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Brenton Clarke

210 Recent Advances for the Design, Analysis and Implementation of Response-Adaptive Designs: Are We There Yet?

Sofia Villar ORCID iD

University of Cambridge, Cambridge, United Kingdom

Abstract

Response-adaptive designs, where treatment allocation dynamically adjusts based on observed outcomes, have a rich history dating back to the 1930s, even predating the first randomized controlled trial (RCT) in 1948. While theoretically appealing, their practical implementation in clinical trials has faced significant challenges. This talk will present three recent advances developed by my group that bridge the gap between theory and practice. Each contribution addresses a critical aspect of response-adaptive designs, encompassing innovative design strategies (introducing a new optimal allocation proportion from a finite sample perspective), novel analytical approaches (exact tests and allocation probability tests), and practical implementation solutions (a mapping strategy to facilitate the seamless integration of response-adaptive designs into clinical trial workflows). These advances, while motivated by the context of clinical trials, have broader implications for improving the efficiency and ethical conduct of experiments in various domains, including digital experimentation.

Morning tea

10:30 - 10:50am Monday, 1st December, 2025

Horizon: Damjan Vukcevic

10:50 - 11:50am Monday, 1st December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Lyle Gurrin

211 You May Peek at Your Data

Damjan Vukcevic ORCID iD

Monash University, Clayton, Australia. ARC Training Centre in Optimisation Technologies, Integrated Methodologies, and Applications (OPTIMA), Melbourne, Australia

Abstract

Data dredging, data snooping, p-hacking. These familiar terms remind us how easily statistical inference can be misused. The pursuit of “statistical significance” can lead to inappropriate multiple testing and selective reporting. One safeguard is to follow a strict pre-specified analysis plan, waiting until all data are collected. A more flexible alternative are methods that allow interim analyses. Taking this idea to the extreme are methods that allow you to “peek” at your data any time at all, or even to decide to collect more.

Such methods trace back to Wald’s sequential probability ratio test and fall under the broader umbrella of sequential analysis. Though rarely covered in standard curricula or applied routinely in practice, they are attracting renewed attention as tools for “safer” statistical inference.

I will outline recent developments in this area and demonstrate how we have used these tools to construct methods to verify the outcomes of preferential elections through efficient post-election audits. This application poses a high-dimensional inference problem, which we can now address in a “stop-whenever-you-like” manner.

Classification

11:50am - 1:05pm Monday, 1st December, 2025
405.201 Haydn Williams LT
Presentation type Contributed Session
Chairs Inge Koch

5 Multi-Layer Scoring Rules for Evaluating Probabilistic Models

Ryan H.L. Ip[1], Muhammad Arif Khan[2], Md Geaur Rahman[3], K.Y.K. Wu[4]

[1]Auckland University of Technology, Auckland, New Zealand. [2]Charles Sturt University, Wagga Wagga, Australia. [3]Charles Sturt University, Port Macquarie, Australia. [4]Singapore University of Social Sciences, Singapore, Singapore

Abstract

Classification is a fundamental task in statistical analysis and machine learning. Over the years, numerous algorithms and models have been proposed, including logistic regression, decision trees, support vector machines, and naïve Bayes. The need to compare the performance of these methods has led to the development of performance evaluation metrics. Commonly used evaluation metrics, such as precision, recall, and F1-score, are all derived from the confusion matrix. However, the reliance on the confusion matrix assumes hard classification decisions, despite the fact that classifiers often provide probability estimates. For instance, two classifiers might be deemed equally effective if they assign the highest probability to the true category, regardless of how sharply or confidently they distinguish between categories. Arguably, moving beyond confusion-matrix-based metrics could better utilise the information embedded in the probability estimates, leading to a more objective comparison. In fields such as weather forecasting, scoring rules have been employed to evaluate forecasters based on their probability estimates—an approach that can be similarly applied to classifiers. This work generalises scoring rules into multi-layer scoring rules (MLSR), enabling the evaluation of classifiers that produce probability estimates across multiple dimensions. Theoretically, we demonstrate that MLSR is strictly proper, a desirable statistical property that ensures integrity. Practically, through both hypothetical and real-world examples, we show that MLSR often provides different and, in many cases, more insightful conclusions compared to confusion-matrix-based metrics and traditional scoring rules.

147 Clustering Skewed Features With Low True Cluster Separation

Muntazir Mehdi ORCID iD, Haydar Demirhan, Sona Taheri

RMIT University, Melbourne, Australia

Abstract

In real-world applications, it is common to find features that are heavy tailed or have skewed feature distributions with outlier or influential observations. Traditional clustering techniques operate poorly with these heavy tailed feature distributions due to the fact that the data distribution is symmetrical. In addition to this challenge, having low true cluster separation between true clusters in the feature space increases complications. Cell grouping, forest fires, maritime search and rescue, urbanization studies, and neuroimaging are only a few of the significant practical domains where these issues arise. Clustering techniques based on Bayesian models are able to provide a precise solution by capturing the centers of weakly separated clusters as well as the skewness in the data, but they exhibit runtime inefficiency because of their Bayesian characteristics. To overcome these issues, we combine generalized multivariate log-gamma distribution with a Dirichlet process mixture model in a Bayesian model-based clustering framework. According to comparative numerical experiments on twenty benchmarking datasets, the proposed approach performs better than traditional and Bayesian model-based clustering algorithms, especially for skewed datasets with poorly true cluster separability. Additionally, compared to its Bayesian counterparts, the suggested method, which is implemented in R, has superior computing efficiency.

152 Joint Clustering of Multiple Datasets with Applications to Flow Cytometry

Inge Koch ORCID iD[1], Lucy Conran[2]

[1]RMIT University, Melbourne, Australia. [2]UWA, Perth, Australia

Abstract

In the analysis of multiple datasets objective comparisons of cluster patterns are required to detect changes that have occurred over time, to discriminate between datasets from different underlying distributions---such as healthy and diseased---and to find variables or biomarkers connected with change. With an increasing number of variables in biological and biomedical applications, good methods are required to provide efficient and reproducible solutions to these problems. Doherty et al (2025) distinguishes between three regimes for clustering multiple datasets and comparing their cluster patterns: pooling, meta-clustering and joint clustering. Although these authors consider model-based clustering approaches only, and their reviews are restricted to flow-cytometry applications, the three regimes have much wider applicability in terms of more general clustering approaches and fields of application.

We explore properties of these three regimes from the perspective of simultaneous clustering and comparison of multiple datasets and examine (1) how some common clustering methods fit into these regimes, and (2) whether they are extendable to clustering of multiple datasets as most clustering approaches are developed for clustering a single dataset at a time. The clustering methods we consider include common nonparametric approaches such as k-means and hierarchical clustering, and model-based entropy-maximisation-type approaches. We also include the newer t-SNE and the histogram-based SOPHE which we recently extended to Multi-SOPHE for clustering multiple datasets. We illustrate these comparisons with an HIV-related dataset. A larger and more comprehensive comparison is the subject of our companion talk (L. Conran and I. Koch: Non-parametric Clustering of Multiple Datasets with Applications to Flow Cytometry).

1. Conran, L. Koch, I., Turlach, B., and Zaunders, J. (2025). Multi-SOPHE, a Joint Clustering and Variable-Selection Approach, Preprint.

2. Doherty, U. P., McLoughlin, R. M., & White, A. (2025). Challenges and Adaptations of Model-Based Clustering for Flow and Mass Cytometry. Wiley Interdisciplinary Reviews: Computational Statistics, 17(1).

Bayesian Statistics 1

11:50am - 1:05pm Monday, 1st December, 2025
403.101 Ken Hall LT
Presentation type Contributed Session
Chairs Khue-Dung Dang

13 CLUE (Collective Level of Uncertainty in Expert Judgement): A Novel Approach to Expert Judgement Elicitation

Chaitanya Joshi[1], Sergeja Slapnicar[2]

[1]University of Adelaide, Adelaide, Australia. [2]University of Queensland, Brisbane, Australia

Abstract

We develop a model to financially quantify cyber risk at organizational level through scenarios in which critical digital assets and critical processes are compromised. While being one of the top business risks, quantifying cyber risk is challenging due to the lack of reliable data, analyses that would appropriately handle expert judgments in replacement of objective data and the fact that the risk is not one, but there can be multiple interrelated risks in an organization. Moreover, any analysis that is relevant for an organization and integrated with its existing cyber risk management needs to take into account its idiosyncrasies by focusing on the relevant threats, considering its digital assets and their vulnerabilities, and the financial value of its processes. We propose a model to evaluate (i) probabilities of threats and interlinked control failures, and (ii) the financial impact of cyber scenarios at the organizational level. The novelty of our approach is in a combination of methods used to a) to elicit experts' collective uncertainty about the likelihood and financial impact of a cyber scenario, and b) to derive the likelihood of scenarios with Bayesian Fault Tree Analysis. We illustrate the applicability of our model in a real-world case, where we quantify the cyber risks faced by an insurance company.

157 Bayesian Hierarchical Modelling of Conveyor Belt Wear Using Functional Data Analysis and Gamma Processes

Ryan Leadbetter ORCID iD[1,2], Aloke Phatak[3,4]

[1]Centre for Transforming Maintenance Through Data Science, Perth, Australia. [2]Australian Antarctic Division, Hobart, Australia. [3]0000-0002-1942-3121, Perth, Australia. [4]Curtin University, Perth, Australia

Abstract

Reliability and maintenance often require the modelling of the wear of a one- or two-dimensional surface to predict remaining useful life. An example of this in the mining and mineral processing industry is a conveyor belt, which companies rely upon to transport ore from mine to port. Hence, unexpected failure of a conveyor belt can cause major downtime and therefore production and profit loss. Measurements of a belt's wearing surface are taken to monitor its degradation and thus the risk of failure. Despite the dire consequences of an unplanned failure, methods to model such degradation data are relatively underdeveloped. In this work, we propose the use of functional data analysis to model the wear profiles of a conveyor belt through time. We show how the functional interpretation of the wear can be extended to the well-established gamma degradation process through the Bayesian hierarchical modelling paradigm. We demonstrate fitting, evaluation, and use of the model using an industry data set from an iron ore conveyor and show how to construct detailed forecasts and failure time distributions with uncertainty intervals for the belt.

193 A Bayesian Approach to Characterise Seasonal Growth of Hass Avocado in Western Australia

Nushrath Najimuddin[1], Daryl Joyce[2], Declan McCauley[3], Helen Thompson[1], James McGree[1]

[1]Queensland University of Technology, Brisbane, Australia. [2]Department of Agriculture and Fisheries, Gatton, Australia. [3]Department of Primary Industries and Regional Development, Manjimup, Australia

Abstract

The aim of this study was to characterise the growth of Hass avocado fruit based on changes in fruit diameter over a single season. Data were obtained from a commercial avocado orchard in southwest Western Australia. Sigmoidal growth models were considered to describe how fruit diameter changes over time. Comparisons between single and double sigmoid growth models were made. A Bayesian hierarchical modelling approach was used to account for variability, including between orchard rows and individual trees, while also incorporating expert knowledge via a formal elicitation process with an avocado horticulturist. The double Gompertz model provided an appropriate fit to the observed growth patterns, effectively capturing the distinct phases of avocado development. The hierarchical structure was able to account for nested sources of variability, reducing the residual variability within the model. The findings advance our quantitative understanding of avocado fruit growth by identifying an effective model that captures developmental phases and underlying variability, providing a quantitative basis for future predictive modelling offering valuable insights for orchard management and yield forecasting.

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98 Tree-Structured Mixtures for Bayesian Spatial Modelling

Clara Grazian

University of Sydney, Sydney, Australia

Abstract

In spatial statistical modelling, the specification of prior distributions is crucial for capturing spatial dependencies and supporting robust inference, particularly in regions where data are sparse or irregularly distributed. We introduce a novel class of spatial priors grounded in distance-driven mixture models with a spanning tree structure. This framework incorporates spatial distances directly into the prior hierarchy by employing tree-constrained clustering, where each node represents a spatial location and connections reflect distance-based similarity. The spanning tree constraint enables efficient partitioning of space, ensuring coherence across spatial units while preserving the capacity to model sharp transitions or nonstationary behaviour. The resulting priors adaptively group spatial locations according to similarities in their latent or observed responses, enabling flexible, data-informed borrowing of strength across space.

Unlike traditional Gaussian Markov random field priors or conditional autoregressive models, our approach does not require pre-specified neighbourhood structures or strong parametric assumptions. Instead, it offers a semi-parametric and interpretable framework that is particularly advantageous when the spatial structure is complex or unknown. The method is computationally scalable, allowing application to high-resolution spatial datasets.

We demonstrate the versatility of the proposed framework through two real-world applications from oceanography and epidemiology.

Longitudinal Studies

11:50am - 1:05pm Monday, 1st December, 2025
402.220 LT

Presentation type Contributed Session
Chairs Cameron Patrick

59 Prediction of Anger Intensity Using Wearable-Derived Physiological Data in Trauma-Affected Populations

Sophie Zaloumis ORCID iD[1], Karen Lamb[1], Olivia Metcalf[1], David Forbes[1], Meaghan O'Donnell[1], Tianchen Qian[2], Tracey Varker[1], Sean Cowlishaw[3]

[1]University of Melbourne, Melbourne, Australia. [2]UC Irvine, Irvine, USA. [3]Monash University, Melbourne, Australia

Abstract

Digital technologies offer potential to assess and predict dysregulated mood and behavior. We aimed to determine if wearable-derived physiological data can predict high anger intensity. Trauma-exposed adults with problem anger were recruited in Australia between 2022 and 2024. Participants were asked to complete pre- and post-study online self-report surveys and 10-day ambulatory assessment, involving continuous Garmin wearable device data and ecological momentary assessments (EMA) captured using smartphones four times daily. Problem anger from EMAs was defined as anger intensity of ≥7. Garmin stress scores within 10 minutes prior to EMAs were considered as continuous variables.

Prediction modelling involved a two-stage approach: summary measures were calculated from continuously measured stress scores and then included as predictors in a generalized linear mixed effects model (GLMM) of problem anger. Twenty-four summary measures were considered, with factor analysis used to select a subset of non-redundant summary measures. Each participant needed ≥4 Garmin stress scores recorded 10 minutes before an EMA to calculate the summary measures. Train/test splits, forward chaining cross-validation and population informed forward chaining were used to assess whether predictive performance will generalise to an independent dataset.

Eighty-four of 98 recruited participants had usable data for modelling, providing 1166 observations with both anger intensity and stress score measured. Factor analysis selected five summary measures. The area under the receiver operating curve for predicting high anger intensity was high using these measures, ranging from 0.76 (95% CI: 0.57-0.92) to 0.90 (95% CI: 0.78-0.98) across cross-validation methods examined.This finding suggests that stress scores from a wearable device can reliably detect anger intensity in the 10 minutes prior to the anger episode. In this study population the predictive performance may be overly optimistic due to the relatively low prevalence of high anger intensity episodes (13.8%). Therefore, this model should be verified using external validation studies.

116 E-Cigarette Use Among Young Australians: Insights from the Longitudinal Study of Australian Children

Amanda Vittiglia ORCID iD[1], Neha Swami[1], Constantine Gasser[1], Michelle Jongenelis[2], Monsurul Hoq[1]

[1]Australian Institute of Family Studies, Melbourne, Australia. [2]University of Melbourne, Melbourne, Australia

Abstract

In Australia, the use of e-cigarettes (or vapes) has increased in recent years, particularly among young adults. Understanding the motivation as well as risk and protective factors is crucial in developing effective prevention strategies to reduce e-cigarette use. International research has predominantly focused on identifying risk factors, while considerably fewer studies have explored protective factors in depth. This research often relied on cross-sectional data whereas longitudinal data can provide a greater understanding of the relationship between protective factors and subsequent e-cigarette use. There are very few longitudinal studies on this issue, particularly in the Australian context where research is limited overall.

The Longitudinal Study of Australian Children (LSAC) is an ongoing study that began in 2004 with a nationally representative sample of 10,000 Australian families. The families, including study children and their parents or caregivers, have been surveyed approximately every two years. LSAC collected data on e-cigarette use for the first time in 2023-24 from young people aged 19-20 and 23-24 years.

In this study we will present detailed demographic statistics on the prevalence of e-cigarette use among young Australians, alongside an exploration of their motivations for e-cigarette use. We will utilising the longitudinal data from previous waves and report on the protective factors and how they affect the likelihood of e-cigarette use.

Given e-cigarette use is a significant public health issue in Australia and internationally, these findings could inform policies aimed at reducing e-cigarette use.

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181 Developing a National Atlas of Risk-Standardised Medication Use for Australian Residential Aged Care

S. Sandun Malpriya Silva ORCID iD, Magdalena Z. Raban ORCID iD, Nasir Wabe ORCID iD, Md Bayzidur Rahman, Amy D. Nguyen ORCID iD, Johanna I. Westbrook ORCID iD

Australian Institute of Health Innovation, Macquarie University, Sydney, Australia

Abstract

Medication management is a key quality indicator in Australian residential aged care (RAC). However, current national indicators such as polypharmacy and antipsychotic use are reported without adjusting for resident-level risk factors, limiting fair comparisons across facilities and regions. This study aimed to develop a National Atlas of Medication Use in RAC that accounts for these differences through appropriate risk adjustment.

This study used daily medication administration data from 2018 to 2022, sourced from an electronic medication management system, covering 46,182 residents in 428 RAC facilities across six Australian states and territories. Resident comorbidities were extracted from free-text fields. To calculate standardised rates, we compared several risk adjustment models: generalised linear models (GLMs) with robust Huber-White standard errors, generalised estimating equations, and generalised linear mixed models. All models adjusted for age, sex, and comorbidities, with non-linear time trends modelled using penalised cubic splines. Expected probabilities were predicted at the resident level and aggregated at the facility, regional, and state levels. Indirectly standardised rates were calculated by comparing observed to expected outcomes. Model calibration and discrimination were assessed to evaluate performance, with GLMs found to be suitable for most indicators at the population level.

The interactive Atlas, built using an innovative R Shiny framework using drilldown functionalities, presents standardised rates for five key medication indicators: psychotropic use, opioid use, sedative load, laxative use, and polypharmacy. Visualisations include geographic heat maps, time series plots, and funnel plots. The tool allows users to explore variation in medication use over time and across locations based on selected indicators.

This Atlas is the first of its kind to use daily resident-level administration data for risk-standardised reporting. It enables more meaningful, data-driven comparisons of medication use across RAC facilities, time periods, and geographic areas, supporting policy, clinical quality improvement, and transparency in aged care.

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184 Respiratory Events and Emergence Delirium – Post Hoc Propensity Score Matched Analysis of Data (RED PANDA)

Anwen Taplin[1,2,3,4], Anne Louise de Barros Garioud[5,6], Daisy Evans[1,2,3,4], Yufei Shang[1,2], David Sommerfield[1,2,3,7], R Nazim Khan[3,4,2], Britta S von Ungern-Sternberg[1,2,3,7]

[1]Department of Anaesthesia and Pain Medicine, Perth Children’s Hospital, Perth, Australia. [2]Perioperative Medicine Team, The Kids Research Institute Australia, Perth, Australia. [3]Institute for Paediatric Perioperative Excellence, The University of Western Australia, Perth, Australia. [4]School of Physics, Mathematics and Computing, The University of Western Australia, Perth, Australia. [5]Department of Anaesthesiology, Juliane Marie Centre, Copenhagen University Hospital – Rigshospitalet, Copenhagen, Denmark. [6]Department of Clinical Medicine, University of Copenhagen, Copenhagen, Denmark. [7]Division of Emergency Medicine, Anaesthesia and Pain Medicine, Medical School, The University of Western Australia, Perth, Australia

Abstract

One in twenty Australian children undergoes anaesthesia every year, with one in seven experiencing at least one perioperative respiratory adverse event (PRAE). Although most events resolve with limited effects, some can lead

to long-lasting harm. Similarly, emergence delirium (ED), characterised by disorientation, agitation and behavioural disturbance, is a common complication affecting children after general anaesthesia. The RED PANDA study aimed to assess whether pre-medications, which are often prescribed for pre-operative anxiety, are related to the risk of PRAE or ED. Data from 39 studies conducted by our team at PCH (and its predecessor PMH) from 2009 to 2024 in children aged 0-16 years old, where both pre-medication and incidence of PRAE or ED was measured, were merged into a single dataset of over 10,000 patients. This dataset was used in separate propensity score matched analyses to compare whether clonidine or midazolam pre-medication is related to risk of PRAE or ED. The propensity score matched analyses accounted for baseline factors related to whether the patient was prescribed a pre-medication as well as known risk factors for PRAE or ED (eg. year of surgery, patient age, surgery type). Multiple imputation was used to account for missing values in either baseline variables or the outcome of PRAE or ED.

Surgeries were matched in a 1:1 ratio in the complete case and multiply imputed datasets. The overall proportion of patients experiencing PRAE or ED in the matched pre-medication and no pre-medication groups were compared using logistic regression. We found that pre-medication with clonidine or midazolam were not associated with the incidence of PRAE or ED.

Mentoring Session: Statistical Consulting Network

11:50am - 1:05pm Monday, 1st December, 2025
407.208 LT
Presentation type Mentoring Session
Chairs Alun Pope

Join members of the Statistical Consulting Network for a friendly and informative mentoring session. Mentors include Alex Jackson, Alysha De Livera, Ben Maslen, Chris Howden, John Henstridge and Kathrin Schemann, who are here to share their wide-ranging experience.

Chair – Alun Pope

Facilitator – Sue Finch

Lunch (SSA Update & Feedback)

1:05 - 2:00pm Monday, 1st December, 2025

The SSA update and feedback session will commence at 1:10 pm in building 407 room 208.

Big, High Dimensional & Modern Statistics 1

2:00 - 3:30pm Monday, 1st December, 2025
405.201 Haydn Williams LT
Presentation type Contributed Session
Chairs Quan Vu

9 Joint Hyperparameter Optimisation and Neural Architecture Search for Simulation-Based Inference: Applications in Phylodynamics

James Bristow ORCID iD, David Hayman ORCID iD, Matthieu Vignes ORCID iD

Massey University, Palmerston North, New Zealand

Abstract

The field of phylodynamics is crucial for understanding the evolutionary drivers and processes of epidemics. More specifically, the early detection and tracing of the spread of infection through genetic surveillance is advantageous when responding to epidemics. However, it is known that traditional Bayesian phylodynamical models may produce biased and unstable parameter estimates when fitted to large, complex, and uncertain phylogenetic trees. Recent advances in simulation-based inference via deep generative models have demonstrated great promise in inferring more stable and less biased parameter estimates. Moreover, dense vectorial representations of phylogenetic trees may be constructed through the application of geometric deep learning, thereby extracting the most informative features from complex evolutionary tree data in an automated fashion. Despite this promise, the incorporation of deep learning models within phylodynamics introduces further complexity, requiring both model hyperparameters and architecture to be carefully calibrated. Here, we demonstrate the novel application of black-box Bayesian optimisation to jointly tune the hyperparameters and neural architecture of a deep generative model coupled with a graph neural network feature extractor. We first compare and evaluate our simulation-based inference approach on synthetic tree data generated from a prior study, which made use of the baseline PhyloDeep library. Second, we perform a showcase study on a phylogenetic tree reconstructed from 200 HIV-1 sequences using a Birth-Death Skyline Serial (BDSS) phylodynamic model. For our showcase, we find comparable results to the prior baseline PhyloDeep study, noting that our approach can scale to larger and more complex phylodynamic models. In addition, the application of geometric deep learning may largely automate the feature extraction process from complex phylogenetic trees, making the interpretation of such models more practical. Last, we demonstrate that the inclusion of Bayesian optimisation allows us to efficiently calibrate the architecture and hyperparameters of our coupled deep learning model.

68 Meta-Analysis of High-Dimensional Data

Jayamini Liyanage ORCID iD, Luke Prendergast ORCID iD, Alysha De Livera ORCID iD

La Trobe University, Melbourne, Australia

Abstract

Meta-analysis of high-dimensional data, such as those in omics, have focused on synthesizing quantitative evidence from studies reporting multiple effect sizes across several correlated outcomes. A challenge for researchers is that there is often a large number of variables of interest, and hence this high-dimensional setting needs further attention above and beyond what is traditionally considered for multivariate meta-analysis. We have previously introduced a multivariate meta-analysis model for high-dimensional metabolomics data (MetaHD), which accommodates the correlation between metabolites, within- and between-study variances, and missing values. MetaHD however, is limited by its computational cost when applied to other omics fields with higher

dimensionality. In this work we introduce fastMetaHD- a memory-efficient, fast version of MetaHD, and compare traditional and more recent meta-analysis methods, evaluating their relative strengths, limitations, and applicability across various high-dimensional contexts. Through a combination of simulation studies and multiple diverse real-world omics datasets, we assess performance in terms of power, top-ranked fraction, type I error, root-mean-square error, and computational efficiency. Based on the findings, we offer practical guidance for selecting appropriate meta-analysis methods depending on research questions and data at hand, thereby making reliable high-dimensional meta-analysis approaches more accessible to applied researchers.

111 From Black-Box to Transparent Artificial Intelligence: A Statistical Framework for Managing Uncertainty in Medical Text Classification

Prabhashrini Manage ORCID iD[1,2,3], Yutong Wu ORCID iD[4], Jinglan Zhang ORCID iD[1,2], Yuefeng Li ORCID iD[1,2]

[1]School of Computer Science, Queensland University of Technology, Brisbane, Australia. [2]Centre for Data Science, Queensland University of Technology, Brisbane, Australia. [3]University of Moratuwa, Moratuwa, Sri Lanka. [4]The Australian e-Health Research Centre, CSIRO, Brisbane, Australia

Abstract

Modern deep learning classifiers are often perceived as black-box systems delivering impressive results, yet their success fundamentally depends on statistical principles. This study highlights that even state-of-the-art artificial intelligence (AI) models require rigorous statistical treatment to address prediction uncertainty—an essential consideration in high-stakes domains such as healthcare.

We propose a two-stage, statistically grounded framework that integrates confidence-based modeling into multi-class text classification. Central to our approach is an uncertain decision boundary, derived from a discriminative confidence distribution over model output probabilities. This threshold, optimized via grid search, partitions test data into “certain” and “uncertain” subsets based on probabilistic evidence.

To enhance reliability on uncertain instances, we introduce **Statistical Term-Class Embedding (STCE)**—a probabilistically grounded method that combines Term Frequency–Inverse Document Frequency (TF-IDF) weighting with log-likelihood estimates to quantify the association between terms and class labels. These term-class correlations are aggregated into a standardized matrix using Z-score normalization, forming a probabilistic embedding space. Fusing STCE with transformer-generated document representations improves class separability in low-confidence regions by explicitly modeling uncertainty-driven variation.

We evaluate our framework using two benchmark medical datasets—Ohsumed (cardiovascular abstracts) and PubMed (cancer-related abstracts). In our experiments, we implement the approach using a Bidirectional Encoder Representations from Transformers (BERT) based classifier to demonstrate practical application. The results show statistically significant gains in accuracy for uncertain data: at a confidence-based threshold (UncertainT) of 0.8, $p = 0.0008$ on Ohsumed; at UncertainT = 0.65, $p = 0.048$ on PubMed, indicating robustness across varying uncertainty levels.

This work positions statistical thinking—not just model complexity—as central to building trustworthy . It illustrates how classical statistical tools, including confidence distributions and hypothesis testing, can demystify deep learning and enhance reliability. The framework also supports educational integration, offering a practical bridge between statistical theory and real-world AI applications.

144 Neural Networks for Parameter Estimation of the Discretely Observed Hawkes Process

Jason Lambe, Feng Chen ORCID iD, Tom Stindl ORCID iD, Jeffrey Kwan ORCID iD

UNSW Sydney, Sydney, Australia

Abstract

When the sample path of a Hawkes process is observed discretely, such that only the total event counts in disjoint time intervals are known, the likelihood function becomes intractable. To overcome the challenge of likelihood-based inference in this setting, we propose to use a likelihood-free approach to parameter estimation, where simulated data is used to train a fully connected neural network (NN) to estimate the parameters of the Hawkes process from a summary statistic of the count data. A naive imputation estimate of the parameters forms the basis of our summary statistic, which is fast to generate and requires minimal expert knowledge to design. The resulting NN estimator is comparable to the best extant approximate likelihood estimators in terms of mean-squared error but requires significantly less computational time. We also propose to use a bootstrap procedure for bias correction and variance estimation. The proposed estimation procedure is applied to weekly count data for two infectious diseases, with a time-varying background rate used to capture seasonal fluctuations in infection risk.

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154 Amortised Neural Analysis of Sampling Distributions

Michael Bertolacci ORCID iD[1], Andrew Zammit-Mangion ORCID iD[2], Raphaël Huser ORCID iD[3]

[1]The University of Western Australia, Perth, Australia. [2]University of Wollongong, Wollongong, Australia. [3]King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia

Abstract

The sampling distribution of a statistic is fundamental to statistical inference. Its analysis typically requires either complex mathematical analysis, or computationally expensive Monte Carlo simulations repeated for each new analysis. We introduce a likelihood-free framework that amortises this computational cost by training neural networks to learn the mapping from statistics to their sampling properties under arbitrary generative models.

Our approach trains neural networks on synthetic data from the generative model to directly predict sampling distribution functionals (means, variances, quantiles) or to construct flexible variational approximations of entire sampling distributions. Once trained, these networks enable rapid computation of quantities that traditionally require thousands of Monte Carlo samples.

The framework applies broadly to estimators, test statistics, and decision rules. We demonstrate how to leverage these neural approximations for core inferential tasks: performing hypothesis tests, estimating standard errors and bias, and computing power functions.

We validate our approach on problems with known analytical solutions and showcase its utility on complex statistics where traditional methods fail. Our experiments reveal that neural amortisation achieves comparable accuracy to Monte Carlo methods while reducing computational time by orders of magnitude at inference time.

Variable & Model Selection 1

2:00 - 3:30pm Monday, 1st December, 2025
403.101 Ken Hall LT
Presentation type Invited Session
Chairs Damjan Vukcevic

43 Optimally Weighted Sufficient Dimension Reduction

Alan Pearse ORCID iD, Howard Bondell

The University of Melbourne, Melbourne, Australia

Abstract

In prediction and classification problems where the response depends on a large number of covariates, sufficient dimension reduction (SDR) methods extract a small number of linear combinations of the covariates that contain all useful information about the complete distribution of the response variable. The parameter of interest is called the central subspace, and the ‘ideal reduction’ is the projection of the covariates into the central subspace. Many techniques in SDR assume a linearity condition on the distribution of the covariates. In practice, it is more common to make the stronger but verifiable assumption that the covariates jointly follow an elliptical distribution. Previous work has shown that, for non-elliptical covariates, the estimation of the central subspace can be improved by reweighting the covariates to an elliptical ‘target distribution’. At the time, this reweighting was presented as a method of producing elliptical covariates, but examination of the theory reveals a more subtle mechanism of action. We show the true mechanism of action in reweighting the covariates and consider its connections to the problem of covariate-shift adaptation in machine learning. Using these insights, we tune the target distribution to enable the most accurate inferences on the central subspace. We illustrate the technique in simulated and benchmark datasets, where we compare unweighted, heuristically reweighted, and optimally reweighted versions of several estimators of the central subspace.

11 Elastic Net for Vector Generalized Linear Models

Wenqi Zhao ORCID iD

University of Auckland, Auckland, New Zealand

Abstract

We present a novel implementation of elastic net regularization for vector generalized linear models (VGLMs) (Yee 2015), extending sparse estimation techniques to the flexible and powerful modeling framework offered by the VGAM package (Yee 2025) in R. Our method incorporates lasso, ridge, and their convex combination into the VGLM setting, which generalizes classical GLMs to accommodate multivariate responses, multiple responses, and structured constraint matrices.

The algorithm is implemented in the new VGAMplus package via the vglmnet function, using a modified iteratively reweighted least squares (IRLS) procedure with pathwise coordinate descent, warm-start initialization, and KKT condition checking for efficient convergence. Unlike traditional penalized GLM methods, our framework supports a broad spectrum of models beyond the exponential family, including ordinal, categorical, and multi-parameter distributions.

We compared our package with leading R packages such as glmnet, ordinalNet, and mpath across a range of models, including Gaussian, binomial, Poisson, ordinal families, negative binomial, and zero-inflated families. We also demonstrate its flexibility through applications to specialized distributions such as the Weibull and Lindley families, as well as bivariate binary models with odds ratios.

This work unifies the structural flexibility of VGLMs with the stability and sparsity of elastic net regularization, providing a general framework for interpretable, scalable modeling in high-dimensional settings. The implementation supports both classical and complex modeling scenarios, making it an ideal tool for statisticians and data scientists working with modern regression problems.

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61 A Large-Scale Data-Driven Study to Understand the Impact of Environment on Rain-Grown Cotton Performance

Qian Feng, Pierce Rafter, Iain Wilson, Warren Conaty, Zitong Li

CSIRO Agriculture and Food, Canberra, Australia

Abstract

In Australia, cotton is traditionally grown as an irrigated crop. However, depending on seasonal outlooks and commodity prices, up to 20% of the annual cotton crop can be produced under rain-grown conditions (i.e. rely solely on rainfall for crop production) without any irrigation. Due to concerns regarding constraints on the availability of irrigation water in the future, one of the best avenues for expansion of the cotton industry may be through increased rain-grown cotton production. A greater emphasis on rain-grown production systems will require the Australian cotton industry to develop an efficient process to breed for new lines of cotton that are well adapted to rain-grown conditions. Genetic selection for high performing plants could fulfil this role but would first require an understanding of the relationship between environmental conditions and plant performance. This understanding will be important not only to account for the confounding effects of environment on plant performance when estimating plant genetic merit, but also to identify and exploit genotype-by-environment interactions. The present study uses the multi-layer group-Lasso, a high-dimensional statistical approach, to quantify the impact of environmental covariates (rainfall, temperature, humidity, and solar radiation data) on fibre yield and fibre quality traits of rain-grown cotton collected over 23 years at 4 locations. We find that the same set of environmental covariates could largely affect multiple traits simultaneously. The results of this study will primarily be used to support future genetic or genomic prediction tools for use in breeding decisions to develop new cotton varieties better suited to rain-grown production systems.

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97 An Application-Specific Approach to Modelling the Probability of Unsuccessful Initial Attack on Wildfires in Victoria, Australia

Kristy Butler[1,2], Elena Tartaglia ORCID iD[2], Jason Rennie[2], Stephen Deutsch[2], Nick McCarthy[1]

[1]Country Fire Authority, Melbourne, Australia. [2]Department of Energy, Environment and Climate Action, Melbourne, Australia

Abstract

Bushfires are a natural hazard that pose a risk to life, property and the environment in Victoria, Australia. Initial attack refers to the initial set of firefighting resources sent to a fire and is crucial to prevent the fire becoming a disaster. On days of elevated fire danger, there are often multiple fires burning at the same time and a need to allocate resources effectively between events. Decisions about how to allocate resources between fires are based on the expected difficulty of controlling the fire. One common way to estimate this expected difficulty is the McArthur Fire Danger Index (FDI), which indicates fire danger based on weather and fuel inputs. Statistical models present the opportunity to directly estimate the probability of a fire remaining uncontrolled after initial attack. Our study models the probability the fire will be unable to be controlled by initial attack based on weather, vegetation, topography and accessibility. We compare the model’s predictions to current common practice of using FDI and discuss the practicality of using the model in operations to assist determining which fires might need an escalated response.

Spatial Modelling

2:00 - 3:30pm Monday, 1st December, 2025
402.220 LT

Presentation type Contributed Session
Chairs Clara Grazian

191 Mapping Kidney Failure in Australia: Statistical Spatial Analysis Approaches and Their Applications

Lijalem Tesfaw[1,2], Mark Tiong[3,4], Nicholas Osborne[5,6,7], Gail Williams[5], Darsy Darssan[1]

[1]School of Public Health, The University of Queensland, Australia., Brisbane, Australia. [2]Department of Statistics, Bahir Dar University, Bahir Dar, Ethiopia. [3]Department of Nephrology, The Royal Melbourne Hospital, Melbourne, Australia. [4]Department of Medicine (RMH), University of Melbourne, Melbourne, Australia. [5]School of Public Health, The University of Queensland, Brisbane, Australia. [6]School of Population Health, University of New South Wales, Sydney, Australia. [7]European Centre for Environment and Human Health (ECEHH), University of Exeter Medical School, Knowledge Spa, Royal Cornwall Hospital,, Truro, United Kingdom

Abstract

Introduction: Understanding spatial variation in the distribution of kidney failure patients is critical for effective health planning and resource allocation. However, the diversity of spatial methods, each with distinct data requirements and software implementation, can make selecting the appropriate approach challenging.

Objectives: This presentation provides a comparative overview of key statistical spatial analytical techniques, with a focus on how they apply to the geographic patterns of kidney failure patients.

Methods: We reviewed different categories of spatial methods, including exploratory spatial data analysis (ESDA), spatial regression (e.g., Spatial Autoregressive or SAR models), Bayesian models (e.g., BYM2 using INLA), and spatial cluster detection. Applications demonstrate using R (package: spdep, INLA, spatstat), SAS (PROC MIXED/GLIMMIX), GeoDa, ArcGIS/QGIS, and SaTScan.

Results: There is considerable spatial variation in the distribution of kidney failure patients across Australia. This variation highlights the importance of selecting analytical approaches that are well-suited to the underlying data characteristics. Each spatial analysis method offers distinct strengths depending on the research goal and spatial structure. Spatial regression effectively adjusts for autocorrelation, while Bayesian approaches are robust in low-count data across small regions. ESDA reveals clustering patterns, and interpolation fills gaps in spatial coverage. Selecting methods misaligned with data characteristics can distort results and limit interpretability.

Conclusion: No single spatial method is universally applicable. Method selection should be guided by the spatial structure, data type, research objectives, and the statistical software. This presentation offers a roadmap to choosing appropriate spatial tools for analysing the geographical patterns of kidney failure patients in Australia and policy-relevant analysis. Detailed examples and software demonstrations will be provided during the session.

82 Modelling Gunfire in Washington, D.C. Using a Spatiotemporal Hawkes Process With Nonseparable Triggering Function

Tom Stindl ORCID iD

UNSW, Sydney, Australia

Abstract

The United States has an epidemic of gun violence, with the risk of firearm-related death being significantly higher compared to other high-income nations. Implementing intervention strategies to reduce firearm-related activity requires an improved understanding of the temporal, spatial, and contagious dynamics of gunfire. A spatiotemporal Hawkes process is introduced to model the space-time clustering of gunfire in Washington, D.C., in 2018 using data obtained from the ShotSpotter technology. The model has three essential features. First, space-time interactions are introduced for contagious gunfire by modeling the triggering function as a mixture of space-time functions. Second, the productivity of contagious gunfire varies across both time and space. Third, the intensity of non-contagious gunfire incorporates demographic and socioeconomic variables at the census block level to better understand the characteristics that lead to persistent gunfire in particular communities. The intensity also includes cyclic trends such as daily, weekly, and long-term effects. An estimation procedure is proposed within the expectation-maximization framework, and the results are discussed and interpreted.

171 Concordance Between Spatial Maps

Stephanie Hogg, Adrian Baddeley

Curtin University, Bentley, Australia

Abstract

This presentation introduces an approach to compare spatial maps or images over a specific geographic area. Spatial images may be generated for a spatial variable or obtained from GIS data, satellite images, or arial photographs. These images may be obtained for the same variable or measure over time, or for different measures for the same spatial region. A method to compare images and determine if trends in distribution are similar is needed. This method should be able discern similarities across different units and scales.

This paper introduces a method based on rank-concordance and rank-variance to determine similarity and differences between images. It can be applied to observations of the same measure over time where absolute measured values vary, for instance seasonal data. It can also be applied to observations from instruments that differ in scale or units or to totally different types of measures (e.g. different gas species) recorded over the same area. Monte Carlo tests of the spatial maps can determine if data is spatially correlated or if areas within images are significantly different.

This approach was applied to spatial data of industrial gas concentrations from the Murujuga Rock Art Monitoring Project (MRAMP) which have been recorded for several years. Spatial maps were generated using spatial smoothing techniques. Pixels in images were ranked and measures of rank concordance and variation applied using functions under development for the spatstat R package. Spatial Monte-Carlo tests (spatstat R package) were also applied to the maps.

This approach provided important insights into seasonal trends and variation in industrial gas concentrations over Murujuga. This approach can be applied generally across a wide range of spatial imagery and applications. It provides a method for exploring and evaluating spatial data without the use of complex models.

55 Utilising Simulations in Statistical Communication: A Worked Case Study Comparing the Implications of Clustered Linear Modelling Techniques

Wesley Billingham ORCID iD[1], Zac Dempsey ORCID iD[1], Robin Cook ORCID iD[1], Elizabeth McKinnon[1], Matthew Cooper ORCID iD[1,2]

[1]The Kids Research Institute Australia, Perth, Australia. [2]UWA Centre for Child Health Research, Perth, Australia

Abstract

As consultant biostatisticians, we are frequently met with questions like “Would it not make sense to use *this* statistical technique over *that*?”—often by researchers who are familiar with an approach they’ve previously used, seen in a related publication, or pursued on the recommendation of a colleague. These suggestions may reflect disciplinary “norms”, or insights gained from reviewing novel techniques. Whilst valuable, they can be based on misunderstandings of the method’s appropriateness to the project’s research question. It is important, as statisticians, to engage with these suggestions seriously and acknowledge the potential merits of alternative approaches, while also being equipped to clarify when and why a proposed method may be inappropriate. Communicating the latter can be difficult, especially when these differences are nuanced and/or complex.

Through a case study, we demonstrate how simulation studies, when thoughtfully designed and paired with clear visualisations, can be a powerful tool for communicating the nuance of and between different analytical methodologies. Inspired by a real-world project, we used simulations to contrast and compare modelling techniques for nested data structures: namely, cluster robust standard errors and mixed effects models. With preference for these widely used approaches often appearing field-specific, we highlight each method's distinct inferential targets and unique characteristics.

Our approach has proved effective in centring the evaluation, for both researchers and fellow statisticians, on a tangible presentation of model parameter variability as opposed to relying solely on either theoretical/technical details or qualitative descriptions. Grounding methodological discussions in simulated, data-driven examples can result in clearer communication, better-informed decisions, and shared understanding across disciplines. More broadly, we encourage this approach in statistical consulting and collaboration, especially when methodological decisions must be made, to bring researchers on board and provide them with new insights to inform their shared decision making.

IPS: Early Career and Student Statistician Network Panel

2:00 - 3:30pm Monday, 1st December, 2025
407.208 LT
Presentation type Invited Session
Chairs Trent Piccicacco

Afternoon tea

3:30 - 4:00pm Monday, 1st December, 2025

IPS: Biostatistics & Bioinformatics

4:00 - 5:30pm Monday, 1st December, 2025
405.201 Haydn Williams LT
Presentation type Invited Session
Chairs Andrew Grant

65 Target Trial Emulation in Observational Studies to Investigate the Role of Vitamin D on the Risk of Serious Falls and Fractures in Older Adults

Temam Beshir Raru ORCID iD[1,2], Julie Pasco[1,3], Mojtaba Lotfaliany[1], Shiva Ganjali[1], Robyn Woods[3], Joanne Ryan[3], John McNeil[3], Alice Owen[3], Michael Berk[1], Mohammadreza Mohebbi[1]

[1]Deakin University, the Institute for Mental and Physical Health and Clinical Translation (IMPACT), School of Medicine, Barwon Health, Geelong, Australia. [2]School of Public Health, College of Health and Medical Sciences, Haramaya University, Harar, Ethiopia. [3]School of Public Health and Preventive Medicine, Monash University, Melbourne, Australia

Abstract

Background

Observational studies remain essential in settings where randomised trials are infeasible or impractical. However, they are subject to design and analytical limitations that undermine causal inference. Target trial emulation (TTE) provides a structured framework that aligns observational analyses with the design and analytical principles of a randomised trial, thereby enabling causal inference. This study applied TTE to estimate the effect of vitamin D supplementation on preventing serious falls and fractures among older adults.

Methods

We explained the protocol for a two-arm pragmatic randomised trial and emulated it using longitudinal data from a prospective cohort, which includes community-dwelling healthy Australians aged ≥70 years. Propensity scores for vitamin D supplementation were estimated using a multivariable logistic regression model with predefined covariates. Inverse probability of treatment weighting (IPTW) was applied to balance confounders across exposure groups. Intention-to-treat analyses were conducted using Fine-Gray sub-distribution hazard regression model to estimate the risk of serious falls and fractures, accounting for death as a competing risk.

Results

We restricted the participants to new users and aligned the eligibility criteria, assignment of treatment, and time zero to minimise prevalent user and immortal time bias. Adequate covariate balance was achieved across the exposure groups, thereby minimising residual confounder bias. The final cohort included 4,656 eligible participants (698 new users, 3,958 non-users). Vitamin D supplementation was not associated with a statistically significant reduction in the risk of serious falls, any fractures, and osteoporotic-related fractures as indicated by the sub-distribution hazard ratio (sHR=1.14; 95%CI: 0.80-1.63), (sHR=1.15; 95%CI: 0.80-1.66), and (sHR=1.38; 95%CI: 0.82-2.32), respectively.

Conclusions

This study illustrates how TTE can prompt causal inference from observational data when randomised trials are not feasible. The employed TTE did not support a causal link between vitamin D supplementation and the prevention of serious falls and fractures in older adults, as the association was not statistically significant.

83 Inferring Time Of Infection Based On Antibody Data Through A Hierarchical Bayesian Framework

Felicia Bongiovanni[1], Eamon Conway[1], Jodie McVernon[2], Ivo Mueller[1]

[1]The Walter and Eliza Hall Institute of Medical Research, Melbourne, Australia. [2]The Peter Doherty Institute for Infection and Immunity, Melbourne, Australia

Abstract

Modelling of the humoral immune response to pathogens can help us better understand protection against disease. Multiple models have been created to describe the production and waning of antibodies against pathogens, with some assuming a simple exponential decay, others a more complex biphasic decay, and some employing a power-law approach. These provide a powerful tool to better understand the serological response to infection, which in turn can be used to translate a single antibody titre to a time of infection, which is usually an unobserved event.

This work will present a hierarchical Bayesian framework to estimate the time of infection of an individual by fitting antibody kinetics model parameters to data. Methods that have already been established to infer exposure times usually involve machine learning algorithms or assume a simple exponential decay of antibodies. Our approach will provide a powerful way to utilise longitudinal serological data to understand the boosting and biphasic decay of antibodies over time. This information will be used to inform population-level parameters, which will then serve to estimate the time of infection for an individual with a single antibody measurement. This has important implications in incidence estimation and understanding the window of protection for an individual.

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126 Multivariable Mendelian Randomization With Weak Instruments

Andrew Grant

University of Sydney, Sydney, Australia

Abstract

Mendelian randomization is an increasingly popular technique used in genetic epidemiology to estimate the causal effect of a genetically predicted exposure on an outcome of interest. Mendelian randomization can be framed as an instrumental variables analysis using genetic variants as instruments. Its popularity has come off the back of a rapid increase in the availability of genetic association data from large-scale genetic studies. Multivariable Mendelian randomization is an extension of the approach which estimates the causal effects of multiple related exposures simultaneously.

Weak instruments present a barrier to obtaining valid causal inference in the instrumental variables framework. Even if a proposed instrument satisfies the instrumental variables assumptions, effect estimates may still be biased if the instrument is not strongly associated with the exposure. This is particularly pertinent for Mendelian randomization, where proposed genetic instruments often explain only a little of the variation in the exposure trait. This is further exacerbated in the multivariable setting, where instruments are required to not only strongly predict each exposure, but to do so conditional on all other exposures in the analysis.

In this talk, I will present methods for performing multivariable Mendelian randomization which are robust to weak instruments. I will compare frequentist approaches with a Bayesian approach, demonstrating the advantages of the latter with respect to its ability to mitigate weak instrument bias while retaining nominal coverage and type I error rates.

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136 Revisiting Optimal Allocations for Binary Responses: Insights from Considering Type-I Error Rate Control

Lukas Pin[1], Sofía S. Villar[1], William Rosenberger[2]

[1]MRC Biostatistics Unit - University of Cambridge, Cambridge, United Kingdom. [2]Department of Statistics, George Mason University, Fairfax, USA

Abstract

This work revisits optimal response-adaptive designs from a type-I error rate perspective, highlighting when and how much these allocations exacerbate type-I error rate inflation — an issue previously undocumented. We explore a range of approaches from the literature that can be applied to reduce type-I error rate inflation. However, we found that all of these approaches fail to give a robust solution to the problem. To address this, we derive two optimal allocation proportions, incorporating the more robust score test (instead of the Wald test) with finite sample estimators (instead of the unknown true values) in the formulation of the optimization problem. One proportion optimizes statistical power and the other minimizes the total number failures in a trial while maintaining a fixed variance level. Through simulations based on an early-phase and a confirmatory trial we provide crucial practical insight into how these new optimal proportion designs can offer substantial patient outcomes advantages while controlling type-I error rate. While we focused on binary outcomes, the framework offers valuable insights that naturally extend to other outcome types, multi-armed trials and alternative measures of interest.

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221 From Design to Compliance: Leveraging Biostatistical Expertise in Research

Kathrin Schemann ORCID iD

The University of Sydney, Sydney, Australia

Abstract

Maintaining strong research integrity and ethical conduct through rigorous study design and analysis is crucial for researchers, particularly when working with complex biological data from animal and human studies. This talk will examine the critical role of biostatisticians in supporting research integrity and ethical conduct throughout the entire research lifecycle and ensuring compliance with evolving regulatory requirements.

We will explore a compelling case for investing in dedicated statistical support to optimise research outcomes and mitigate risks associated with suboptimal study designs, at the individual and institutional level. This presentation will highlight how biostatisticians can help streamline workflows, enhance study design and reporting rigor and foster a culture of best-practice statistical methodology across research departments, including Animal Ethics Committees (AECs) and their regulator, researchers, and research support staff, to strengthen the robustness of research pipelines. Ultimately, this dedicated statistical support enhances research return on investment and animal welfare while minimising resource expenditure on studies unlikely to detect meaningful effects.

This session provides valuable insights for biostatisticians and bioinformaticians looking to enhance research integrity, promote ethical scientific practice, and maximise their impact at both the individual research project and institutional levels.

Epidemiology 1

4:00 - 5:30pm Monday, 1st December, 2025
403.101 Ken Hall LT
Presentation type Contributed Session
Chairs Thomas Fung

23 Improving Solid Cancer Incidence Estimates in Atomic Bomb Survivors: The Role of Migration in Catchment Area-Based Registries

Hanna Lindner ORCID iD, Richard Sposto ORCID iD, Hiromi Sugiyama ORCID iD, Ritsu Sakata ORCID iD

The Radiation Effects Research Foundation, Hiroshima, Japan

Abstract

Since 1958, the Radiation Effects Research Foundation's Life Span Study (LSS) has tracked cancer incidence in over 120,000 atomic bomb survivors from Hiroshima and Nagasaki. All participants resided within the regional cancer registry catchment areas at study entry, but many have since migrated. Because cancer diagnoses outside the regional catchment areas are not captured, migration introduces bias into incidence rates, background rates in ERR/EAR models, and absolute risk estimates. Simulations based on LSS data suggest that ignoring migration can lead to underestimation of cancer incidence by as much as 35%, underscoring the need for appropriate adjustment.

While full residential histories are not available, we reconstruct migration histories for a sub-cohort using multiple RERF data sources, including address records and proxy indicators. We model time to out- and return-migration using Poisson regression for grouped person-year data with smoothing splines, stratified by city, sex, attained age, and calendar year. From these hazard functions, we compute smooth probability surfaces describing the likelihood of residence within the catchment area, which are applied as an adjustment factor in cancer incidence analyses. We provide a theoretical justification to show that the mechanism of adjustment is valid under a rare disease assumption.

A recent 10-year extension of cancer incidence follow-up (2010–2019), necessitated updating migration estimates. Reanalysis shows that individuals exposed at young ages - especially Nagasaki males - were the most likely to migrate starting in the 1960s, with little subsequent return. Those who entered the cohort at older ages tended to remain in the catchment area for life. Since most of the new incident cancer cases in the extended follow-up occur among those exposed early in life, incorporating migration adjustments is particularly critical to ensure accurate estimation of radiation-related cancer risks.

39 A Hierarchical Copula-Based Sparse VECM for Cause-of-Death Mortality Modelling: Forecasting and Connectedness Analysis

Hasna Afifah Rusyda ORCID iD, Yanlin Shi ORCID iD, Han Lin Shang ORCID iD

Macquarie University, Sydney, Australia

Abstract

As technology advances, increasing life expectancy continues to raise global concerns about longevity risk. Understanding the dynamics of underlying mortality rates requires a thorough examination of the evolution and interdependencies among cause-of-death (COD) rates. In this paper, we propose a copula-based hierarchical modeling framework for seven major COD groups: cancer, diabetes, external causes, influenza, mental disorders, nephritis, and vascular diseases. Our methodology unfolds in two stages. First, we evaluate three hierarchical structures differing in how COD-specific rates aggregate to total mortality, a process known as reconciliation.

Utilizing the Lee-Carter model, we estimate temporal trends and analyze them further through a sparse vector error correction model (VECM) augmented by hierarchical Archimedean copulas. The optimal hierarchical structure is selected by minimizing out-of-sample forecast errors, resulting in two distinct COD groups: those with increasing trends (diabetes, mental disorders, nephritis) and those with decreasing trends (cancer, external causes, influenza, vascular diseases). Second, using U.S. mortality data from 1970 to 2018, we apply the sparse VECM framework to study the interdependencies of temporal patterns of COD mortality rates. The sparsity is determined by the previously chosen two-group structure. The interdependencies are studied via the time connectedness, which is based on the generalized forecast error variance decomposition. Among the COD groups, we find that diabetes and vascular diseases serve predominantly as transmitters, whereas cancer and external causes primarily function as receivers. These findings offer crucial insights for demographers, actuaries, healthcare professionals, and policymakers tasked with assessing longevity risk and managing future mortality-related liabilities.

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87 Survival Estimates of Individuals Living With Multiple Primary Cancers: A Pioneering Analysis Using Australian Cancer Database (ACD)

Aarti Gulyani ORCID iD[1], Nicolas Hart[1,2], Murthy Mittinty[1]

[1]Flinders University, Bedford Park, Australia. [2]University of Technology Sydney, Moore Park, Australia

Abstract

Title: Survival Estimates for Individuals with Multiple Primary Cancers (MPC): A Pioneering Analysis Using Australian Cancer Database (ACD)

Background – ACD records only first occurrence of distinct primary cancers. There are individuals diagnosed with more than one primary cancer over their lifespan. Relative Survival (RS) is currently reported for single primary cancers (SPC) only. Survival estimates for individuals with MPC are non-existential.

Aim - Estimating RS of individuals with MPC compared to SPC by cancer sites.

Method - Queensland cancer data spanning the period from 1982 to 2020 (38 years) was sourced from ACD. Mortality data for Australian population by single age, gender and single year was extracted from Australian Bureau of Statistics. RS estimates at 5, 10, 15, 20 years were calculated using Ederer II method. Age-standardized RS (ASRS) were also calculated.

Results – Of 616,786 individuals reported, 70,096 (11.4%) had more than one primary cancer, comprising the MPC cohort. Males and those aged 55-74 years were more likely to be diagnosed with MPC. Prostate, lung, melanoma, colon and breast were leading secondary cancers. The median survival was higher for MPC (13.7 years) than SPC (9.3 years). Five-, 10-, and 15-year RS was notably higher in MPC (90%, 80%, 71%) than in SPC (66%, 61%, 59%). Age and gender specific RS were higher for MPC than SPC cohort. There were differences by cancer site over time. Individuals with Prostate cancer who later developed another cancer (e.g. colon or bladder) showed better 5-year but worse 10-year survival than prostate cancer alone. However, individuals with melanoma or breast cancer plus additional cancers showed improved 5 or 10-year survival than single cancer cases. Similar results were found for ASRS.

Conclusion – Patients with MPC exhibit better survival overall compared to single cancers. Detecting cancers early, at a treatable stage could be reason for better survival estimates.

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164 Development of a Novel Perioperative Respiratory Risk Management Optimisation Tool Using Statistical Evidence

Daisy Evans ORCID iD[1,2,3,4], Anwen Taplin[1,2,3,4], R Nazim Khan[1,2,4], Berwin Turlach[2,4], Aine Sommerfield[1,2,3,5], Neil Hauser[1,2,3,5], David Sommerfield[1,2,3,5], Britta S von Ungern-Sternberg[1,2,3,5]

[1]Perioperative Medicine, The Kids Research Institute Australia, Perth, Australia. [2]Institute for Paediatric Perioperative Excellence, The University of Western Australia, Perth, Australia. [3]Department of Anaesthesia and Pain Medicine, Perth Children's Hospital, Perth, Australia. [4]School of Physics, Mathematics and Computing, The University of Western Australia, Perth, Australia. [5]Medical School, The University of Western Australia, Perth, Australia

Abstract

Each year 18,000 children undergo general anaesthesia at Perth Children's Hospital, of whom approximately 15% experience perioperative respiratory adverse events (PRAE). Although most PRAE pass without long-term consequences, some lead to catastrophic injuries or even death. We aim to develop a clinical decision support tool to assist anaesthetists to make appropriate perioperative patient management decisions to reduce the incidence of PRAE. Existing PRAE prediction tools perform poorly in datasets other than their training data.

Using data from over 12,000 patients, we are in the process of developing a risk prediction tool for PRAE using Bayesian Network modelling. Bayesian Network models are ideal for clinical decision support tools, since they can easily be visualised in a Directed Acyclic Graph (DAG) and can incorporate possible interventions into their design. Through close collaboration between clinical experts and statisticians, we will ensure that both clinical utility and rigorous statistical methods are prioritised.

Based on data analysis and expert consultation, we have developed a causal DAG for the outcome of PRAE in paediatric general anaesthesia using the GeNIe software from BayesFusion. We will temporally validate this model using a new patient dataset from 2025, and based on these results will recalibrate it if necessary. Future work will involve planning clinical trials to test the impact of translation of this tool to the clinical setting.

The successful development and implementation of this tool could improve anaesthesia safety for children at PCH and around the world.

169 Patient Demographic Changes During COVID-19 Affect Risk Factors for Healthcare-Associated Infection in Intensive Care Unit

Zhuan Kee Leong ORCID iD[1], Ewilly Jie Ying Liew ORCID iD[2], Yeh Han Poh ORCID iD[3], Dianne Cook ORCID iD[1], Michael Lydeamore ORCID iD[1]

[1]Monash University, Melbourne, Australia. [2]Monash University, Subang Jaya, Malaysia. [3]Hospital Sungai Buloh, Sungai Buloh, Malaysia

Abstract

The COVID-19 pandemic disrupted hospital operations and patient flows, especially in intensive care units (ICUs). In particular, the demographics of patients admitted during the pandemic differed greatly from the norm. These demographic shifts pose a challenge to conventional risk factor models for healthcare-associated infections (HAI). Using retrospective ICU data from a major referral hospital in Malaysia during and after the pandemic period, we analysed the trends in patient characteristics and infection incidence. Two logistic regression models were used. Model 1 included age, time at risk, mechanical ventilation, post-COVID period, and their interactions with the post-COVID period. Model 2 was identical, except that age was centered by the weekly median to account for temporal shifts in patient demographics. A key finding was that the age profile of ICU patients shifted during the pandemic peak, coinciding with an outbreak of HAI. In Model 1, younger age appeared to be associated with increased risk of HAI, along with associations with mechanical ventilation, post-COVID period, and the interaction between post-COVID period and time at risk. However, in Model 2 the association of relative age with HAI disappeared while the other associations remained consistent across both models. This suggests that the observed relationship may be confounded by changes in who was being admitted to the ICU during the pandemic rather than individual-level risks. Shifts in patient profiles over time created statistical associations that could mislead infection surveillance and response strategies. This work shows that observed risk factors could reflect systemic changes to underlying patient characteristics, rather than individual-level risks associated with infection. Incorporating contextual changes over time into the analysis is important to avoid biased inference and to inform robust infection control strategies.

Time Series Applications

4:00 - 5:30pm Monday, 1st December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Wesley Burr

16 Time Series Fuzzy Bayesian Logistic Regression for Wildfire Event Prediction

Georgios Charizanos ORCID iD, Haydar Demirhan ORCID iD, Duygu Icen ORCID iD

RMIT, Melbourne, Australia

Abstract

Wildfires are devastating occurrences with far-reaching implications for rural communities. Predicting such events is paramount for early intervention and protection against such events. In this research, we propose a new method for predicting wildfire event probability called Time Series Fuzzy Bayesian Logistic Regression (TS-FBLR) framework. This method is designed to incorporate seasonal elements in the model, while also handling issues of class imbalance, complete separation and multicollinearity. Most wildfire datasets are impacted by significant levels of class imbalance in the binary response variable, while separation and multicollinearity are often present due to strong correlations between predictors. The TS-FBLR framework is applied on 12.5 years of monthly data extracted from 16 forests in Australia, North America and the Mediterranean. We also consider a time series SMOTE application with two deep learning methods for benchmarking. Six performance indicators are used to analyse classification performance of the benchmarking methods. Across all performance indicators, TS-FBLR significantly outperforms the deep learning methods, providing strong classification accuracy on wildfire events. We then showcase a practical approach for forecasting such events on unseen data by sampling 12 more months of data from 6 forests using TS-FBLR coefficients. The results show that the proposed method accurately predicts probabilities for wildfire events during the same high-risk time periods that actual events happen, while also correctly classifying the non-event observations. These findings highlight the TS-FBLR framework as a robust and practical tool for accurate wildfire event prediction and early risk assessment in diverse forest regions.

38 Smart Maintenance for Hidden Deterioration Using RUL Predictions

Soudabeh Shemehsavar ORCID iD[1], Khayrat Nawaya[2]

[1]Murdoch University, Perth, Australia. [2]University of Tehran, Tehran, Iran, Islamic Republic of

Abstract

Hidden deterioration presents significant challenges for maintenance strategies, as it can lead to unexpected failures if not effectively managed. In complex industrial systems degradation often occurs latently, meaning that visible signs like distortion or contamination do not directly reflect the underlying degradation process.

To address this, advanced modelling and predictive techniques are required to estimate the true state of the system using indirect indicators and to schedule maintenance accordingly.

We propose a maintenance policy based on the Remaining Useful Life (RUL) for systems characterized by a two-dimensional stochastic process: a latent degradation level and an observable marker process. The marker is not only correlated with the latent degradation but also tracks its progression over time. Our policy uses observations of the marker and failure data to schedule inspections and estimate latent degradation levels dynamically. This results in an adaptive preventive maintenance threshold that evolves over time. A numerical study is presented to demonstrate the effectiveness of the proposed model in predicting degradation and enhancing maintenance performance, even in scenarios where the correlation between the observable marker and the latent degradation is weak.

165 Enhancing Stock Return Forecasts with Network-Based Shock Diffusion

Shahab Nankali ORCID iD, Laleh Tafakori ORCID iD, Xiaolu Hu ORCID iD, Mahdi Jalil ORCID iD

RMIT University, Melbourne, Australia

Abstract

Forecasting stock returns is complicated by the pervasive interconnections among assets, which traditional univariate time series models often fail to capture. We propose a novel framework that improves predictive accuracy by integrating a network diffusion model into return forecasting. Specifically, we estimate baseline forecasts using ARIMA models and define shocks as the deviations between predicted and observed returns. These shocks are then propagated through a directed network of stocks, where edges represent return-based similarity measures, under a threshold mechanism that transmits shocks when they exceed capacity-adjusted limits. The resulting cumulative shocks quantify the broader contagion effects arising from inter-asset dependencies. By updating the initial shocks with their network-diffused components, we generate adjusted forecasts that incorporate latent spillover dynamics. Preliminary evidence suggests this approach substantially reduces forecast error compared to standard time series models and provides a more realistic representation of systemic influences in financial markets. This methodology offers a transparent and flexible tool to improve return forecasts and manage portfolio risk by explicitly modeling information transmission.

45 Record-Breaking Evidence: Detecting Non-Stationarity in Australian Daily Temperature Extremes Using Record-Based Statistics

Tinula Kariyawasam ORCID iD, Inge Koch, Gary Bryant

RMIT University, Victoria, Australia

Abstract

Record-breaking values, which surpass all previous observations, offer a simple yet powerful statistic for communicating climate change. However, daily record behaviour across Australia has rarely been tested against the assumption of stationarity. We present a distribution-free, record-based framework that couples classical record theory with recent permutation-based hypothesis tests and change-point detection to diagnose shifts in extreme temperatures.

Using 74 years (1950 – 2023) of Australian daily temperature data from 53 high-quality stations, we examine upper records in both daily maximum and minimum temperatures. Results show widespread departures from stationarity: 94% of stations reject the null hypothesis for at least one of the temperature measures, with minimum-temperature records typically shifting almost a decade earlier than maximum-temperature records (median change points 1984 vs 1993). Spatial analysis reveals a marked coastal–inland contrast: coastal minimum records are strongly correlated ($r > 0.6$) with local sea-surface temperature anomalies, whereas inland stations exhibit larger deviations in maximum records, implicating distinct physical drivers.

By quantifying when and where Australian temperature records depart from stationarity, this study provides statistical evidence for local climate risk assessments.

Health and Wellbeing

4:00 - 5:30pm Monday, 1st December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Yunwei Zhang

29 Evaluating the Impact of Social Media on Health-Related Behaviour in Health Promotion Campaigns

Atousa Ghahramani

Victoria University, Melbourne, Australia

Abstract

Social media has become a powerful tool for disseminating health-related information and influencing public health behaviours at scale. While previous studies have highlighted a correlation between user engagement and potential behaviour change, concerns persist regarding the credibility of online content and the challenges of evaluating behavioural outcomes within digital platforms. This often leads to an underestimation of social media's impact on shaping public health actions.

This research addresses these challenges through a mixed-methods investigation into user interactions with health promotion content on social media. Analytical techniques include co-occurrence network analysis and association rule mining to examine hashtag usage during health promotion campaigns, lexicon-based sentiment analysis to assess public attitudes, and actor–activity network analysis to map engagement patterns in social media networks. A source credibility scoring system was developed to evaluate the trustworthiness of sleep health content, and regression analysis was applied to explore the relationship between source credibility and engagement.

A major outcome of this research is the development of a validated multi-layered model comprising Content, Engagement, Source, Network, and Outcome layers. This framework enables the assessment of behavioural intent through digital proxies such as sentiment, engagement rates, and source credibility. By integrating behavioural science with social media analytics, this research offers a replicable framework for evaluating and guiding evidence-based health promotion interventions in complex online environments.

This research advances theoretical and practical understanding of how health-related behaviours are shaped in digital ecosystems, offering actionable insights for researchers, practitioners, and policymakers to improve the effectiveness and credibility of future health promotion campaigns.

148 Causal Machine Learning for Personalised Workplace Mental Health: Evidence from Australian Longitudinal Data

Md Ehsanul Haque Tamal ORCID iD, Kamrul Hassan, Domenic Gasbarro ORCID iD, Khurshid Alam ORCID iD

Murdoch University, Perth, Australia

Abstract

Background: Despite growing interest in workplace flexibility policies, research lacks causal evidence on which interventions work best for whom for mental-health and productivity (MHP). Prior research suffers from endogeneity concerns and limited theoretical integration of boundary management and temporal flexibility theories. Moreover, traditional econometric approaches lack critical treatment effect heterogeneity.

Methods: Using 23 waves of Household, Income and Labour Dynamics in Australia (HILDA) data (2001-2023, 3673 individuals and 46,811 person-years), we implement cutting-edge causal machine learning methods to identify heterogeneous effects of workplace flexibility policies on mental-health and productivity. Our novel approach combines double/debiased machine learning (DML) with comprehensive robustness checks to discover treatment effect heterogeneity without pre-specified functional forms on MHP. We develop an integrated boundary-temporal flexibility framework, using statistical analysis to identify optimal policy portfolios based on individual boundary management preferences. The analysis employs longitudinal mixed-effects models with ML-assisted causal estimation, controlling for time-invariant confounders while capturing complex non-linearities

Results: DML techniques uncover significant treatment effects heterogeneity that conventional analytical approaches fail to detect. Employees who prefer integrated work-life boundaries demonstrate 60% greater mental-health benefits from flexible policies (schedule and location flexibility, decision-making autonomy) compared to employees who prefer separated work-life domains (effect size: 0.12 vs 0.05 standard deviations, $p<0.001$). Cross-validation confirms robust causal relationships, with significant productivity gains (22% wage increases) for optimally matched worker-policy combinations.

Conclusions: Our ML analysis reveals that customised workplace flexibility policies work much better than generic approaches. Results inform evidence-based policy design, suggesting Australian workplaces could achieve 19% greater mental-health improvements through data-guided policy targeting. The methodology provides a template for using administrative data to optimise workplace interventions, with implications for industrial relations policy and organisational design.

Policy Implications: Findings support legislative initiatives requiring employers to offer flexible work arrangements while enabling personalised implementation based on worker preferences and organisational contexts.

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182 Financial Circumstances of Young Australians: Insights From the Longitudinal Study of Australian Children

Neha Swami

Australian Institute of Family Studies, Melbourne, Australia

Abstract

Financial wellbeing is critical for life success. It impacts an individual’s academic success, psychological and physical health, and overall life satisfaction. There is increasing concern that the current cost-of-living crisis and rising unemployment may have a disproportionate and enduring effect on young Australians. However, recent evidence on young people's economic experiences from national samples is limited.

This study uses data from the Longitudinal Study of Australian Children (LSAC). The study sample includes more than 4,000 young people in Wave 10, aged 19-20 and 23-24 years in 2023-24. The data includes measures of education, employment, study, budgeting, saving, loans, debt stress, financial support from family, housing, and homelessness. In this study, we present descriptive statistics based on these measures exploring the diverse financial experiences of young people.

We found that 95% of young people were either employed or studying. Financial challenges were common with more than 50% experiencing debt stress or relying on financial assistance from parents. Around 7 in 10 young people were renting, and around half rented from their parents. We will also present data exploring the association between entrenched disadvantage and experiences of financial challenges.

These findings highlight that many young people experience financial stress and are supported by their parents. These insights could inform polices aimed at alleviating financial stress and improve young people’s overall wellbeing.

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31 Developmental Pathways from Childhood Conduct Problems to Adolescent Psychological Distress: A Structural Equation Modelling Approach with Peer Problems as a Mediator

Nahida Afroz ORCID iD[1,2], Enamul Kabir ORCID iD[1], Khorshed Alam ORCID iD[1]

[1]University of Southern Queensland, Toowoomba, Australia. [2]Comilla University, Cumilla, Bangladesh

Abstract

Background: The developmental pathways linking early conduct problems (CP) to later psychological distress (PSYD) remains insufficiently understood, particularly mediating role of peer problems (PP). This study examines how CP influence adolescent PSYD directly and indirectly through PP using longitudinal data from an Australian cohort.

Methods: Data were drawn from Waves 6–8 (ages 10–15) and Wave 9C2 (ages 17–18) of the Longitudinal Study of Australian Children (LSAC), including 1,850 participants with complete data. CP and PP were assessed using the Strengths and Difficulties Questionnaire (SDQ), while PSYD was measured via the Kessler PSYD Scale (K10). Structural Equation Modelling (SEM) with Weighted Least Squares Mean and Variance Adjusted (WLSMV) estimation tested direct and indirect pathways, with PP as a mediator.

Results: Adolescents with persistent or severe CP and PP were more likely to report severe PSYD at ages 17–18. SEM showed that PP significantly mediated the relationship between CP and PSYD, with direct effects of CP largely nonsignificant. The strongest indirect effect was observed for severe CP at Wave 8 (β = 0.147, p = 0.004), followed by borderline CP at Wave 8 (β = 0.107, p = 0.006), severe CP at Wave 7 (β = 0.094, p = 0.004), and borderline CP at Wave 6 (β = 0.027, p = 0.006). PP also showed strong temporal stability across waves (Wave 6 to 7: = 0.548, p < 0.001 at ages 10–13; Wave 7 to 8: = 0.531, p < 0.001 at ages 12-15). The model showed good fit (CFI = 0.953; RMSEA = 0.025), supporting the longitudinal mediation framework.

Conclusions: This study underscores the critical role of PP in the developmental trajectory from early CP to later PSYD. Early identification and intervention targeting both behavioural and peer relationship difficulties are essential to reduce long-term mental health risks in adolescence.

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64 Typologies of Domestic and Family Violence Perpetrators: A Data-Driven Study

Fan Cheng ORCID iD

NSW Bureau of Crime Statistics and Research, Sydney, Australia

Abstract

Domestic and family violence (DFV) remains a critical social issue in Australia, yet interventions often lack specificity due to the heterogeneous nature of perpetrators. This study classifies DFV perpetrators into distinct groups based on their demographic characteristics, risk of reoffending, risk of revictimisation, and interactions with the New South Wales criminal justice, child protection, housing, and homelessness systems, using administrative data for 196,697 individuals charged with at least one DFV offence between 10 March 2008 and 1 January 2023. The analysis employs a two-step statistical approach: first, *Factor Analysis of Mixed Data* reduces the wealth of available information into four principal components that explain approximately one-quarter of the observed variation, with these components primarily reflecting prior involvement as a defendant in criminal courts, police contact as a victim, engagement with the child protection system as a child, and experiences of diversion under the *Young Offenders Act 1997 (NSW)* or childhood abuse; second, *K-prototype clustering* integrates these principal components with demographic characteristics—including age, gender, Aboriginality, and socioeconomic status—as well as reoffending and revictimisation rates, to categorise perpetrators into eight distinct groups. The results reveal that three of these groups consist predominantly of female perpetrators while four consist predominantly of males, highlighting important gender dynamics; groups with a higher proportion of individuals from low socioeconomic areas also feature a larger share of Aboriginal people and demonstrate higher rates of both reoffending and revictimisation, whereas groups of female perpetrators present a higher risk of revictimisation and a lower risk of reoffending compared to male perpetrator groups. This typology provides critical insights for the design and delivery of targeted interventions aimed at reducing DFV, and the findings underscore the value of integrating multi-system administrative data to inform evidence-based policy and practice in this domain.

Poster Viewing

5:30 - 6:00pm Monday, 1st December, 2025
Track Poster
Presentation type Poster

Library, level 7 (building 105)

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36 Maternal Exposure to High Ambient Temperature and Risk of Stillbirth: A Whole-of-Population Study in South Australia, 2000-21

Chelse Dyer, Adriana Milazzo ORCID iD, Lynne Giles ORCID iD

Adelaide University, Adelaide, Australia

Abstract

Background: The impacts of climate change, including the increasing frequency and intensity of high ambient temperature, are associated with adverse health outcomes. Recently, pregnant women have been identified as an at-risk group. Evidence suggests that high temperature increases the risk of stillbirth, but how any association differs by social disadvantage remains unclear.

Objectives: To investigate the effect of maternal exposure to high ambient temperature on the risk of stillbirth, overall, and by social disadvantage.

Methods: Maternal characteristics and birth outcomes from 1 January 2000 to 31 December 2021 were obtained from the South Australian Perinatal Statistics Collection and linked to gridded daily maximum temperatures. We implemented a cohort study design of 417,207 births including 2,776 (0.7%) stillbirths. We considered potential nonlinear effects of temperature and used natural cubic splines with 3 degrees of freedom, and considered temperature using lags of 0 through to 7 days before delivery to assess the effect of maximum temperature on the risk of stillbirth. Separate models were fit for each lag day, with knots at the 10th, 50th and 90th percentiles of maximum daily temperature. Potential confounders included in the models were maternal age, maternal country of birth, birth setting, plurality, and year of birth.

Results: Maternal exposure to higher temperatures in the week prior to delivery showed little effect on the odds of stillbirth, and there was minimal difference in effects across the gradient of social disadvantage.

Conclusions: Our results inform the expanding field of research on high ambient temperature and stillbirth. It is possible that there are critical periods of exposure earlier in pregnancy that adversely affect risk of stillbirth. We also recommend studies consider the effect of cumulative exposure to high temperature in different gestational epochs to identify a susceptible time-period for stillbirth.

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48 Exploring the Role of Generative AI in Undergraduate Statistics Education: A Mixed-Methods Investigation

Mitra Jazayeri ORCID iD

La Trobe University, Melbourne, Australia

Abstract

Generative AI tools are reshaping higher education across disciplines, including statistics, where many students struggle with conceptual understanding and applied problem-solving. This presentation outlines an evolving

research agenda that investigates the role of Generative AI in supporting student learning in statistics at the undergraduate level.

Our research adopts a mixed-methods approach to explore how students engage with AI tools during learning and assessment, and how this engagement may influence their understanding, confidence, and ability to transfer knowledge. Drawing on survey instruments and designed assessment tasks, we aim to capture patterns of use, perceived utility, and motivational factors related to AI-supported learning.

As part of this broader project, we also aim to map the emerging research landscape around AI in statistics education using bibliometric tools. This will provide a structured overview of current trends, influential contributions, and thematic directions in the field.

An important outcome of this work is the development of a conceptual tool to support other researchers and educators conducting similar studies. This tool is designed to streamline the design of crossover experiments, collection of student feedback, and integration of bibliometric insights, ultimately enhancing research accessibility and reproducibility.

Our presentation will share the scope and aims of this research program and reflect on its implications for curriculum design, digital literacy, and student engagement. Findings from specific studies within the project will be shared in future publications.

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53 Discovering New Adverse Drug Reactions to Zoloft Using the FDA Adverse Event Reporting System

Karen Lu, Indu Bala

Adelaide University, Adelaide, Australia

Abstract

Background:

Zoloft is one of the most prescribed antidepressants in the world. But the adverse effects (AEs) on the label were based on results from 3606 clinical trial participants, which may not be enough to detect important but rare adverse reactions. By analyzing data from the Food and Drug Administration (FDA) Adverse Event Reporting System (FAERS), which contains a wider array of information about adverse effects, it is possible to potentially discover new adverse drug reactions (ADRs) for Zoloft.

Method:

Analysis began by downloading all FAERS data from 2024, processing it into R using the package faersquarterlydata, filtering for indications Zoloft is prescribed for. Afterwards two new datasets are created, filtered to include or exclude Zoloft. AEs with both a proportional reporting ratio (PRR) above 2 and reporting odds ratio (ROR) with a lower 95% confidence interval bound above 1 are tentatively considered potential new ADRs specific to Zoloft.

Results:

Only data from 2024 was used to test the methodology. 798 patients prescribed Zoloft reported 2,507 AEs, while 6417 patients with the same indications not prescribed Zoloft reported 23,349 AEs. Comparing demographic data between these two groups showed that Zoloft users were significantly younger, and indication proportion significantly differed between populations ($p=0.000499$). Using this method, the most prominent potential ADRs were eyelid ptosis, Bell’s palsy, and abnormal gastrointestinal sounds. Since gastrointestinal disorders are already listed on the label as an ADR, this supports the method’s ability to discover ADRs.

Conclusion:

The FAERs database can be used to discover new ADRs. This can provide evidence for future clinical trials to further explore the prevalence of these ADRs. If this research successfully provides support for the existence of new ADRs, I aim to create a version of the program that can automate this process for any drug in the database.

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92 When COVID-19 Specific Vaccines Became an Intercurrent Event

Francesca Orsini ORCID iD

Murdoch Children Research Institute, Melbourne, Australia

Abstract

Introduction and Objective(s)

The COVID-19 pandemic created significant challenges for clinical trials, including unexpected intercurrent events (ICEs) that influenced both data collection and interpretation. The BRACE trial (BCG vaccination to Reduce the impact of COVID-19 in healthcare workers following Coronavirus Exposure) was a phase III, multicentre, randomised placebo-controlled trial investigating whether BCG vaccination reduced COVID-19 incidence in healthcare workers. The trial was stopped early due to the rollout of COVID-19-specific vaccines, which limited its ability to determine the efficacy of BCG. This work aims to describe potential estimands of interest in the BRACE trial and explore their impact on estimated treatment effects.

Methods and Results

The estimand of interest was the difference in the proportion of participants with symptomatic COVID-19 by 6 months between the BCG and placebo groups, estimated using a flexible parametric survival model (Royston-Parmar). In the primary analysis, the ICE of receiving a COVID-19 vaccine was addressed using a hypothetical strategy, censoring participants at the time of their first COVID-19 vaccine dose—reflecting a scenario where COVID-19 vaccination had not occurred. A supplementary analysis used a treatment policy strategy, retaining follow-up data post-COVID-19 vaccine. Estimated treatment effects were similar across both strategies, with narrower 95% confidence intervals in the treatment policy analysis due to the inclusion of more follow-up data.

Conclusions

The ICE of COVID-19 vaccine receipt did not materially change the estimated efficacy of BCG. This case study illustrates the value of the estimand framework in trial design and analysis. By clearly aligning the analysis with the research question, estimands provide clarity on the treatment effect being estimated, particularly in trials disrupted by external events.

139 Developing Robust Predictive Models: A Nested Cross-Validation and Bootstrapping Framework for Optimal Model Selection and Stability Assessment

Xia Li[1], Mark McEvoy[2,3], Georgia Petrou[2]

[1]La Trobe University, Melbourne, Australia. [2]La Trobe University, Bendigo, Australia. [3]University of Newcastle, Newcastle, Australia

Abstract

Predictive modeling in complex data environments requires not only high predictive accuracy but also robust performance across multiple evaluation criteria. This study presents a comprehensive framework for developing and evaluating predictive models using nested cross-validation (CV) and bootstrap strategies across multiple candidate algorithms, with a focus on selecting the optimal model while ensuring stability in variable importance, predictive accuracy, and calibration.

The procedure begins with an outer loop of nested cross-validation to rigorously estimate generalization performance, while the inner loop is dedicated to hyper-parameter tuning and variable selection. Multiple models—including regressions, regularized regressions and tree-based ensembles—are compared based on a combination of assessment indices such as mean absolute percentage error (MAPE), root mean squared error (RMSE), and area under the curve (AUC), etc. depending on the outcome type. Model selection prioritizes not only the best-performing model but also the one with consistent performance across resampling iterations.

We introduce and quantify prediction instability and calibration instability, reflecting variability in predicted outcomes and accuracy across folds. Calibration instability, measured by variations in calibration curves and

intercept/slope across folds, is also used to assess model reliability. Additionally, variable ranking stability is evaluated using measures or selection frequency to determine the consistency of key predictors across resamples.

This framework provides an approach to model selection, going beyond average metrics to evaluate the stability of model behaviour. The proposed procedure enhances model interpret-ability, guards against over-fitting, and improves generalisation—making it well-suited for applications in clinical prediction, economics, and other fields where decision reliability is critical.

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168 Statesynth: A Bayesian State Space Approach to Time-Varying Treatment Effect Estimation

Filip Reierson

Monash University, Melbourne, Australia

Abstract

The synthetic control method (SCM) is widely used for estimating treatment effects in comparative case studies, but Abadie, Diamond, and Hainmueller (2010) advise against its use when pre-treatment fit is poor. While researchers facing poor fit may turn to less constrained methods like dynamic linear regression, these approaches extrapolate in the sense that the synthetic control weights may no longer be positive and sum to one. This research addresses situations where poor fit arises from uncaptured weight dynamics, rather than lack of a solution in the convex hull. We propose “Statesynth”, a Bayesian state space model that preserves the fundamental weights structure of the original SCM while allowing for temporal flexibility. Our approach models the treated unit’s counterfactual outcome as a time-varying weighted combination of control units, where weights follow a softmax transformation of latent states that evolve according to a random walk. This formulation maintains the constraint that weights sum to one and remain non-negative, preserving the no-extrapolation property of traditional SCM. The model is estimated using a Gibbs sampler combined with a particle filter for state estimation and backward sampling for joint posterior inference. Inverse-gamma priors on variance parameters enable closed-form conditional distributions, while a particle filter handles the non-linear observation equation introduced by the softmax transformation. Our Bayesian framework provides theoretically justified uncertainty quantification for both the latent weights and treatment effect estimates. We explore the method’s performance through simulation studies and discuss its possible use.

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178 Copula-Based Modeling for the Maintenance of Systems with Latent Degradation

Connor Stewart-Green, Soudabeh Shemehsavar, Graeme Hocking

Murdoch University, Perth, Australia

Abstract

Maintenance costs are a significant concern across a wide range of industries, often driven by the continuous degradation of equipment and the need for repairs. Among the various strategies developed to address these challenges, Condition-Based Maintenance (CBM) has emerged as one of the most cost-effective. CBM relies on monitoring observable indicators of system health to inform repair decisions. However, a major limitation of this approach is its inability to detect damage that remains latent within a system, remaining undetected until a breakdown occurs, commonly referred to as latent degradation.

Since this type of degradation is inherently unobservable, we must rely on correlated observable indicators, known as markers, to provide indirect information about Latent Degradation. The statistical relationship between these markers and latent degradation is typically modeled using bivariate stochastic processes. However, specifying a suitable multivariate stochastic framework that both possesses a known joint distribution and conforms to known conditional distributions remains a central challenge.

Copula theory offers a promising solution. Introduced by Sklar, copula functions enable the construction of multivariate joint distributions by coupling univariate marginal distributions. In this study, we investigate the application of copula-based models for bivariate Lévy processes to better characterize the joint dynamics between

observable markers and latent degradation. Our aim is to improve inference capabilities for the latent degradation process, thereby contributing to more robust maintenance strategies in industrial settings.

183 The Prognostic Role of Comorbidities on Breast Cancer Survival: A Stage-Stratified Analysis

Mildred Mmbone ORCID iD[1], Kerrie Mengersen[1], Peter Baade[2], Helen Thompson[1], Susanna Cramb[1], Jessica Cameron[1], Kou Kou[2]

[1]Queensland University of Technology, Brisbane, Australia. [2]Cancer Council Queensland, Brisbane, Australia

Abstract

Purpose

Breast cancer survival has improved markedly due to advances in screening and treatment. As survival continues to improve, prognostic models have become increasingly important to guide clinical management and tailor treatment strategies. Traditionally, these models have focused primarily on tumour-related factors, including stage, grade, hormone receptor status, and mode of detection, with very limited studies assessing the role of comorbidities. The aim of this study is to evaluate the prognostic effect of comorbidities on breast cancer–specific survival and to examine whether their impact differs between early (stage I–II) and late-stage disease (stage III–IV).

Methods

A total of 3,323 women aged 20–79 years diagnosed with invasive breast cancer in Queensland between 2010 and 2013 were identified through the Queensland Cancer Register. Demographic and comorbidity information was collected via a questionnaire, while clinical data were obtained from registry linkage and medical record review. Prognostic modelling was done using flexible parametric survival models and missing data imputed through multiple imputation.

Results

The impact of comorbidities on survival was observed primarily among patients with late-stage disease, whereas no significant associations were seen in early-stage patients. In the late-stage group, bronchitis emerged as a significant predictor of poorer breast cancer–specific survival. Other variables associated with worse survival included clinical subtype, mode of detection, tumour grade, and a history of breast or ovarian cancer. Cumulatively, these variables explained approximately 26% of the variation in survival outcomes and achieved a Harrell’s C-statistic of 0.70 (95% CI 0.65–0.76).

Conclusion

Chronic bronchitis was predictive of poorer survival among late stage patients. This underscores the need to consider comorbidities in treatment planning for advanced breast cancer, in line with existing ESMO recommendations. These results help address this gap by providing evidence that can inform more refined treatment strategies and support individualized care planning.

128 On Distinguishing Properties Of Tensors Within Science

Paul Dewick ORCID iD

University of Canberra, Canberra, Australia

Abstract

Tensors traditionally began their journey within the fields of mathematics and physics and were later introduced into other scientific disciplines as they developed, such as data science and statistics. An early example of a tensor system is Einstein’s general theory of relativity (1915), which leads us to understand that a gravitational field is simply a set of space–time–dependent coefficients of a metric. This had a huge impact within the field of physics along with the use and application of tensors. Tensors can be described as multidimensional arrays that are generalizations of vectors and matrices to higher dimensions. Tensors can also be described as quantities that

obey certain transformation laws. In the literature, tensors have different meanings, applications, and uses that can confuse the underlying understanding of tensors and how they are used throughout scientific disciplines, including statistics. The distinguishing properties of tensors are required to fill the gap in the many and varied descriptions and understandings of tensors within the literature, highlighting the key differences between tensors and methods in identifying the different types of tensors, and is consistent with the use of tensors and tensor definitions within many disciplines of science to avoid confusion in applications and misinterpreting the mechanics and results.

Welcome Drinks & Reception

6:00 - 7:00pm Monday, 1st December, 2025

Library, level 7 (building 105)

Foreman Lecture: Phillip Gould

9:00 - 10:00am Tuesday, 2nd December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Lynne Giles

214 Modernising National Statistics in Australia

Phillip Gould

Australian Bureau of Statistics, Canberra, Australia

Abstract

Official statistics underpin evidence-based policy and democratic accountability. But declining survey response rates in Australia and around the world pose challenges for traditional approaches to statistical production. Rising costs, the potential for non-response bias, and survey fatigue signal that we cannot simply work harder at what we have always done.

This address explores the transformation underway at the Australian Bureau of Statistics through three elements of modernisation: sustainable data sourcing, smarter production, and democratised access. It will explore how administrative data integration can facilitate better analysis whilst reducing burden on respondents, the ways that artificial intelligence is modernising statistical production, and how expanded data access is giving broader research communities the ability to generate evidence-based insights.

Morning tea

10:00 - 10:30am Tuesday, 2nd December, 2025

Keynote: Joanne Potts

10:30 - 11:30am Tuesday, 2nd December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Ben Harrap

213 Bridging the Gap Between Data and Decisions in the Age of AI: What It Means to Be a Consultant Statistician Today

Joanne Potts

The Analytical Edge, Blackmans Bay, Australia

Abstract

Recently I was introduced at a workshop as “a *formally trained* statistician,” as if to distinguish me from… what exactly? Data analysts? Database engineers? Machine modellers? AI enthusiasts? Statisticians were never common, but lately it feels like an endangered title — often met with surprise, even disbelief, that one might actually hold a PhD in statistics. Over my career as a consulting statistician, some challenges have been eternal: messy datasets, poor data management, and quality control that’s more aspiration than practice. But new challenges have emerged. The rise of AI tools has changed the consulting landscape, for better and for worse. Clients now arrive armed with model outputs and ChatGPT summaries, brimming with confidence that the analysis is complete — yet often missing the critical thinking needed to (1) check whether the analysis is correct, and (2) turn numbers into meaningful advice. A little knowledge, combined with powerful tools, can be a dangerous mix. Am I simply getting older and grumpier, or are people getting bolder with less knowledge? This talk offers a candid look at consulting in the age of AI, and some reflections on how we can keep bridging the gap between data and wisdom — without losing our sanity (or our sense of humour) along the way.

IPS: Bayesian Statistics

11:30am - 1:00pm Tuesday, 2nd December, 2025
405.201 Haydn Williams LT
Presentation type Invited Session
Chairs Scott Sisson

37 Variance Reduction in Sequential Monte Carlo With Knots

Joshua Bon[1], Anthony Lee[2]

[1]University of Adelaide, Adelaide, Australia. [2]University of Bristol, Bristol, United Kingdom

Abstract

Sequential Monte Carlo (SMC) algorithms, or particle filters, are widely used for approximating intractable integrals, particularly in Bayesian inference and state-space models. We introduce a new variance reduction technique, the knot operator, which improves the efficiency of particle filters by incorporating potential function information into part, or all, of a transition kernel. We show that the knot operator guarantees an asymptotic variance reduction in all cases but those that are trivial, where there is no change. Knots induce a partial ordering of Feynman-Kac models implying an ordering of the asymptotic variance of various particle filters, offering a new approach to algorithm design. Particle filters with knots can be implemented with little to no additional computation cost for a large class of models, including those where the fully adapted auxiliary particle filter (FAAPF) is tractable. With knots, we revisit the FAAPF to show how an asymptotic variance reduction can be guaranteed, even though previous research has suggested it is not possible. We also explore examples related to state-space models and SMC samplers.

79 Bayesian Group Variable Selection via Penalised Credible Region

Weichang Yu[1], Khue-Dung Dang[2]

[1]The University of Melbourne, Melbourne, Australia. [2]The University of Western Australia, Perth, Australia

Abstract

We propose a Bayesian method for grouped variable selection in high-dimensional regression models. Most existing Bayesian methods are subjected to either high computation costs due to long MCMC runs, or yields ambiguous variable selection results due to non-sparse solution output. Our proposed method, GroupPenCr, is built upon the penalised credible region framework, which allows efficient computations of a sequence of sparse solutions via existing algorithms. We focus on settings where the number of predictors grows with sample size n. For this problem, we propose using global-local shrinkage priors and performing GroupPenCr on the mean-field variational Bayes posterior instead. This allows our method to avoid the computational hassle of implementing MCMC on ultra high-dimensional data. Furthermore, we show that this approach is also variable selection consistent. Through extensive simulations, we show that GroupPenCr can outperform common methods for Bayesian group variable selection.

133 When Bayes Goes Bad: Weakly-Regularized Covariate Adjustment Biases Prevalence Estimates

Swen Kuh[1], Lauren Kennedy[1], Qixuan Chen[2], Andrew Gelman[2], Marnie Downes[3], John Carlin[3]

[1]School of Computer and Mathematical Sciences, University of Adelaide, Adelaide, Australia. [2]Columbia University, New York City, USA. [3]Murdoch Children's Research Institute, Melbourne, Australia

Abstract

Bayesian inference allows us to build models that account for multiple sources of uncertainty. Computational advances like Stan have made it easier than ever to combine multiple modelling components in order to propagate uncertainty through to a final estimate. However, this can easily lead to complex models that produce surprising or counter-intuitive estimates.

In this talk we face the challenge of diagnosing the difference between a valid yet surprising inference and an invalid inference caused by an inappropriate model. This work is motivated by an observed artefact that arose when modeling the prevalence of SARS-CoV-2 (COVID-19) specific antibodies in metropolitan Melbourne, Australia in Nov--Dec 2020 based on a serosurvey of blood donors. To fully answer the problem we delve into Bayesian rare events bias, measurement error models, priors and model feedback.

Time Series Methods

11:30am - 1:00pm Tuesday, 2nd December, 2025
405.205
Presentation type Contributed Session
Chairs Rob Hyndman

28 Continuous-Time Nowcasting with Gaussian Process

Xiaolei Wang ORCID iD, Tomasz Woźniak ORCID iD, Weichang Yu ORCID iD

University of Melbourne, Melbourne, Australia

Abstract

We propose a Gaussian Process (GP) factor model for high-frequency, high-dimensional forecasting. Multivariate time series are represented by a small number of latent factors, whose dynamics are modelled in continuous time using GPs. Traditional forecasting models rely on discrete-time dynamics, limiting their ability to capture information between fixed observation intervals. In contrast, our approach treats time series as functional data, enabling forecasts at arbitrary time points. By embedding GP-based continuous-time factors within the dynamic factor model (DFM) framework, we address the curse of dimensionality while capturing short-term correlations. To ensure scalability, we design an efficient simulation smoother-based sampler. Forecasting accuracy is evaluated against the analogous DFM. Our results demonstrate that the proposed method offers competitive predictive performance while providing richer insights into high-frequency fluctuations, establishing the GP factor model as a promising tool for real-time policy analysis and decision-making.

44 On Simulation and Statistical Analysis of Multifractional Time Series

Andriy Olenko ORCID iD[1], Antoine Ayache ORCID iD[2], Nemini Samarakoon ORCID iD[1]

[1]La Trobe University, Melbourne, Australia. [2]University of Lille, Lille, France

Abstract

Multifractional processes extend the concept of fractional Brownian motion by replacing the constant Hurst parameter with a time-varying Hurst function. This allows for modelling data and systems with changing dynamics and to modulate the roughness of trajectories over time. The talk introduces a new class of multifractional models, the Gaussian Haar-based multifractional processes (GHBMP), which is based on the Haar wavelet series representations. The resulting processes cover a significantly broader set of Hurst functions compared to the existing literature, enhancing their suitability for both practical applications and theoretical studies. The theoretical properties of these processes will be discussed. Then, it will be demonstrated how GHBMP can be easily implemented for simulations with various Hurst functions. The applicability and statistical analysis will be illustrated, even for Hurst functions exhibiting discontinuous behaviour. The Rmfrac package was developed to perform simulations of and statistical analysis of multifractional time series. An overview of the Rmfrac package and its main functionalities will be given.

The talk is based on the author's joint papers:
[1] On Construction, Properties and Simulation of Haar-Based Multifractional Processes, (2025), submitted.

[2] Rmfrac: R Package for simulation and analyses of multifractional processes, (2025), submitted.

156 Coherent Estimation and Criminal Justice Program Evaluation in Hierarchical Time Series

Thomas Fung ORCID iD[1], Joanna Wang[2]

[1]Macquarie University, Sydney, Australia. [2]University of Technology Sydney, Sydney, Australia

Abstract

Crime time series data can often be naturally disaggregated based on various attributes of interest, such as crime type or geographical location. When modelling this type of data, the current practice in crime science is to model each series at the most disaggregated level, as it helps to identify more subtle changes. However, authorities and stakeholders often focus on the bigger picture, leading researchers to either simply sum the fitted value series or model the aggregated series independently. This practice often leads to poorer performance at the higher levels of aggregation as the most disaggregated series typically exhibit a high degree of volatility, while the most aggregated series tends to be smoother and less noisy. We will demonstrate how the hierarchical and grouped time series structure can be utilised to provide coherent estimates for all disaggregate and aggregate series while also reconciling them to enhance forecast and criminal justice program evaluation by using all the available information. We will utilise NSW and US crime data alongside the COVID lockdown as the intervention effect for illustrative purposes.

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172 Smoothing, Estimation and Signal Detection for Generalized Uniformly Modulated Processes

Wesley Burr[1], Skye Griffith[2], Glen Takahara[2]

[1]Trent University, Peterborough, Canada. [2]Queen's University, Kingston, Canada

Abstract

Among nonstationary time series, the class of Uniformly Modulated Processes (UMPs) are a particularly tractable group, having evolutionary power spectra that are the outer product of modulating functions of time and the spectrum of a stationary process. Generalized UMPs are those which can be composed as additive combinations of UMPs with additional minor restrictions, and provide a useful tool for decomposition of nonstationary time series in general.

In this presentation, we will discuss recent results on smoothing and estimation of the evolutionary power spectra for the class of 2-GUMPs, with some discussion of the possibility of extending the results more generally to k-GUMPs. We additionally apply the new estimation procedure to the questions of detection of signals ("line components") in nonstationary backgrounds as a generalization of Thomson's harmonic F-test.

These GUMP models are useful in a variety of engineering and science applications, including the analysis of seismic data, and we conclude with a brief summary of their applicability to multi-receiver seismic event data.

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190 Statistical Performance of the Common Trend Test in Controlled Interrupted Time Series Analysis

Sean Gannon[1,2], Michael Waller[1], Louise Marquart-Wilson[1]

[1]University of Queensland, Brisbane, Australia. [2]QIMR Berghofer Medical Research Institute, Brisbane, Australia

Abstract

Interrupted Time Series (ITS) studies are quasi-experimental methods used to assess the impact of interventions, such as public health measures or policy changes, on system performance or outcomes over time. Controlled ITS (CITS) studies offers a potential improvement over ITS, incorporating an appropriate control group to control for the

effect of confounding variables that are common to both groups. CITS analysis assumes that any secular trends influencing the level and/or trend of the outcome have the same influence on both groups. Testing that the control and intervention groups share a common trend prior to the intervention can be assessed using the common trend test (CTT).

The CTT assesses for a difference in trends between groups, $\beta_{1,}$ estimated using regression of the difference of the outcomes between the control and intervention group at each timepoint. The statistical performance of the CTT for continuous outcomes using ordinary least squares (OLS) and four covariance matrix estimators was assessed using simulation studies. Performance of the CTT using simulated preintervention data was evaluated for a range of scenarios typically seen in published CITS studies including varying length of the preintervention phase, first-order autocorrelation, and effect sizes violating the common trend assumption.

Simulation results show that OLS parameter estimates of β_1 are unbiased regardless of positive autocorrelation in the error terms, although positive autocorrelation increases the variance of the estimates. When there is no autocorrelation in the error terms, only the default covariance estimator is appropriate for inferential procedures for time series lengths between 6 and 36. Sandwich covariance estimators exhibit a downward bias, resulting in confidence interval coverage rates well below the nominal 95%. Type II error rates for mild violations of the common trend assumption were quite high for both the default and sandwich covariance estimators in preintervention phases shorter than 12 observations.

Experimental Design / Theory

11:30am - 1:00pm Tuesday, 2nd December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Guy Nason

32 An Exact Test of Two Section Error Variance Heteroscedasticity in Mixed Two Way Layout Incorporating Covariates Experimental Designs Applicable to Agricultural Variety Frost Trials

Angelika Pilkington

Murdoch University, Perth, Australia

Abstract

Two-way layouts are common in grain industry research where it is often the case that there are one or more covariates. It is widely recognized that when estimating fixed effect parameters, one should also examine for possible extra error variance structure. An exact test for heteroscedasticity, when there is a covariate, is illustrated for a data set from frost trials in Western Australia. While the general algebra for the test is known, albeit in past literature, there are computational aspects of implementing the test for the two-way mixed model when there are covariates.

The exact F test is shown to have greater power than the industry standard, and because of its exact size, is preferable to use of the Restricted Maximum Likelihood Ratio Test (REMLRT) based on the approximate asymptotic distribution in this instance. Formulation of the exact test considered here involves creation of appropriate contrasts in the experimental design. This requires choices of observations corresponding to an index set in the linear model for the two-way layout. Comparisons of size and power of the exact F test and REMLRT then ensue. A graphical method to help determine l when it is not known but an ordering can be assumed for the blocks is also presented.

The exact F test has natural extensions when there are unbalanced data, and more than one covariate may be present. Results can be extended to Balanced Incomplete Block Designs. Future work seeks the robustification the exact F test.

49 Reconciling Least Squares and Robustness in the Two-Way Layout Fixed Effects Experimental Design with an Example from R.A.Fisher's book The Design of Experiments.

Brenton R. Clarke ORCID iD

Murdoch University, Perth, Australia

Abstract

The competition between least squares and robust methodologies documented in many books including Hampel et al. (1986) and Clarke (2018), has led many either to dismiss least squares analysis or to dismiss robust methods, and bury their heads in the sand when it comes to the alternative methodology. In Clarke (1983) the justification of the influence curve of Hampel (1974), later called the influence function in Hampel et al (1986), was revealed through conditions sufficient for Fréchet differerentiability of M-estimators, a critical requirement for robustness. An example of a Fréchet differentiable and robust estimator uses the Tukey bi-square estimator of location of Beaton and Tukey (1974), coupled with the MADN estimator of scale, both later being replaced by the MM-estimator. See Clarke (2018, pp169-170).

The practical example of the two-way layout of Fisher (1949) illustrates least squares methodology in Analysis of Variance specifically for the randomized complete block design. In this paper we compare least squares and MM-estimation using the Tukey bi-square. Why this is enlightening is revealed in Clarke and Godolphin (1992, pp. 2520-2522) and Clarke and Monaco (2004), this data being subject to possible swapping of observations as suggested by Daniel (1976, p.159), and also, large residuals for the variety Trebi, not necessarily outliers, but leading one to doubt least squares analysis, without further remodeling.

Comparing least squares and MM-estimation approaches is complicated by the restraints placed on estimated variety effects in the model. The R library 'robustbase' provides MM-estimators with the restraints, for example, of the first row and column effects being set to zero. Using the sum-to-zero restraints to make comparisons achieved by appropriate transformations, we show both least squares and robust inferences essentially agree for this data. This reinforces the results resolving what potentially could have been a schizophrenic like situation.

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160 Design and Spatial Analysis of On-Farm Strip Trials: From Principles to Simulation Evaluations

Zhanglong Cao[1], Barbara Kachigunda[1], Julia Easton[1], Mark Gibberd[1], Suman Rakshit[1,2]

[1]Curtin Biometry and Agricultural Data Analytics, Centre for Crop and Disease Management, Curtin University, Perth, Australia. [2]School of Electrical Engineering, Computing and Mathematical Sciences, Curtin University, Perth, Australia

Abstract

This study explores statistical strategies for the design and analysis of on-farm experiments (OFE), grounded in established principles of experimental design and spatial modelling. Through simulation studies, we assess various approaches, including design layouts combined with Geographically Weighted Regression (GWR) for continuous response variables and Linear Mixed Models (LMM) for categorical treatments. Our results align with key findings in the literature, emphasising the importance of appropriate design and modelling choices for effectively capturing spatial heterogeneity. Additionally, we compare two trial types—large-strip trials and stacked replicated trials—and highlight the significance of data granularity, which informs data collection strategies for industry partners. The findings advocate for appropriate trial designs for different purposes, and increased within-trial sampling, rather than reliance solely on average values to better reflect local variability and enhance inference accuracy. This work underscores the critical role of tailored statistical methods in improving the reliability and practical applicability of OFE outcomes.

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161 Sequential Testing of Complex Hypotheses via Adaptive Martingales and Combinatorial Optimisation

Alexander Ek ORCID iD[1,2], Michelle Blom ORCID iD[3,2], Philip Stark ORCID iD[4], Peter Stuckey ORCID iD[1,2], Damjan Vukcevic ORCID iD[1,2]

[1]Monash University, Clayton, Australia. [2]ARC Training Centre in Optimisation Technologies, Integrated Methodologies, and Applications (OPTIMA), Melbourne, Australia. [3]University of Melbourne, Parkville, Australia. [4]University of California, Berkeley, USA

Abstract

Sequential testing enables hypothesis testing without fixing the sample size in advance, aiming to minimise data collection while maintaining statistical power and control over false positives. Unlike fixed-sample methods, sequential procedures must prevent premature stopping “when the data look good” while allowing flexibility in deciding when to stop.

Our motivating application is the **statistical auditing of preferential elections**, where we test (using a sample of ballots) whether the reported winners match those who would win under a full manual count. Such audits are increasingly important when electronic counting or automation limits transparency. To reduce costs, it is crucial to keep the number of ballots examined as small as possible; sequential methods help by stopping early when evidence is strong, and continuing otherwise.

Auditing preferential elections involves **testing complex hypotheses**, representing all configurations of voter preferences that would alter the outcome. We previously introduced AWAIRE, the first method to rigorously audit such elections without requiring external data (e.g., digitised preferences). AWAIRE tests the composite “grand” null hypothesis by decomposing it into a **union of intersections of simpler null hypotheses**, using **adaptively weighted nonnegative supermartingales** to evaluate evidence against each intersection while maintaining sequential validity.

However, AWAIRE’s scalability was limited to elections with only a few candidates. We overcome this by introducing a **combinatorial optimisation** approach that partitions the grand null using a tree structure, applying a **branch-and-bound** strategy to progressively and efficiently reject large portions. We also extend the collection of simpler null hypotheses and introduce safe heuristics for discarding uninformative ones without compromising type I error control.

We demonstrate the efficiency and scalability of our approach on real and synthetic election data, enabling practical audits of elections with up to 55 candidates. Beyond advancing election auditing, our work showcases the power of interdisciplinary collaboration between statistics and computer science.

Spatial and Spatiotemporal Methods and Theory

11:30am - 1:00pm Tuesday, 2nd December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Kerrie Mengersen

90 Spatio-Temporal Modeling with Fourier Neural Operators

Pratik Nag ORCID iD, Andrew Zammit-Mangion, Sumeetpal Singh, Noel Cressie

University of Wollongong, Wollongong, Australia

Abstract

Spatio-temporal process models are often used for modelling dynamic physical and biological phenomena that evolve across space and time. These phenomena often exhibit environmental heterogeneity and complex interactions that are difficult to capture using traditional statistical process models such as Gaussian processes. This work proposes the use of Fourier neural operators (FNOs) for constructing spatio-temporal models for prediction and forecasting. An FNO is a flexible mapping of functions that approximates the solution operator of non-linear partial differential equations (PDEs) in a computationally efficient manner. It does so using samples of inputs and their respective outputs, without requiring explicit knowledge of the underlying PDE. Here, we introduce FNO-based models within a hierarchical statistical framework, and through extensive simulations we compare their predictions to those from state-of-the-art spatio-temporal-prediction methods. Using sea surface temperature data over the Atlantic Ocean and precipitation data across Europe, we demonstrate the ability of FNO-based hierarchical statistical modelling to capture complex real-world spatio-temporal dependencies, and to provide accurate and valid probabilistic forecasts.

140 Stochastic Modelling and Statistical Analysis of Spatial and Long-Range Dependent Data

Ravindi Nanayakkara ORCID iD

La Trobe University, Melbourne, Australia

Abstract

In statistical theory, observations are generally assumed to be independent. However, this assumption is often an approximation and does not hold for real-world data, where dependencies between observations are common. Consequently, significant interest arose over the past two decades in random processes modelling such dependencies. These models found broad applications in analyzing temporal and spatial data across various fields including cosmology, geoscience, finance, genomics, and many more. The main objective of this study was to investigate long-range and spatial dependencies in such data and to develop stochastic models and statistical methods for effective time series and spatial analysis.

First, the multifractality and multifractionality of spherical random fields were studied with cosmological applications. The primary motivation was to investigate non-Gaussianity and other anomalies in the spatial data. In the literature, three multifractal models with explicit Rényi functions were identified for spherical random fields. This research proposed new theoretical models, conducted numerical studies, and demonstrated a methodology to compute the Rényi function and multifractal spectrum, revealing minor multifractality in Planck cosmic microwave background radiation (CMB) data. For multifractionality, the Hölder exponent is used to measure the roughness of random fields. The pointwise Hölder exponent values were estimated for one- and two-dimensional sky regions,

using the HEALPix ring and nested orderings, respectively. The analysis indicated multifractional behavior in the CMB data. The proposed methodology was applied to identify anomalous regions in inpainted CMB maps.

Finally, the study explored simultaneous estimation for cyclic long-memory processes using semiparametric models with spectral singularities at non-zero frequencies. A generalized filtered method of moments estimated both singularity locations and long-memory parameters, with asymptotic normality established for key statistics. Wavelet transforms were included as a special case within the methodology.

Key words: spherical random fields, multifractality, multifractionality, Rényi function, Hölder exponent, cosmic microwave background radiation, asymptotic normality, cyclic long-memory.

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166 Multi-Level Cluster(ed) Processes for Spatial Point Patterns

Bethany Macdonald, Tilman Davies, Martin Hazelton

University of Otago, Dunedin, New Zealand

Abstract

Spatial point patterns can arise from a vast array of application areas including epidemiology, ecology and geoscience. In cases where points are positively correlated we use clustered processes, such as the log-Gaussian Cox process and Neyman-Scott processes, to model these. In many application areas clustering can occur at multiple scales. For example, disease cases might cluster loosely around major urban centres but then consist of tighter clusters within these large clusters, reflecting localised outbreaks. Similar patterns can occur in ecology with structure of family groups. However, standard clustered processes fail to capture this additional structure in the data.

Inhomogeneous clustered processes have been proposed which incorporate covariates, but in some cases covariates are unavailable, do not properly reflect the population at risk, or simply lack the flexibility necessary. We propose multi-level clustered processes which model this additional structure through the inclusion of extra stochastic components, allowing clustering to be flexibly modelled at different scales.

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194 Characterising the Spectral Variability of the Black Hole at the Centre of our Galaxy using Multi-Output Gaussian Processes

Shih Ching Fu ORCID iD

Curtin University, Perth, Australia

Abstract

Multi-Output Gaussian Process (MOGP) regression extends univariate Gaussian Process regression to cases where multiple response variables should be modelled together. For example, these may be response variables that co-vary because they share an underlying origin. This makes MOGP regression perfectly suited for modelling the multi-band data collected by astronomical facilities such as the Atacama Large Millimeter/submillimeter Array (ALMA). In this instance, the multiple outputs correspond to the brightness of emissions from a celestial source as detected at different electromagnetic (EM) frequencies. Here, the celestial source is Sagittarius A* (Sgr A*), the supermassive black hole (SMBH) at the Galactic centre of the Milky Way, and we assume that its changing brightness over time, known as its light curve, is a stochastic process. Using a Bayesian hierarchical model, we demonstrate how MOGPs can be used to infer the relationship between the brightnesses observed at different frequency bands, known as the spectral index. We compare the results of different modelling decisions, such as the choice of cross-covariance structure, and despite light curves being notoriously sparse and unevenly sampled, we obtain a reasonable estimate of the spectral index. Astronomers are interested in the spectral index because it offers clues to the mechanism behind the EM emission and, accordingly, the astrophysical properties of Sgr A* itself.

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10 Multinomial Multilevel Time-Series Modelling With Application to Current Smoking Status in Australia

Bernard Baffour, Sumonkanti Das, Mu Li, Alice Richardson

Australian National University, Canberra, Australia

Abstract

BACKGROUND:

Despite decades of successful public health strategies that have significantly reduced smoking rates in Australia, smoking remains the leading cause of preventable disease and death. While the proportion of current smokers has steadily declined and the proportion of never-smokers has risen, there has been no consistent trend for ex-smokers. Moreover, these shifts have not occurred evenly across all geo-demographic groups. To support the continued monitoring of smoking trends and inform targeted public health interventions, this study estimates smoking status trends across small domains over a 20-year period (2001–2022). These domains are defined by seven age groups (18–24, 25–29, 30–39, 40–49, 50–59, 60–69, and 70+ years), two genders, and eight Australian states and territories.

METHODS:
The study used micro-data from eight rounds of the Australian National Drug Strategy Household Survey, conducted triennially from 2001 to 2022, to derive direct estimates of smoking status (current, ex-, and never smokers). These estimates informed the development of multinomial multilevel time-series models within a hierarchical Bayesian framework using small-area estimation techniques. The models borrow strength both cross-sectionally and temporally across various aggregation levels—age-sex-state, age-sex, and state-sex—to ensure consistency. Aggregated trend estimates were obtained by weighting predictions at the detailed domain level.

RESULTS:
Ex-smokers were used as the reference group due to their relatively stable proportions, facilitating interpretation of regression coefficients. The models revealed significant random intercepts and slopes at the state-age-sex level and temporal random effects at the state-sex and age-sex levels, modelled through a first-order random walk. These components were essential in achieving stable and consistent trend estimates.

CONCLUSIONS:
This modelling approach provides comprehensive trend estimates across multiple domains, identifying population segments lagging in smoking reduction. The findings can inform targeted interventions, helping Australia meet national health targets more effectively.

Lunch

1:00 - 2:00pm Tuesday, 2nd December, 2025

Social Afternoon

2:00 - 5:00pm Tuesday, 2nd December, 2025

As part of ASC2025, we're excited to offer a range of optional social activities. These activities provide a chance to relax, explore Perth and its surroundings, and connect with fellow delegates in a more informal setting.

Social activities are booked separately from the main conference registration and will incur an additional cost.

See https://www.asc2025.net/social-activities for further details.

Keynote: Tanya Garcia

9:00 - 10:00am Wednesday, 3rd December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Garth Tarr

212 Powering Through: Robust and Efficient Estimation With Right-Censored Covariates

Tanya Garcia ORCID iD

University of North Carolina at Chapel Hill, Chapel Hill, USA

Abstract

In Huntington’s disease research, a current goal is to understand how symptoms change prior to a clinical diagnosis. Statistically, this entails modelling symptom severity as a function of the covariate "time until diagnosis," which is often heavily right-censored in observational studies. Existing estimators that handle right-censored covariates have varying statistical efficiency and robustness to mis-specified models for nuisance distributions (those of the censored covariate and censoring variable). On one extreme, complete case estimation, which utilizes uncensored data only, is free of nuisance distribution models but discards informative censored observations. On the other extreme, maximum likelihood estimation is maximally efficient but inconsistent when the covariate's distribution is mis-specified. We propose a new "SPARCC" estimator (for Semi-parametric Right-Censored Covariate) that is robust and efficient. When the nuisance distributions are modelled parametrically, the SPARCC estimator is doubly robust, i.e., consistent if at least one distribution is correctly specified, and semiparametric efficient if both models are correctly specified. When the nuisance distributions are estimated consistently via nonparametric or machine learning methods, the estimator is consistent and semiparametric efficient. We show empirically that the proposed estimator, implemented in the R package **sparcc**, has its claimed properties, and we apply it to study Huntington's disease symptom trajectories using data from the Enroll-HD study.

Morning tea

10:00 - 10:30am Wednesday, 3rd December, 2025

Keynote: Martin Hazelton

10:30 - 11:30am Wednesday, 3rd December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Nicola Armstrong

215 Fibre Sampling: Where Applied Statistics Meets Algebraic Ring Theory

Martin Hazelton ORCID iD

University of Otago, Dunedin, New Zealand

Abstract

Consider the following problems: inference for interzonal traffic flows in transport engineering, estimation of non-compliance rates in biosecurity surveillance, and analysis of animal capture-recapture counts in ecology. All are highly practical applications of statistics, and all share a common structure, whereby the counts of interest (e.g. interzonal vehicle flow, numbers of non-compliant mail items, patterns of animal sightings) are observed only indirectly, in summarised or corrupted form.

Inference in such situations takes the form of a statistical linear inverse problem. This typically requires sampling from the fibre (i.e. solution set) of values for the latent variables that are consistent with the indirect observations. Fibre sampling for count data is a gnarly problem, because it is difficult to ensure connectivity of a random walk over the fibre. In principle this issue can be addressed by employing Markov bases computed using algebraic ring theory, but the resulting samplers often mix very poorly. This talk explores why this is the case, and describes some new sampling schemes with much improved performance.

IPS: Time Series

11:30am - 1:00pm Wednesday, 3rd December, 2025
405.201 Haydn Williams LT
Presentation type Invited Session
Chairs Adam Sykulski

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14 Forecasting the Age Structure of the Scientific Workforce in Australia

Rob Hyndman ORCID iD, Kelly Nguyen

Monash University, Clayton, Australia

Abstract

I will demonstrate how to forecast the age structure of a workforce using demographic growth-balance equations that incorporate functional data components for graduate influx, international mobility, retirement, and mortality. This model adapts the stochastic population framework proposed by Hyndman and Booth (IJF, 2008), substituting the birth component at age 0 with a graduation process at any age, and integrating retirement dynamics alongside a standard mortality process. It accommodates variable graduate influx, international mobility, and aging, through functional data components.

The model is illustrated by forecasting the future age structure of scientific disciplines in Australia, utilising a variety of data sources. Data limitations are addressed through cohort interpolation and constrained penalised regression splines in order to obtain a workable data set. A combination of local and global ARIMA models is employed to address the temporal dynamics in the time series components.

By forecasting age structures in disciplines such as Physics and Astronomy, Mathematical Sciences, Chemical Sciences, Earth Sciences, and Biological Sciences, the model informs strategic workforce planning and policy development.

This forecasting framework not only aids in strategic planning and policy formulation within the scientific community, but also offers a blueprint for other professional sectors seeking to understand workforce dynamics amid demographic shifts.

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216 Forecasting UK Consumer Price Inflation With Random Generalised Network Autoregressive Processes (RaGNAR)

Guy Nason ORCID iD

Imperial College, London, United Kingdom

Abstract

The Consumer Price Index (CPI) is a key economic statistic that informs stakeholders about inflation. Accurate inflation assessment and good forecasts are vital to understand and control the economy, such as attempted by central banks in setting interest rates. The CPI averages various indices of prices of a basket of underlying items taken from across the economy and versions of it are computed in many countries. The Bank of England was recently criticised for its forecasts, but the 2024 Bernanke Review noted that "the forecast errors made by the Bank and those made by external forecasters are barely distinguishable." Recently, generalised network autoregressive (GNAR) models have proved to be powerful forecasters in many situations. We develop RaGNAR to forecast monthly UK CPI using GNAR models by averaging forecasts over many randomised networks and models. RaGNAR outperforms both traditional benchmark and pre-trained probabilistic time series forecasting models across all horizons and delivers materially more accurate forecasts than the Bank of England across four- to six-month horizons. This is somewhat surprising given the comprehensive nature of the Bank’s forecasting process and the speed and efficiency of RaGNAR. RaGNAR also permits us to identify components that strongly influence the CPI during different periods.

Bayesian Statistics 2

11:30am - 1:00pm Wednesday, 3rd December, 2025
405.205
Presentation type Contributed Session
Chairs Lauren Kennedy

54 Efficient Direct Sampling Schema for Gaussian Hierarchical Models with Proportional Scale Priors

Gentry White ORCID iD[1], Filip Juricev-Martincev ORCID iD[2]

[1]Queensland University of Technology, Brisbane, Australia. [2]Australian Bureau of Statistics, Brisbane, Australia

Abstract

Gaussian hierarchical models are ubiquitous in fields such as spatial statistics, machine learning or time-series analysis. Oftentimes sampling from the posterior density requires costly high-dimensional MCMC schema. We introduce an efficient and scalable direct-sampling schema for any model with a Gaussian likelihood and a latent Gaussian prior that have proportional variances and a fixed precision. Under these conditions, the latent field's full conditional attains closed form, whilst the variance parameter can either be drawn via any univariate sampler or via direct draw if a conjugate prior was used. This method attains significantly higher effective sample size and lower error than a classical MCMC schema. We demonstrate the method on thin-plate splines, CAR spatial models, geographically weighted regression, and self-exciting models showing it produces independent posterior draws with minimal computational overhead. This schema generalises conjugate Gaussian inference and offers an effective alternative to Gibbs or Metropolis schemes in large-scale hierarchical applications. We further demonstrate that this method is highly scalable when combined with Nyström low-rank approximation.

107 Variational Transdimensional Inference

Scott Sisson[1], Laurence Davies[2], Daniel MacKinlay[3], Rafael Oliveria[3]

[1]UNSW Sydney, Sydney, Australia. [2]University of Wollongong, Wollongong, Australia. [3]Data61, CSIRO, Australia

Abstract

This talk will discuss how to implement a variational approximation to a transdimensional (multi-model) posterior distribution. Rather than construct a standard variational approximation of each model, and separately across the distribution of model probabilities, we instead attempt this using a single variational target that spans the entire transdimensional model space. The approach introduces a CoSMIC normalising flow, and embeds this within a variational transdimensional inference (VTI) algorithm. We'll look at Bayesian mis-specified variable selection problem, and observationally note some robustness of estimated VTI model probabilities to poor within-model prior specification.

110 Black Holes and Bayesian Physical-Statistical Models: a Multivariate Mixture Regression Model for Astronomical Data of Black Hole Jets

Arash Bahramian ORCID iD[1], Sara Motta ORCID iD[2], Aloke Phatak[3]

[1]Curtin Institute of Radio Astronomy, Perth, Australia. [2]Istituto Nazionale di Astrofisica, Merate, Italy. [3]Centre for Transforming Maintenance Through Data Science, Perth, Australia

Abstract

Bayesian physical-statistical models are a vital part of astronomical sciences because domain knowledge and physical principles can be used to inform the model structure and prior information when modelling astronomical data. Generally, astronomical data are noisy, sparsely sampled observational data with non-trivial measurement uncertainty. In this context, carefully considered physical-statistical models informed by the limitations of the telescopes and our understanding of the cosmos open a new window to understanding some of the most exotic phenomena in the universe such as black holes. Black holes have no light of their own, meaning the most effective way to study these elusive objects is to observe them as they interact with other stars. During these interactions, the gravity of the black hole pulls matter from the surface of the star and a fraction of the hot stellar plasma falling towards the black hole can get catapulted away as high velocity "jets". Characterising the energy of these jets (through their motion) is a key method to study behaviour of matter near black holes. Astronomers generally approach this by taking a sequence of astronomical images over a period of time (typically weeks to months) to infer the movement of the jets through space. However, these astronomical images are often noisy and are taken sparsely and at uneven rates. Furthermore, the jets themselves behave like moving expanding clouds of smoke. This combination of noisy data and complex astronomical phenomena makes inference about the jet movement challenging. In this talk, I describe a Bayesian physical-statistical model informed by the inner workings of the telescope and the physics of the black holes to address some of these challenges. I also show results from applying the model to observations of jets from a black hole in our Galaxy, revealing previously unknown behaviour in jet launching of black holes.

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115 Recursive Variational Gaussian Approximation with the Whittle Likelihood for Linear Non-Gaussian State Space Models

Bao Anh Vu ORCID iD, David Gunawan ORCID iD, Andrew Zammit-Mangion ORCID iD

University of Wollongong, Wollongong, Australia

Abstract

Parameter inference for linear and non-Gaussian state space models is challenging because the likelihood function contains an intractable integral over the latent state variables. While Markov chain Monte Carlo (MCMC) methods provide exact samples from the posterior distribution as the number of samples go to infinity, they tend to have high computational cost, particularly for observations of a long time series. When inference with MCMC methods is computationally expensive, variational Bayes (VB) methods are a useful alternative. VB methods approximate the posterior density of the parameters by a simple and tractable distribution found through optimisation. This work proposes a novel sequential VB approach that makes use of the Whittle likelihood for computationally efficient parameter inference in linear, non-Gaussian state space models. Our algorithm, called Recursive Variational Gaussian Approximation with the Whittle Likelihood (R-VGA-Whittle), updates the variational parameters by processing data in the frequency domain. At each iteration, R-VGA-Whittle requires the gradient and Hessian of the Whittle log-likelihood, which are available in closed form. Through several examples involving a linear Gaussian state space model; a univariate/bivariate stochastic volatility model; and a state space model with Student's t measurement error, where the latent states follow an autoregressive fractionally integrated moving average (ARFIMA) model, we show that R-VGA-Whittle provides good approximations to posterior distributions of the parameters, and that it is very computationally efficient when compared to asymptotically exact methods such as Hamiltonian Monte Carlo.

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201 Bayesian Inference of a Nearest Neighbor Gaussian Process Model for Pooled Genetic Data

Imke Botha[1], August Hao[2], Nick Golding[2], Daniel Weiss[2], Jennifer Flegg[1]

[1]University of Melbourne, Melbourne, Australia. [2]The Kids Research Institute Australia, Perth, Australia

Abstract

Large scale genetic studies often report the total allele counts of distinct genetic markers, but not their co-occurrence. Inferring haplotype frequencies from this pooled data is a challenging problem, particularly as the number of haplotypes increases geometrically with the number of genetic markers. Previous work applied a Gaussian process model in this setting, but only considered 3 genetic markers due to computational limitations. In this work, we propose a nearest neighbor Gaussian process model and develop an efficient sequential Monte Carlo squared algorithm for Bayesian inference of the model parameters. The resulting model is applied to genetic data relating to antimalarial drug resistance in Africa as a test case of the methods.

Methods for High Dimensional and Network Data

11:30am - 1:00pm Wednesday, 3rd December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Harriet Mason

4 Visualise Non-linear Dimension Reductions as Models in a High-Dimensional Data Space

Piyadi Gamage Jayani Lakshika ORCID iD[1], Cook Dianne ORCID iD[1], Harrison Paul ORCID iD[1], Lydeamore Michael ORCID iD[1], Thiyanga S. Talagala ORCID iD[2]

[1]Monash University, Melbourne, Australia. [2]University of Sri Jayewardenepura, Nugegoda, Sri Lanka

Abstract

Non-linear dimension reduction (NLDR) techniques such as t-SNE and UMAP provide a low-dimensional representation of high-dimensional data by applying a non-linear transformation. NLDR often exaggerates random patterns, sometimes due to the samples observed. However, NLDR views have an important role in data analysis because, if done well, they provide a concise visual (and conceptual) summary of high-dimensional distributions. The NLDR methods and (hyper)parameter choices can create wildly different representations, making it difficult to decide which is best, or whether any or all are accurate or misleading. To help assess the NLDR and decide on which, if any, is the most reasonable representation of the structure(s) present in the high-dimensional data, we have developed an algorithm to show the 2D NLDR model in the high-dimensional space, viewed with a tour. The algorithm hexagonally bins the data in the NLDR view and triangulates the bin centroids to create the wireframe. The high-dimensional representation is generated by mapping the centroids to high-dimensional points by averaging the values of observations belonging to each bin. The corresponding lines in the NLDR view also connect the corresponding high-dimensional points. The resulting model representation is overlaid on the high-dimensional data and viewed using a tour, a movie of linear projections. From this, one can see if the model fits everywhere, or better in some subspaces, or completely mismatches the data. Also, we can see how different methods may have similar summaries or quirks. The process is what Wickham et al. (2015) coined as viewing the “model in the data space," and the NLDR view would be considered viewing the “data in the (NLDR) model space”. This work will interest analysts in understanding the complexity of high-dimensional data, especially in bioinformatics and ecology, where NLDR is prevalent. The algorithm is available via the R package quollr.

122 Is This Normal? A New Projection Pursuit Index to Assess a Sample Against a Multivariate Null Distribution

Dianne Cook ORCID iD

Monash University, Melbourne, Australia

Abstract

Many data problems contain some reference or normal conditions, upon which to compare newly collected data. This scenario occurs in data collected as part of clinical trials to detect adverse events, or for measuring climate change against historical norms. The data is typically multivariate, and often the normal ranges are specified by a multivariate normal distribution. The work presented in this paper develops methods to compare the new sample against the reference distribution with high-dimensional visualisation. It uses a projection pursuit guided tour to produce a sequence of low-dimensional projections steered towards those where the new sample is most different from the reference. A new projection pursuit index is defined for this purpose. The tour visualisation also includes

drawing of the projected ellipse, which is computed analytically, corresponding to the reference distribution. The methods are implemented in the R package, `tourr`.

This is joint work with Annalisa Calvi and Ursula Laa

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199 Maximum Likelihood Estimation for Network Models from Ambiguous Egocentric Data

Pavel Krivitsky ORCID iD

University of New South Wales, Sydney, Australia

Abstract

Egocentric sampling is a common approach to collecting network data in a wide variety of settings, in which a sample of actors in the network ("egos") is asked to provide information about their connections ("alters"). Some designs allow each ego's alters to be unambigously identified; in particular, they can observe if two egos had nominated the same alter and/or if an alter is also an ego. Other designs---such as those that only ask about alters' demographic information to preserve confidentiality---are more ambiguous, in that each alter described can be one of several actors in the network. It is often of interest to fit whole-network models such as exponential-family random graph models (ERGMs) to such data. This facilitates both statistical inference, to identify and test social forces affecting network formation; and simulation, to obtain realisations of networks consistent with observed data. For the unambiguous design, maximum likelihood estimation using missing data techniques is straightforward, if computationally intensive (Handcock and Gile, 2010). For the ambiguous design, it is often possible to use standard survey inference techniques to obtain estimators for an ERGM's sufficient statistics and hence obtain pseudo- maximum likelihood (PMLE) estimates (Krivitsky and Morris, 2017). In this work, we derive and implement computational techniques for maximum likelihood estimators for the ambiguous egocentric data. Simulation studies and applications are used to explore their properties and compare them to those of PMLE, as well as quantifying the information lost compared to unambiguous egocentric data.

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173 Statistical Insights into Graph Pangenomes: Comparative Frameworks for Structural Variant Analysis in Plants

Kristina Gagalova ORCID iD[1], Mario Fruzangohar ORCID iD[2], Mark Derbyshire ORCID iD[1], Julian Taylor ORCID iD[2], David Edwards ORCID iD[3], Adam Sparks ORCID iD[1]

[1]Curtin University, Perth, Australia. [2]University of Adelaide, Adelaide, Australia. [3]University of Western Australia, Perth, Australia

Abstract

Graph pangenomes have emerged as a computational framework to capture the full spectrum of genomic diversity across populations. Unlike linear reference genomes that represent a single individual, graph-based pangenomes enable the simultaneous representation and comparison of multiple genomes within a single, non-linear structure. This method enables a richer, population-level perspective of genomic variation, particularly in plant species where structural variants such as insertions, deletions, inversions, and duplications are prevalent and biologically meaningful.

We focus on recent statistical developments and methodological innovations in the construction and analysis of graph pangenomes, with a spotlight on plant genomics. We study how structural variants, often missed by linear approaches, can be accurately identified and annotated when leveraging graph representations. These variants play critical roles in adaptation, domestication, and phenotypic plasticity in plants, making them a key target for population-level inference.

We further compare several state-of-the-art methods for building and analyzing graph pangenomes, including the PanGenome Graph Builder (PGGB) and Mini-Graph cactus, which employs whole-genome alignment to create unbiased, reference-free graphs. To characterize the quality of the resulting graphs, we use Gretl, a recently developed tool that provides quantitative metrics of graph quality. Additionally, we evaluate tools and pipelines for structural variation discovery and genotyping within graph-based representations, such as the VG toolkit, ODGI, and variant analysis frameworks including Practical Haplotype Graph (PHG), PanGenie, and GraphTyper.

We examine the underlying data structures used by graph-based pangenome tools to evaluate their compatibility with statistical modelling frameworks. As a case study, we focus on structural variants in *Brassica napus* (canola), aiming to understand how genomic structural diversity contributes to agronomic traits and adaptive potential. Gaining insight into how genomic information is encoded within graph structures is crucial for bridging algorithmic advances with statistical inference and for enabling downstream applications such as population-level analyses and selection scans.

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153 Shifted Affine GARCH Models: Applications to Value at Risk

Aaron Gibson-John, Gopalan Nair, Darfiana Nur

University of Western Australia, Perth, Australia

Abstract

The modelling of financial risk requires capturing not only time-varying volatility but also the asymmetric and heavy-tailed behaviour often seen in financial return series. While standard GARCH models provide a strong foundation, they tend to underperform when forecasting extreme losses relevant to Value at Risk (VaR). Recent studies have shown that introducing a variance shift into the Heston–Nandi GARCH (HN-GARCH) model can enhance its ability to capture tail risk and deliver more reliable risk forecasts. Building on this idea, we explore whether this shifting technique can be extended to a broader class of affine GARCH models—particularly those with non-Gaussian innovations. We focus on models incorporating Inverse Gaussian and Gamma-distributed errors, as these distributions are better suited to capturing skewness and kurtosis in financial data.

A key component of this work involves calibrating these shifted models via simulation and maximum likelihood estimation. We investigate parameter recovery and identifiability across repeated trials, aiming to understand the conditions under which the shifting technique improves model fit and interpretability. We also apply both shifted and unshifted models to in- and out-of-sample data from major indices such as the S&P 500, Dow Jones Industrial Average, and NASDAQ. By comparing their predictive power for one-step-ahead VaR estimates, we aim to evaluate the practical benefits of the variance shift in real-world settings. In this talk we will present findings from both the simulation and empirical analysis, highlighting the potential of shifted affine GARCH models as more effective tools for risk management under non-Gaussian market dynamics.

Tjanpi Award Talks

11:30am - 1:00pm Wednesday, 3rd December, 2025
407.208 LT
Presentation type Invited Session
Chairs Joanne Potts

Talks for the Tjanpi Award for best student paper in Environmental Statistics

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219 Model-Robust Bayesian Design for Monitoring Submerged Shoals

Dilishiya De Silva ORCID iD

Queensland University of Technology, Brisbane, Australia

Abstract

Optimal sampling strategies are critical for surveys of deeper coral reef and shoal systems due to the significant costs and logistical challenges of accessing these remote and poorly understood ecosystems. Standard diver-based sampling methods, effective in shallow reefs (<15 m), are not feasible for deeper submerged shoals (>15 m), as survey regions can extend to depths of over 100 m. To address this, we develop a Bayesian design strategy to optimise sampling for deeper submerged shoals. Bayesian designs are typically found by maximising an expected utility with respect to the joint distribution of parameters and responses under an assumed statistical model. Unfortunately, specifying such a model a priori is difficult, as knowledge of the data-generating process is typically incomplete. Our approach tackles this challenge by developing Bayesian designs that are robust to model uncertainty. This is achieved via a generalised additive modelling framework where prior information about the additive component is specified to capture discrepancies between what is assumed and the underlying data-generating process. We demonstrate this approach through the design of a future survey of a submerged shoal off the north-west coast of Australia, assessing the robustness of our designs in comparison to alternatives. To conclude the presentation, extensions of this research will be discussed, including high-dimensional Bayesian spatio-temporal designs using mean field variational Bayes and assessing the transferability of Bayesian designs across different shoals.

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220 From Multivariate Power to Underwater Fish Videos: Advancing Ecological Inference through Simulation, AI, and Distance Sampling

Ben Maslen ORCID iD

University of New South Wales, Sydney, Australia

Abstract

Ecological research increasingly demands robust statistical tools that can handle complexity, uncertainty, and imperfect data. In this talk, I present a multivariate power analysis framework designed for ecological studies involving community-level abundance data. This approach leverages a Gaussian copula model with factor analytical structure to simulate realistic ecological communities, incorporates expert-informed effect directions across taxa, and dramatically reduces computational burden through a critical value-based approach. The methods are described through a developed R package ‘ecopower’, which allows researchers with pilot data to answer a wide range of design questions to assist them in planning their studies.

Building on this theme of improving ecological inference, I also explore recent work on underwater video data analysis of fish assemblages. I discuss the role movement can play in improving AI predictions and supplement these predictions with Monte Carlo simulations to compare the statistical properties of common metrics ecologists

use to summarise this data, highlighting biases and differences in power. Further, by extracting distance information from stereo remote underwater videos (RUVs), I adapt terrestrial distance sampling techniques to estimate fish density while correcting for imperfect detection, using equipment and software that marine ecologists are already frequently implementing, but in new ways. Together, these methods offer ecologists new pathways to design, analyse, and interpret data with greater precision and ecological relevance.

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224 Optimal Prediction of Positive-Valued Spatial Processes Under the Asymmetric Power-Divergence Loss Function

Alan Pearse ORCID iD[1], Noel Cressie[2], David Gunawan[2]

[1]The University of Melbourne, Melbourne, Australia. [2]University of Wollongong, Wollongong, Australia

Abstract

Many environmental-science problems involve the prediction of positive-valued spatial processes in a setting where under-prediction and over-prediction can carry very different consequences. Optimal spatial prediction based on the symmetric squared-error loss (SEL) function, which underpins many established spatial-statistical methods (e.g., kriging), may not be appropriate in this setting. In this presentation, we develop inference for positive-valued spatial processes under the asymmetric power-divergence loss (PDL) function. We derive the corresponding optimal power-divergence (OPD) spatial predictors and their properties relevant to spatial inference. For example, we show that the OPD spatial predictors are biased, but the degree and direction of this bias can be carefully controlled via the power parameter of the PDL functions. Hence, the parameter of the PDL function can be tuned in precise and interesting ways. A general method to derive prediction intervals from loss functions also provides uncertainty quantification for OPDs. The special case of OPD spatial prediction of log-Gaussian spatial processes is considered in detail. For an illustrative application, OPD spatial prediction is used to map soil zinc concentrations on a floodplain of the Meuse River in the Netherlands. This is joint research with Noel Cressie and David Gunawan.

Lunch (R-Ladies and Women in Stats)

1:00 - 2:00pm Wednesday, 3rd December, 2025

The R-Ladies and Women in Stats session will commence at 1:10 pm in building 407 room 208.

IPS: Computing & Visualisation

2:00 - 3:30pm Wednesday, 3rd December, 2025
405.201 Haydn Williams LT
Presentation type Invited Session
Chairs Dianne Cook

129 Automated Assessment of Residual Plots with Computer Vision Models

Patrick Li ORCID iD[1], Dianne Cook ORCID iD[2], Emi Tanaka ORCID iD[1], Susan VanderPlas ORCID iD[3], Klaus Ackermann ORCID iD[2]

[1]The Australian National University, Canberra, Australia. [2]Monash University, Melbourne, Australia. [3]University of Nebraska-Lincoln, Lincoln, USA

Abstract

Plotting the residuals is a recommended procedure to diagnose deviations from linear model assumptions, such as non-linearity, heteroscedasticity, and non-normality. The presence of structure in residual plots can be tested using the lineup protocol to do visual inference. There are a variety of conventional residual tests, but the lineup protocol, used as a statistical test, performs better for diagnostic purposes because it is less sensitive and applies more broadly to different types of departures. However, the lineup protocol relies on human judgment which limits its scalability. This talk presents a solution by providing a computer vision model to automate the assessment of residual plots. It is trained to predict a distance measure that quantifies the disparity between the residual distribution of a fitted classical normal linear regression model and the reference distribution, based on Kullback-Leibler divergence. Evaluation using data from a prior human-subject experiment shows the model exhibits lower sensitivity than conventional tests but higher sensitivity than human visual tests. Several examples from classical papers and contemporary data will be presented to demonstrate the method's practical utility in automating residual diagnostics.

202 A Unified Approach to Extracting Publicly Available Australian Agroecological Georeferenced Data for Analysis

Adam Sparks ORCID iD

Curtin Biometry and Agricultural Data Analytics, Centre for Crop and Disease Management, Curtin University, Bentley, Australia

Abstract

Researchers often have the need to gather data for their analyses that include publicly available data sets. However, these data sets often have differing formats and leave the researcher to go find ways of accessing the data through downloading in a browser, using a specialised package or rolling their own code in their software language of choice. I created {fifo} to allow Australian researchers to find and extract data for analysis streamlining the process by using a unified interface and a unified format for outputs that are easily joined to make analysis-ready data sets in R. {fifo} was born out of a need to be able to quickly and easily gather environmental data to support the analysis of plant pathogenic isolates that were collected for a population genetics study and was also used to support modelling work in the same research project. Since the conclusion of this initial project, development of the package has continued, adding new sources of data that provide valuable resources to Australian researchers that need to easily access data for their research. The package was built by using a combination of existing API client packages and making {fifo} an API client itself. Care was taken to ensure that the arguments for accessing data can use the same language and inputs across several data sets to ease access for researchers. Currently nine public data sets are supported, with researchers' own spatial data also being supported and adding access to more data sets is straightforward. Several methods exist to access data, but to my

knowledge there are few others that simplify and streamline the access and joining of data like {fifo} does. Streamlining the gathering and unifying the formatting of data lowers the barrier for researchers to be able to complete the work that they care about most.

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204 mixgb: Multiple Imputation Through XGBoost

Yongshi Deng ORCID iD[1,2], Thomas Lumley ORCID iD[1]

[1]University of Auckland, Auckland, New Zealand. [2]Oritain Global Limited, Auckland, New Zealand

Abstract

Multiple imputation is widely used for handling missing data. Traditional methods often require proper model specification to perform well, and can be computationally demanding for large and complex datasets. In this talk, I will introduce our R package **`mixgb`**, which provides a scalable and automated solution for multiple imputation by leveraging XGBoost, subsampling, and predictive mean matching. I will discuss how to evaluate multiple imputation methods using simulation studies and I will demonstrate how our visual diagnostic package **`vismi`** can be used to assess imputation quality.

Big, High Dimensional & Modern Statistics 2

2:00 - 3:30pm Wednesday, 3rd December, 2025
405.205
Presentation type Contributed Session
Chairs Martin Hazelton

18 Testing for Replicated Signals Across Multiple Studies with Side Information

Ninh Tran, Dennis Leung

University of Melbourne, Melbourne, Australia

Abstract

Partial conjunction (PC) p-values and side information provided by covariates can be used to detect signals that replicate across multiple studies investigating the same set of features, all while controlling the false discovery rate (FDR). However, when many features are present, the extent of multiplicity correction required for FDR control, along with the inherently limited power of PC p-values—especially when replication across all studies is demanded—often inhibits the number of discoveries made. To address this problem, we develop a p-value-based covariate-adaptive methodology that revolves around partitioning studies into smaller groups and borrowing information between them to filter out unpromising features. This filtering strategy: 1) reduces the multiplicity correction required for FDR control, and 2) allows independent hypothesis weights to be trained on data from filtered-out features to enhance the power of the PC p-values in the rejection rule. Our methodology has finite-sample FDR control under minimal distributional assumptions, and we demonstrate its competitive performance through simulation studies and a real-world case study on gene expression and the immune system.

80 Utilising Genetically Predicted Gene Expression to Identify Risk Factors for Dyslexia

Danielle Adams[1,2], Murray Cairns[1,2]

[1]University of Newcastle, Newcastle, Australia. [2]Hunter Medical Research Institute, Newcastle, Australia

Abstract

Dyslexia is a type of learning disability which effects word reading accuracy, fluency and comprehension. Dyslexia affects approximately 10% of Australians and is the most common cause of reading, writing and spelling difficulties. Additionally, it is a highly heritable disorder with estimates from twin studies indicating a heritability of approximately 70%. A genome wide association studies (GWAS) which estimates the association between common variants and disease has recently been published for dyslexia utilising 51,800 self-reported cases and 1,087,070 controls. Current treatment for dyslexia involves learning support tools and teaching techniques, while no pharmaceutical targets are available. Identifying genes associated with changes in disorder risk could reveal biological mechanism related to dyslexia. GWAS can be utilised in combination with expression quantitative trait loci to model the effects of genetically regulated expression on dyslexia through a transcriptome wide association study (TWAS). Across 14 tissue types (N=100-558) from the brain and whole blood we identified 112 unique signals which were associated with dyslexia after multiple testing correction. Conditional analysis which is used to adjust for the predicted expression of nearby genes further refined this to 56 conditionally independent genes. Although these genes are associated with dyslexia, further evidence is required to determine if they may causally influence disorder pathology. Bayesian finemapping is an approach which identifies credible sets of genes with a 90% probability of containing the true causal gene if such as gene exists within the region investigated. We identified 21 genes with strong evidence ($PIP > 0.8$) of potentially causally influencing dyslexia. Triangulated these

results highlights 3 genes (*COMMD7*, *COG5*, *GNAQ*) which were both TWAS conditionally independently associated and possessed strong evidence from finemapping. Exploring the role of these genes and how they may functionally relate to dyslexia could reveal novel insights into disorder pathophysiology.

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170 ROSE: Rank-based Order Statistics Ensemble for High-Dimensional Feature Screening

Ibrahim Joudah ORCID iD[1], Samuel Muller ORCID iD[1,2], Houying Zhu ORCID iD[1]

[1]Macquarie University, Sydney, Australia. [2]The University of Sydney, Sydney, Australia

Abstract

High-dimensional data, where the number of features far exceeds the number of observations, poses significant challenges for statistical modeling and machine learning. In this context, feature screening plays a crucial role by eliminating irrelevant variables while retaining those that are truly influential, a guarantee known as the sure screening property, which ensures that as the sample size increases, all truly influential features are preserved. Ensemble approaches have emerged to combine multiple screening methods and leverage their complementary strengths. However, conventional ensembles require that every constituent method satisfies the sure screening conditions, an assumption often too restrictive for complex real-world datasets. In this work, we introduce ROSE, a novel Rank-based Order Statistics Ensemble that aggregates feature ranks across multiple screening methods based on their order statistics. Under the ROSE framework, each method only needs to satisfy the sure screening conditions for a subset of the truly influential features, provided that the union of these subsets covers the entire set of truly influential features. This implies that if the sure screening conditions of any of the constituent methods are satisfied, the ensemble satisfies sure screening as well. Simulation studies demonstrate that ROSE effectively captures non-linear relationships, maintains robustness against outliers in both the response and predictors, and detects joint linear associations, consistently outperforming both individual methods and traditional ensembles. Applications to omics cancer data further illustrate ROSE’s high potential in complex high-dimensional settings, particularly for cancer classification.

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207 On the Intrinsic Evaluation of Compressed Text Embeddings: A Multitask Benchmarking Framework

Nathan Inkiriwang ORCID iD[1], Garth Tarr ORCID iD[1], Necva Bolucu ORCID iD[2], Maciej Rybinski ORCID iD[3]

[1]University of Sydney, Sydney, Australia. [2]CSIRO, Sydney, Australia. [3]Universidad de Malaga, Malaga, Spain

Abstract

Text embeddings are fundamental to modern Natural Language Processing, yet their high dimensionality leads to substantial storage and computational costs. While compression effectively addresses these challenges, selecting optimal compression methods remains a significant open problem, largely due to the limitations of existing evaluation techniques, which are often either computationally expensive or not generalisable.

Our work introduces a novel, comprehensive, and scalable intrinsic evaluation framework designed specifically for compressed text embeddings. Our framework leverages multiple task-agnostic metrics to systematically assess the preservation of crucial embedding properties, including local neighborhood fidelity, global geometric structure, and spectral information content. A key contribution is the formulation of EOS$_k$, a novel eigen-overlap score that quantifies the alignment between dominant eigenspaces. This metric is particularly robust to embedding anisotropy, a common characteristic of contemporary embedding models.

Empirical analyses across various embedding architectures, downstream tasks, and compression techniques demonstrate that our framework's intrinsic metrics reliably predict downstream performance. More importantly, it yields interpretable insights into how different models preserve essential local versus global structural information under compression. This framework offers a practical, efficient, and interpretable alternative to traditional, resource-intensive extrinsic evaluations for compressed embeddings, with broad implications for deploying NLP systems in resource-constrained environments.

Variable & Model Selection 2

2:00 - 3:30pm Wednesday, 3rd December, 2025 402.220 LT

Presentation type Contributed Session

Chairs Berwin Turlach

30 Robust Variable Selection Methods with Cox Model

Yunwei Zhang ORCID iD

Murdoch University, Perth, Australia. Macquarie University, Sydney, Australia

Abstract

The increasing availability of high-dimensional omics data with censored survival outcomes introduces significant methodological challenges, particularly in the context of variable selection. The inherent skewness and potential contamination in survival time distributions motivate the use of robust variable selection techniques. While recent developments have extended robust methods from linear models to Cox proportional hazards models, their adoption in practice remains limited, potentially due to a lack of systematic evaluation and awareness of their benefits.

In this talk, I will present a comparative review study evaluating the variable selection performance of twelve robust and non-robust penalised Cox models. Our study reveals the intricate relationship among covariates, survival outcomes, and modeling approaches, demonstrating that even minor deviations from ideal model assumptions can substantially influence selection accuracy and model stability.

Our empirical results reinforce the practical utility of robust methods: robust penalised Cox models consistently outperform their non-robust counterparts in the presence of outliers, while maintaining comparable efficiency and accuracy under clean data conditions. These findings advocate for broader implementation of robust approaches in high-dimensional survival analysis and provide methodological insights into the complex relationship between correlated covariates and censored outcomes.

35 False Discovery Rate Controlled Robust Variable Selection under Cellwise Contamination

Xiaoya Sun ORCID iD[1], Houying Zhu ORCID iD[1], Garth Tarr ORCID iD[2], Samuel Muller ORCID iD[3]

[1]School of Mathematical and Physical Sciences, Macquarie University, Sydney, NSW, Australia. [2]School of Mathematics and Statistics, The University of Sydney, Sydney, NSW 2006, Australia. [3]Faculty of Science and Engineering, Macquarie University, Sydney, NSW, Australia

Abstract

The increasing complexity and dimensionality of modern datasets make cellwise outliers a frequent and expected phenomenon, complicating tasks such as variable selection. While many robust variable selection methods have been developed, few effectively handle high-rate cellwise contamination, and it remains an open problem to what extent such methods can control error rates. We introduce GRALF, the Gaussian-Ranked Adaptive Lasso with False discovery rate (FDR) control, a robust method designed for high-dimensional variable selection under cellwise contamination, taking advantage of FDR control to increase statistical reliability. GRALF builds on the framework of GR-ALasso, an effective robust variable selection method using the Gaussian-Rank estimator in the Adaptive Lasso, and integrates FDR control by generating fake counterparts of the Gaussian-Rank transformed variables and estimating the number of false discoveries in the optimization process. Theoretical FDR control is established based on GR-ALasso in infinite sample settings. Simulation studies demonstrate GRALF's desirable performance in terms of empirical power and FDR control under various conditions. Real data analysis further validates the effectiveness of GRALF.

69 Stability Selection via Variable Decorrelation

Mahdi Nouraie ORCID iD[1], Connor Smith ORCID iD[1], Samuel Muller ORCID iD[1,2]

[1]Macquarie University, Sydney, Australia. [2]The University of Sydney, Sydney, Australia

Abstract

The Lasso is a prominent algorithm for variable selection. However, its instability in the presence of correlated variables in the high-dimensional setting is well-documented. Although previous research has attempted to address this issue by modifying the Lasso loss function, this presentation introduces an approach that simplifies the data processed by Lasso. We propose that decorrelating variables before applying the Lasso improves the stability of variable selection regardless of the direction of correlation among predictors. Furthermore, we highlight that the irrepresentable condition, which ensures consistency for the Lasso, is satisfied after variable decorrelation under two assumptions. In addition, by noting that the instability of the Lasso is not limited to high-dimensional settings, we demonstrate the effectiveness of our method for low-dimensional data. Finally, we present empirical results that indicate the efficacy of our method across different variable selection techniques, highlighting its potential for broader application. The DVS R package is developed to facilitate the implementation of our methodology.

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143 Using a supervised principal components analysis for variable selection in high-dimensional datasets reduces false discovery rates

Insha Ullah[1], Kerrie Mengersen[2], Anthony Pettitt[2], Benoit Liquet[3]

[1]ANU, Canberra, Australia. [2]QUT, Brisbane, Australia. [3]Laboratory of Mathematics and its applications, CNRS 5132, UPPA, Pau, France

Abstract

High-dimensional datasets, where the number of variables p is much larger than the number of samples n, are ubiquitous and often render standard classification techniques unreliable due to overfitting. An important research problem is feature selection, which ranks candidate variables based on their relevance to the outcome variable and retains those that satisfy a chosen criterion. This article proposes a computationally efficient variable selection method based on principal component analysis tailored to a binary classification problem or case-control study. This method is accessible and is suitable for the analysis of high-dimensional datasets. We demonstrate the superior performance of our method through extensive simulations. A semi-real gene expression dataset, a challenging childhood acute lymphoblastic leukemia gene expression study, and a GWAS that attempts to identify single-nucleotide polymorphisms (SNPs) associated with rice grain length further demonstrate the usefulness of our method in genomic applications. We expect our method to accurately identify important features and reduce the False Discovery Rate (fdr) by accounting for the correlation between variables and by de-noising data in the training phase, which also makes it robust to mild outliers in the training data. Our method is almost as fast as univariate filters, so it allows valid statistical inference. The ability to make such inferences sets this method apart from most current multivariate statistical tools designed for today's high-dimensional data.

Estimation and Clustering

2:00 - 3:30pm Wednesday, 3rd December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Ross Maller

19 Performance of Model Selection Criteria

Shafiqur Rahman, Syfun Nahar

University of Papua New Guinea, Port Moresby, Papua New Guinea

Abstract

Model selection criteria play an important role in applied Economics, Econometrics and Statistics. At present a large inventory of model selection criteria are available in the literature including AIC, BIC, generalized cross validation (GCV) criterion, Hannan and Quinn's criterion (HQC), Hocking's S_p, joint information criterion (JIC), and Mallows C_p criterion for choosing an appropriate model from several competing alternative models for a particular data set. The performance of any criterion varies from situation to situation. None of them is better in all situations. So, there is a need to investigate which one is performing better in which situation. If we can express all existing criteria in one common form, then it will be easy to compare their performances among themselves. Keeping this in mind we propose a new model selection criterion named 'Generalized Criterion' (GC) based on residual sum of squares with a multiplicative penalty function. Almost all-existing criteria are special cases of GC in terms of probability of correctly selecting the true model. The proposed GC is used to derive general analytical formula for finding the probability of correct selection and testing consistency of all existing model selection criteria. Also, GC can be used to construct new model selection criteria which may perform better than existing model selection criteria.

40 Logarithmic Methods for Coefficient of Variation Estimation with Past Data

Poonam Singh

Banaras Hindu University, Varanasi, India. University of technology, Sydney, Australia

Abstract

The difficulty in estimating the coefficient of variation (CV) is precisely assessing relative variability among datasets with different scales. This is important in areas where generating accurate comparisons can have a big influence on decision-making, like public health, economics, and agriculture. Using a memory-type statistical technique, we present a logarithmic estimator for the CV in this work, encompassing both historical and present data.

A simulation study's findings show that, in comparison to conventional techniques, the estimator performs noticeably better when historical data is included. Furthermore, two real-world case studies that demonstrate the logarithmic estimator's superiority and practical advantages in real-world applications are provided.

119 Possibilistic and Entropy-based Fuzzy Clustering Against Noise Contamination and Overlapping Cluster Centres

Gizem Eker ORCID iD, Haydar Demirhan ORCID iD

RMIT, Melbourne, Australia

Abstract

The presence of noise distorts the results of clustering algorithms. Removal of noise or influential observations does not provide a satisfactory solution as it affects the quality of the results if genuine observations are unintentionally removed. Moreover, when the true cluster centres are close to each other, clustering algorithms have difficulty in separating the estimated clusters. The real life applications range from habitats of species to types of soil where each application can have overlapping cluster centres due to each cluster sharing many common characteristics.

Soft clustering algorithms have the advantage of handling noise over hard clustering methods. The fuzzy c-means (FCM) algorithm and its extensions present viable solutions though they have challenges such as overlapping cluster centres. The FCM algorithm with a noise cluster is a robust fuzzy approach against the outlier observations. However, it is not studied in conjunction with the recently developed possibilistic fuzzy methods, which potentially improve the robustness of the fuzzy clustering algorithms against the outliers. In this study, we develop the noise extensions of the Possibilistic FCM, Improved Possibilistic FCM and generalized entropy-based possibilistic FCM algorithms. We benchmark the proposed algorithms against the mainstream and recent fuzzy clustering algorithms to evaluate the performance of the proposed algorithms under various outlier contamination settings. The benchmarking study is conducted based on sample size, cluster numbers, contamination rate and centre proximity. Under the condition of high pairwise overlap, proposed Possibilistic FCM with noise cluster performs better than Possibilistic FCM. Proposed Improved Possibilistic FCM with noise cluster performs better than Improved Possibilistic under high noise contamination and pairwise overlap. Proposed Possibilistic FCM with noise cluster performs better at centre proximity against Possibilistic FCM.

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175 From Performance Metrics to Statistical Rigor in ML Evaluation

Julio Romero ORCID iD[1], Joanne Thandrayen ORCID iD[2], Shuangzhe Liu ORCID iD[1]

[1]University of Canberra, Canberra, Australia. [2]Australian National University, Canberra, Australia

Abstract

Machine learning (ML) has become central in data science applications across fields, with models such as random forests, logistic regression, and XGBoost widely adopted for classification and prediction tasks. Despite their popularity, a critical issue persists in current research: the comparison and assessment of model performance often rely solely on point estimates of accuracy, AUC, or other performance metrics, without incorporating inferential statistical tools to evaluate uncertainty or generalisation.

This paper argues that robust model comparison requires systematic use of inferential statistics, including probability distributions, confidence intervals, hypothesis testing and effect size measures. Such tools enable researchers to assess whether observed differences in models' performance are statistically significant or simply due to sampling variability, thus strengthening the validity of conclusions drawn from ML studies.

This research is structured in three parts. First, a systematic review of recent literature is conducted to evaluate the reporting standards of inferential statistics in ML model comparisons. Preliminary results indicate that a substantial proportion of publications neglect these crucial metrics, leading to potentially misleading claims about model superiority. Second, a structured framework is proposed for model comparison based on established inferential statistical methods, providing guidelines for incorporating hypothesis tests, confidence intervals, and effect sizes in ML performance reporting. Third, the assessment paradigm is expanded by integrating explainability and interpretability analyses to ensure that model evaluation considers both predictive power and practical understanding, which are essential for trustworthy deployment in real-world settings.

Overall, this work emphasises the need for rigorous statistical assessment in ML research to improve reproducibility, transparency, and informed decision-making. Adopting the proposed approach can advance the field towards more robust, interpretable, and reliable applications of data science and machine learning models.

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81 Convergence of Functionals of the Allele Frequency Spectrum

Ross Maller[1], Soudabeh Shemehsavar[2]

[1]ANU, Canberra, Australia. [2]Murdoch Uni, Perth, Australia

Abstract

For a general class of models, including those constructed from gamma and stable subordinators by Kingman (thus, including the Ewens sampling model) and the two-parameter extension by Pitman and Yor, we derive representations of the distribution of components of the allele frequency spectrum vector (AFS), conditional on the number of alleles, calculated from a sample of genes.

The derived formulae are applied to obtain a variety of limiting results for the normed and centered AFS. Besides the normal, the resulting asymptotic distributions include stable laws with truncated L\'evy measures and the Dickman distribution.

Afternoon tea

3:30 - 4:00pm Wednesday, 3rd December, 2025

IPS: Environment & Ecology

4:00 - 5:30pm Wednesday, 3rd December, 2025
405.201 Haydn Williams LT
Presentation type Invited Session
Chairs Joanne Potts

73 Ice Sea the Future: Statistical Machine Learning to Predict Antarctic Sea Ice With Quantified Uncertainty

Jacinta Holloway-Brown ORCID iD[1], Elizabeth Shine[2], Ariaan Purich[3], Ryan Heneghan[4]

[1]University of Adelaide, Adelaide, Australia. [2]Australian Bureau of Statistics, Canberra, Australia. [3]Monash University, Melbourne, Australia. [4]Griffith University, Brisbane, Australia

Abstract

Antarctic sea ice has reached record lows in recent years. These changes are unprecedented in the observational record. It is important to model these short-term, unusual changes due to the role sea ice plays in regulating the global climate system and potential sea level rise.

We use the Stochastic Spatial Random Forest (SS-RF) method to accurately predict Antarctic sea ice extent a year in advance. This is a peer reviewed, open source machine learning method which is accurate and effective for monitoring land cover change in satellite images. It is fast, non-parametric and generalisable to many types of big data. Importantly, this method explicitly quantifies uncertainty by producing posterior probabilities of land cover which are mapped. This is highly valuable as sea ice extent change is inherently uncertain, and increasingly so as the climate is changing.

222 The Fast and the Fourier-ous: Non-parametric Autocovariance Modeling With Spline Kernels

Lachlan Astfalck ORCID iD

University of Western Australia, Perth, Australia

Abstract

Modeling autocovariance functions (ACFs) is fundamental in time-series, spatial, and spatio-temporal statistics. Standard parametric models can struggle with complex, non-separable dependence, and non-parametric approaches must ensure the ACF remains positive semi-definite. In this talk, I present a new family of non-parametric, closed-form ACFs that are provably dense in a broad class of continuous processes, provide optimally efficient functional representations, and extend naturally to multivariate and multidimensional settings. By avoiding rigid assumptions such as separability, the method captures realistic space-time interactions and can handle irregularly observed data. Illustrations are given from oceanographic spatio-temporal applications.

223 Robust Spatial Smoothing: Balancing Global Spatial Trends While Preserving Rare Extreme Values Using Influence Diagnostics and Winsorization

Mohomed Abraj ORCID iD, Adrian Baddeley ORCID iD

Curtin University, Perth, Australia

Abstract

Spatial smoothing is a widely used statistical technique that enables the visualisation of spatial patterns and trends. The Nadaraya-Watson (NW) kernel smoother is commonly applied due to its simplicity and minimal assumptions. However, this method is highly sensitive to extreme values, particularly when they occur at sampling locations that are remote or environmentally distinct. Such cases can lead to distorted or misleading smoothed surfaces. Winsorization is an efficient method for reducing the impact of spurious outliers. However, not all outliers can be removed in a spatial context, as they may carry important spatial information and be influential. This study investigates the impact of high leverage and high influence points in spatial setting and proposes an approach to improve the robustness of NW spatial smoothing. The concept is adapted from established leverage and influence diagnostics in regression models. The proposed method was applied to elemental composition data collected from rocks in Murujuga, Western Australia. The standard NW method produced poor smoothing results, higher prediction errors, and significant information loss, making interpretation difficult. In contrast, the proposed method identified high leverage and high influence points and was compared with alternative approaches, including the spatially weighted mean, spatially weighted median, log transformation, probability integral transformation, trimmed mean, and winsorization. The method that replaces high influence points using winsorization performed best in terms of cross-validation and provided more robust smoothing results. Overall, this study demonstrates that robust spatial smoothing, incorporating influence diagnostics and winsorization, can effectively balance global spatial trends while preserving rare extreme values, providing more reliable and interpretable spatial environmental analyses.

Genetics and Biomarkers

4:00 - 5:30pm Wednesday, 3rd December, 2025
405.205
Presentation type Contributed Session
Chairs Tanya Garcia

67 A Novel Mixture of Expert Model for Phenotype Prediction Using Genome Scale Data With Application in Plant Breeding

Dung Ngoc Nguyen, Zitong Li

CSIRO, Canberra, Australia

Abstract

Genomic prediction (GP) is an emerging data-driven approach in plant and animal breeding that aims to predict phenotypes or breeding values using genomic information, and potentially other sources such as environmental data. A wide range of statistical and machine learning methods have been developed for GP, including linear mixed models, penalized regression, Bayesian variable selection, random forests, and deep learning. However, most existing methods assume a homogeneous genetic background, which often does not reflect the complexity of real-world breeding programs, where data frequently originate from multiple populations or families.

To address this limitation, we propose a novel mixture of experts (MoE) regression model for genomic prediction. The MoE framework, implemented as a neural network architecture comprising two key components: a gating network that probabilistically assigns individuals to latent clusters based on genomic features, and expert functions that model the relationship between genotypes and phenotypes within each cluster. We develop an efficient Expectation-Maximization (EM) algorithm for parameter estimation. The proposed method is evaluated on both simulated datasets and a large-scale real-world cotton genomic prediction dataset. Comparative results with Bayesian multiple regression and residual neural network models highlight the benefits of explicitly modelling genetic heterogeneity in improving predictive accuracy.

77 Leveraging Biostatistics to Identify Causal Mechanisms in Chronic Respiratory Disorders

Tess Geraghty[1,2], Danielle Adams[1,2], Murray Cairns[1,2]

[1]The University of Newcastle, Callaghan, Australia. [2]Hunter Medical Research Institute, New Lambton Heights, Australia

Abstract

Chronic respiratory disorders are a major contributor to the burden of non-communicable diseases worldwide. Despite this, there are no known cures for these disorders and they drastically reduce quality of life. There are numerous data from clinical studies revealing that metabolic dysregulation and the associated inflammatory response contribute to the pathogenesis of respiratory disorders. Notably, disorders including asthma, chronic obstructive pulmonary disease (COPD), and sleep apnoea have been linked to metabolic conditions including obesity and type II diabetes in a bi-directional manner. While these systems have been identified as tractable therapeutic targets, these relationships are largely supported through observational studies, and thus, further work is needed to confirm that metabolic and inflammatory mechanisms are causally linked to respiratory outcomes. With the advent of the genome wide association study (GWAS) in the early 2000s it became possible to compare genomes between biological traits, allowing for a hypothesis free approach to map genetic variations to disease aetiology and possibly identify new targets for treatment. This project aimed to use statistical modelling to identify associations between 602 metabolic and immunological traits, alongside brain measure traits, and chronic respiratory disorders. Linkage disequilibrium score regression and latent causal variable modelling were employed

to test for genetic correlation and partial genetic causality between two recently published metabolite GWAS databases and the ENIMGA consortium, and asthma, COPD, and sleep apnoea GWAS from the FinnGen Project, respectively. This revealed 19, 45, and 42 partially genetically causal traits for asthma, COPD, and sleep apnoea GWAS, respectively, when testing significantly genetically correlated trait pairs. Together, these results provide further evidence for the causal relationship between metabolic and immunological regulation, and chronic respiratory disorders.

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88 From Theory to Practice: Using AI to Empower Non-Statisticians with Sound Statistical Methods

Marijke Welvaert

Australian Red Cross Lifeblood, Sydney, Australia

Abstract

A common barrier to good statistical practice in applied settings is the gap between theoretical methodology and its practical implementation, particularly for non-statisticians or analysts without advanced training. This challenge is amplified when working with large datasets, complex models, and limited computational resources.

We present a case study showcasing how artificial intelligence (AI), specifically large language models like ChatGPT, can bridge this gap by translating complex statistical methodology into accessible, usable code. Using Wright et al.'s (2017) case-control subsampling method for fitting generalised linear mixed models to large clustered binary data, we demonstrate how an AI-assisted dialogue can produce a fully functional R workflow from a published paper, without requiring expert-level statistical programming or derivation.

The resulting code incorporates principled subsampling, offset correction, and mixed modelling using familiar R packages (`lme4`, `mgcv`), enabling robust model exploration on modest hardware. This approach offers a practical tool for statistical consultants and applied analysts to empower collaborators in health, business, and government to apply advanced methods responsibly, even when direct statistical support is limited.

By guiding users to incorporate key theoretical adjustments (e.g. sampling offset terms), AI can help prevent common missteps and promote better model-based inference. We argue that integrating AI into the statistical consulting workflow can enhance reproducibility, foster learning-by-doing, and scale the reach of statistical expertise. This talk explores the opportunities and boundaries of such tools in supporting evidence-based decision making across disciplines.

Environment & Ecology 1

4:00 - 5:30pm Wednesday, 3rd December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Trent Piccicacco

33 Integrated Species Distribution Models: A Single-Index Approach

Quan Vu

Australian National University, Canberra, Australia

Abstract

In ecology and fisheries, species abundance data are often collected using a number of surveys that have different characteristics. Spatial statistical models can be used to integrate information from these datasets to improve both interpretability and prediction of the abundance. In this talk, we introduce the single-index integrated species distribution model, based on a single index representing the latent spatial distribution of the species abundance, and survey-specific link functions representing the different catchability properties of each survey. We demonstrate the use of the model through an analysis of scallop data collected from two surveys: a bottom trawl survey which covers a wide spatial domain but is less efficient, and a hydraulic dredge survey which is more efficient and spatially targeted. Results show that our model offers meaningful interpretations of covariate effects, spatial distribution, and survey catchability differences, and achieves superior predictive performance compared to contemporary species distribution models.

This is joint work with Francis Hui, Alan Welsh, Samuel Muller, Eva Cantoni and Chris Haak.

51 Continental-Scale Bayesian Analysis of Acacia Flowering Phenology: A Novel Framework Integrating Phylogenetic Signal and Circular Statistics

Owen Forbes ORCID iD, Peter Thrall, Andrew Young

CSIRO, Canberra, Australia

Abstract

Climate change is driving widespread shifts in plant phenology globally, with critical implications for ecosystem functioning and species interactions. While most phenological studies focus on temperate deciduous species at local scales, Australia's diverse Acacia genus offers unique opportunities to understand phenological responses to variations in rainfall and temperature across continental gradients. Leveraging digitised herbarium collections, we present the largest phylogenetically-informed analysis of acacia flowering phenology undertaken to date, spanning 548 species across Australia's full climatic spectrum.

We developed a novel phenological modelling approach using a Bayesian hierarchical framework that makes several key innovations. First, we extend traditional phylogenetic least squares (PGLS) approaches by incorporating phylogenetic signal as a structured random effect within a fully Bayesian context, enabling simultaneous estimation of phylogenetic covariance alongside spatial and temporal dependencies. Second, we employ von Mises circular distributions to model flowering time, naturally accommodating seasonal cyclicity while avoiding linear boundary artifacts. This modular approach also integrates spatial autocorrelation structure, time-lagged climate covariates, and phylogenetic relationships within a unified probabilistic framework. Our analysis of 20,998 herbarium specimens spanned 110 years (1910 – 2020) across Australia, using climate data from the Australian Gridded Climate Data, with Köppen climate zones included in the model to stratify climate-phenology relationships.

This framework successfully quantifies species-specific climate sensitivities while accounting for phylogenetic constraints. Expected effect of time-lagged temperature and rainfall effects vary significantly across Köppen climate zones and species. Species vulnerability patterns could help identify taxa which could be potentially at higher risk under projected climate scenarios. This modular Bayesian approach demonstrates the research potential of natural history collections for continental-scale ecological inference, providing a statistical template for global phenological studies addressing critical climate change impacts.

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60 Combining Spacetime Geostatistical Analysis and Machine Learning to Improve Precipitation Isoscape Interpolation

Candy Duff ORCID iD

Charles Darwin University, Darwin, Australia

Abstract

The stable isotopic composition of precipitation ($\delta^{18}O$ and $\delta^{2}H$) has wide-ranging applications in water resource assessment, informing groundwater modelling, determining transit-times in catchments, and other ecological studies. Precipitation isoscapes are thematic maps that illustrate the isotopic signature of rainfall and are an invaluable tool for understanding the spatial patterning of stable precipitation isotopes around the world. Researchers have used many different interpolation methods to create isoscapes from simple deterministic models, using a weighted average, to stochastic methods like machine learning, universal kriging and cokriging. Recently, spatio-temporal geostatistical methods have proved to be better predictors than traditional methods such as generalised least squares regression. However, to the best of our knowledge, combining spatio-temporal geostatistical methods with machine learning has not yet been applied. This study aims to determine the best interpolation methods for precipitation isoscape interpolation, addressing key statistical challenges such as data sparsity, irregular sampling intervals and many predictors (i.e., high-dimensional data). By combining the strengths of Random Forest – a robust machine learning method, known for its handling of nonlinear relationships and high-dimensional datasets – with geostatistical methods such at universal spacetime kriging, the hybrid approach captures the global and localised trends. The study also explores model interpretability and computational efficiency, critical considerations for applying these advanced methods in environmental science. Simple regression models, geostatistical models, and geostatistical models incorporating machine learning are compared using mean absolute error (MAE); predicted residual error sum of squares (PRESS); Akaike information criterion (AIC); and Bayesian information criterion (BIC). The best models were used to produce isoscapes that illustrate the intra-seasonal patterns of precipitation isotopes in tropical northern Australia. This interpolation provides a robust framework enhancing predictive performance and spatial coherence for environmental data (e.g., reflecting realistic geographic patterns such as gradual depletion with distance from the coast), particularly in data-scarce regions.

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84 Modelling Species Diversity for Citizen-Science Bird-Count Data

Jeremy Silver ORCID iD[1], Graham Hepworth ORCID iD[1], William Steele[2], Al Danger[2]

[1]University of Melbourne, Melbourne, Australia. [2]Melbourne Water, Melbourne, Australia

Abstract

Bird count data from the Greater Melbourne region over a 25-year period were recorded by volunteer observers, based on 20-minute surveys. Counts for 326 different bird species were reported either as counts (positive integers) or simply as “present”, if the precise count could not be determined. This work investigated whether there has been an increase in the presence and abundance of noisy miners, an endemic species known for its aggressive and territorial behaviour, and whether this could partly explain apparent declines in some species.

The dataset was augmented with weather, geospatial and ecological information, and a mixed-effects modelling framework was adopted. Seasonal effects and long-term trends were accounted for via regression splines. The influence of different observers and locations were represented as random effects. Generalised additive mixed-effects models were fitted for (a) the presence/absence of noisy miners, (b) the abundance of noisy miners, and (c) species richness for particular groupings of birds (relating to foraging patterns). In modelling species richness, noisy miner abundance was included as a predictor; we used multiple imputation to account for the missing data when this species was reported as “present” but no count was recorded.

There was a small, positive association found between noisy miner abundance and all-species diversity – a surprising result. A stronger negative association was found for some species groupings. For many species groups, diversity tended to decrease with increasing distance from the nearest native vegetation feature, and with increasing wind speed; seasonal variation in diversity and trends across the study period were also identified. The work provided several statistical and computational challenges, including the organisation of geospatial information from a range of sources.

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149 Quantifying the Relationship Between Cattle Behaviour and Daily Activity Using Triaxial Accelerometer Data

Shuwen Hu[1], Antonio Reverter[2], Reza Arablouei[2], Greg Bishop-Hurly[2], Neil Bagnall[2], Aaron Ingham[2]

[1]RMIT, Melbourne, Australia. [2]CSIRO, Brisbane, Australia

Abstract

An animal's energy expenditure is partially dependent on its activity or movement levels and driven by the amount of time it spends engaging in different behaviours. In this study, we investigate the relationships between five core cattle behaviours and measures of total daily activity based on data from collar-mounted triaxial accelerometers. This accelerometer data, which represents the intensity of animal movement, was used to compute an activity metric quantifying the total motion within a window of time in units of gravitational force (g). We also used pre-trained accelerometer-based machine learning models to classify cattle behaviour into grazing, ruminating, walking, resting, drinking or other behaviour classes. Our objective was to investigate how different behaviours contribute to overall daily activity, a straightforward yet effective measure of movement. Sixty Brahman steers were studied for one month in an Australian pasture, with average daily activity values ranging from 5.44g to 23.69g. The average daily time spent grazing, ruminating, resting, walking, drinking and other was 8.97 (±1.12), 7.78 (±1.03), 5.83 (±1.05), 1.00 (±0.40), 0.18 (±0.12) and 0.24 (±0.11) hours, respectively (mean ± standard deviation). To investigate the associations between behaviours and activity, we employed a linear mixed-effects model incorporating the four behavioural components derived from principal component regression to address collinearity among behaviours. Our analysis revealed that grazing and walking are positively correlated with activity scores (r = 0.44 and 0.3, respectively), whereas ruminating and resting are negatively correlated with activity scores (r = -0.18 and -0.39, respectively). Furthermore, our linear mixed model accounted for 76.2% of the total variation in activity, highlighting significant individual variations among cattle. This study underscores the importance of understanding how different behaviours contribute to overall activity levels, which in turn affects energy expenditure in cattle.

Missingness, Adherence and Estimands

4:00 - 5:30pm Wednesday, 3rd December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Michael Dymock

76 Comparison of Methods for Estimating Treatment Efficacy When There is Missing Data in a Binary Compliance Variable

Cameron Patrick ORCID iD[1,2], Katherine Lee ORCID iD[1,2], Margarita Moreno Betancur ORCID iD[1,2], Thomas Sullivan ORCID iD[3,4]

[1]University of Melbourne, Melbourne, Australia. [2]Murdoch Children's Research Institute, Melbourne, Australia. [3]South Australian Health and Medical Research Institute, Adelaide, Australia. [4]University of Adelaide, Adelaide, Australia

Abstract

Background: Randomised controlled trials (RCTs) often target a treatment policy (or intention-to-treat) estimand. For many interventions compliance with the randomised treatment assignment is imperfect. In these cases, it can also be of interest to estimate the efficacy of the treatment: the effect of the treatment when taken as intended. The literature so far has focussed on the case where compliance is fully observed for every participant. However, it is common for RCTs to feature missing data, and often the compliance variable is not fully observed. The aim of this study was to examine the performance of common methods for handling missing data in a binary compliance variable when used with methods for estimating treatment efficacy.

Methods: A simulation study was conducted, examining various scenarios and missingness mechanisms chosen based on two case study RCTs. The missing data methods considered were complete case analysis, multiple imputation, and inverse probability of missingness weighting. The estimators considered were standardisation (the parametric g-formula), inverse probability of treatment weighting, and instrumental variables. The commonly used, but invalid "naive per-protocol" estimator was also compared.

Results: We found that inverse probability of missingness weighting performed well when used for all methods for estimating efficacy, with little bias and nominal confidence interval coverage. In contrast, multiple imputation was unbiased but showed some under-coverage of confidence intervals. Complete case analysis showed substantial bias in scenarios where compliance was missing at random, with the worst bias found in the combination of complete case analysis and instrumental variables. Naïve per-protocol produced biased estimates in all scenarios. We recommend researchers consider carefully the likely mechanism behind missing compliance data and choose appropriate statistical methods to handle it.

86 Trial Estimands: Using the composite strategy to handle intercurrent events when the endpoint is continuous

Anneke Grobler, Katherine Lee

Murdoch Children's Research Institute, Melbourne, Australia

Abstract

Introduction

The ICHE9(R1)-guideline identified five strategies for handling intercurrent events(ICEs) when defining the estimand. The composite strategy, incorporating the ICE into the outcome, is recommended when the ICE provides information about treatment effect. While implementation is straightforward for binary outcomes, it is unclear how to apply the composite strategy to continuous outcomes.

Methods

We describe approaches for implementing the composite strategy with continuous outcomes and consider the implications for the estimand definition and statistical analysis, namely 1.)Convert the continuous outcome to a binary outcome; or 2)Assign a value on the continuous scale, such as the worst outcome, to participants with the ICE. We highlight the advantages, disadvantages and statistical considerations for each and demonstrate their impact on treatment effect estimates using a palliative care clinical trial.

Results

Option 1 is easy to implement but it relies on defining a cut-point, changes the estimand, and results in loss of information. Option 2 preserves the continuous nature of the outcome, but can mean assigning an arbitrary value to the ICE. When using Option 2, ideally the summary measure should not be influenced by the value assigned. Possible summary measures include a difference in proportions (assessed with a chi-square test), difference in means or trimmed means (estimated using linear regression), medians (estimated using quantile regression), proportional odds ratio (estimated using a proportional odds model).

In our case study, if there were no ICEs, the planned summary measure is difference in means. Applying the composite strategy, we analysed our trial with the methods discussed. Given different estimands, methods could not be compared directly, but we concluded from all methods that one treatment group had higher pain relief.

Discussion

When the outcome is continuous using the composite strategy handlingICE raises questions regarding what summary measure and statistical approach to use.

91 Applying the Estimand Framework to Handle Death as a Primary Outcome and Intercurrent Event in the POLAR Trial: An International Neonatal Respiratory Study

Francesca Orsini ORCID iD

Murdoch Children Research Institute, Melbourne, Australia

Abstract

Objective(s)
The POLAR trial is a phase III/IV, international, multicentre randomised controlled trial conducted in delivery rooms of tertiary NICUs across an international neonatal networks. It compares two respiratory support strategies for preterm infants (<29 weeks postmenstrual age, PMA) at birth: a dynamic positive end-expiratory pressure (PEEP) algorithm (8–12 cmH2O, individualised) versus a static PEEP (5–6 cmH2O) consistent with current Neonatal Resuscitation Program guidelines. The primary aim is to assess whether the dynamic PEEP strategy increases survival without bronchopulmonary dysplasia (BPD). Secondary outcomes include rates of common neonatal morbidities. Death is both a component of the primary outcome and an intercurrent event, complicating analysis and interpretation of secondary endpoints.

Method(s)
The Statistical Analysis Plan is developed using the ICH E9(R1) estimand framework to explicitly address death as both an outcome and intercurrent event. Truncating events like death preclude the existence of subsequent outcomes (e.g. BPD, duration of ventilation,..). Different strategies will be applied depending on the outcome nature and timing. Composite strategies will be used where clinically appropriate (‘death or BPD’) to ensure death is reflected as a poor outcome. For time-limited outcomes meaningful only while alive (e.g. duration of ventilation), a while-alive strategy with clear censoring or truncation rules will be applied. For outcomes assessed at fixed time points, a principal stratum strategy will estimate effects among infants who would survive regardless of treatment assignment. These strategies ensure clinically relevant estimands that appropriately handle death and other intercurrent events.

Results
Implementing the estimand framework in the POLAR trial enhances clarity in defining treatment effects despite complexities from death as an intercurrent event. This approach allows robust, interpretable analyses of survival and neonatal morbidity outcomes. Sensitivity analyses will evaluate robustness of conclusions under alternative assumptions, supporting meaningful clinical decision-making in this vulnerable population.

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162 Dealing with Missing Data in Studies of Paediatric Postoperative Pain: A Comparison of Methodologies Using Simulated Data

Daisy Evans ORCID iD[1,2,3,4], Zahra Vink[1], Harry Smallbone[1,2,5], Anwen Taplin[1,2,3,4], Aine Sommerfield[1,2,3,5], Thomas Drake-Brockman[1,2,3,5], David Sommerfield[1,2,3,5], Neil Hauser[1,2,3,5], Berwin Turlach[2,4], Britta S von Ungern-Sternberg[1,2,3,5], R Nazim Khan[1,2,4]

[1]Perioperative Medicine, The Kids Research Institute Australia, Perth, Australia. [2]Institute for Paediatric Perioperative Excellence, The University of Western Australia, Perth, Australia. [3]Department of Anaesthesia and Pain Medicine, Perth Children's Hospital, Perth, Australia. [4]School of Physics, Mathematics and Computing, The University of Western Australia, Perth, Australia. [5]Medical School, The University of Western Australia, Perth, Australia

Abstract

Missing data are common in clinical trials. Often, the likelihood of a record being incomplete is related to variables that are of interest to the study, and this introduces bias. Approaches to handling missing data vary, and frequently researchers neglect to comment on missing data at all.

The Bee Pain Free study was a randomised controlled trial comparing swallowing honey to a placebo for treating children’s pain after tonsillectomy. Participants were followed up for 2 weeks after surgery, with around 20% missing responses by day 14. Through the example of the Bee Pain Free study, we compare different missing data handling techniques.

Data were simulated based on the Bee Pain Free study, and missing data patterns were imposed. Different methods for handling the missing data were tested: Complete Case Analysis (CCA); Last Observation Carried Forward (LOCF); Multiple Imputation by Chained Equations (MICE); Multiple Imputation with Denoising Autoencoders (MIDAS).

Data imputed using each method were analysed using linear mixed-effects modelling for pain scores over time. The bias in parameter estimates relative to the “true” data (before missingness was imposed) were compared, along with the computing time and memory needed for each method. We also considered the interpretability for clinicians, since statistical performance is moot if the findings are not understood and translated into practice by clinicians.

We found that, although simple to implement, LOCF produced estimates with greater bias than ignoring the presence of missing data altogether. MICE required more computing time and memory, but produced less biased estimates than single-imputation techniques. Although MIDAS took less time than MICE, the implementation in R was difficult to use, and its accessibility must be improved to make it a feasible option for clinical trials analysis.

Our findings provide guidance on the best methods for handling missing data in future paediatric repeated-measures randomised trials.

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132 Measuring Adherence For Assessment Of Complier Average Causal Effects In Randomised Controlled Trials Of Behavioural Interventions: Learnings From Two Clinical Studies

Yee Foong Mok, Anurika De Silva, Peixuan Li, Rong Chen, Karen Lamb

University of Melbourne, Parkville, Australia

Abstract

In randomised controlled trials (RCTs), primary analysis typically follows the intention-to-treat (ITT) principle. Here, participant outcomes are analysed according to intervention assignment, regardless of adherence. While ITT preserves randomisation and reflects more realistic ‘real-world’ effects, other estimates such as the Complier Average Causal Effect (CACE) are of interest. CACE estimates the intervention effect among participants who adhered to, or would have adhered if offered, the assigned intervention. Capturing adherence is key to CACE estimation. However, measuring adherence is challenging, particularly for behavioural interventions. Here, we reflect on approaches adopted in two recent behavioural RCTs.

The first compared anxiety and depression among people with post-traumatic stress disorder, randomised to either a psychosocial education mobile app (intervention) or mood-monitoring app (active control). Intervention adherence was defined as completion of ≥5 of 6 active app modules. This measure offered simplicity, data completeness, and clear links to the intervention. However, it lacked information about the application of module learnings (e.g., managing emotions, improving sleep) and provided no adherence measure for the control.

The second in development will compare health status at 12 months among major trauma patients randomised to either a multidisciplinary early-intervention program for pain management, or usual care. Both treatments are heterogenous and depend on patient needs, complicating adherence measurement. We proposed a combination of appointment tracking, clinician assessment of patient engagement, and self-report of material received and learnings. However, defining a binary measure of minimum adherence for CACE may be challenging due to possible missing data and recall bias.

No framework or consensus on best practice exists for comprehensive measurement of adherence in RCTs. There is a need to balance simplicity with reflections on minimum level of engagement for the intervention to be effective. Further research is needed to improve adherence capture for behavioural interventions given limitations of current approaches.

Conference dinner

7:00 - 10:00pm Wednesday, 3rd December, 2025

The Windsor Hotel, 112 Mill Point Road, South Perth WA 6151

Keynote: Katharina Schueller

9:00 - 10:00am Thursday, 4th December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs Ayse Bilgin

Joint ASC/SES keynote

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217 Why Do We Still Need Statistics (More Than Ever) in Times of AI?

Katharina Schueller

STAT-UP Statistical Consulting, Munich, Germany

Abstract

In times of rapidly advancing AI, we rely more than ever on solid statistical reasoning to create trustworthy evidence. AI systems can process data, but they cannot judge whether the data are valid, representative, or “fit for purpose” for answering real-world questions. True AI literacy therefore presupposes data literacy – and data literacy in turn requires statistical literacy. Only with fundamental statistical competencies can we assess how well data reflect underlying phenomena, what conclusions can legitimately be drawn, and where the limits of inference lie. Statistics remains essential as the foundation for evidence-based decision-making in an AI-driven world.

Morning tea

10:00 - 10:20am Thursday, 4th December, 2025

IPS (EDU 25) SES Panel

10:20 - 11:50am Thursday, 4th December, 2025
405.201 Haydn Williams LT
Presentation type Invited Session
Chairs Shuwen Hu, Ayse Bilgin

93 Statistically Speaking: What We Wish We Knew When We Started Teaching

Ayse Aysin Bilgin ORCID iD[1], Lucy Conran[2], Karol Binkowski ORCID iD[1], Amanda Shaker ORCID iD[3], Gizem Intepe ORCID iD[4]

[1]Macquarie University, North Ryde, Australia. [2]The University of Western Australia, Perth, Australia. [3]La Trobe University, Melbourne, Australia. [4]Western Sydney University, Sydney, Australia

Abstract

Teaching statistics and data science presents a unique blend of challenges and opportunities, especially for early career educators navigating both pedagogical and disciplinary expectations. This panel will enable sharing reflective key insights gained through experience, including the importance of demystifying mathematical content to make it accessible without sacrificing rigor. Each panel member will discuss strategies for engaging students with real, messy data to foster authentic statistical thinking and will highlight the role of digital tools—such as R, and others—in enhancing learning and assessment. The panel discussions will also address the complexities of academic identity in research-intensive environments and offer practical advice for designing meaningful assessments, initiating education-focused research, and adapting instruction for diverse student backgrounds. Through personal anecdotes, including moments of failure that led to growth, this panel aims to support and inspire educators committed to cultivating inclusive, engaging, and reflective statistics education.

This is a panel session.

Official Statistics 1

10:20 - 11:50am Thursday, 4th December, 2025
405.205
Presentation type Contributed Session
Chairs Andrew McMahon

46 Unsupervised Machine Learning for Anomaly Detection using Administrative Data Sources

Jerom Vanderstappen

Australian Bureau of Statistics, Canberra, Australia

Abstract

The Australian Bureau of Statistics (ABS) has been investigating the application of machine learning techniques to identify and treat anomalies in large administrative datasets for use in official statistics. Previous work has demonstrated the benefits in adopting machine learning for anomaly detection, including increased efficiency, reduced manual intervention, informed targeting of anomalies, and the ability to detect emergent anomalous behavior. Here we present a case study using the Business Activity Statement (BAS) dataset provided by the Australian Taxation Office (ATO) applying two unsupervised learning algorithms, Local Outlier Factor (LOF) and Extended Isolation Forest (EIF). The BAS dataset captures over 10 years of Australian business tax reporting including GST, PAYG withholding and other obligations, for businesses typically registered with an annual turnover of $75,000 or more. These models are trained using time series data on an industry-by-industry basis across various primary variables, introducing challenges in data management, feature selection, and model(s) evaluation. The focus of this presentation will be on the development and training of machine learning algorithms in the context of anomaly detection, with a discussion of the associated challenges in explainability, interpretability, representativeness and model maintenance.

56 Modelling Australian Residential Insurance Coverage

Scott Wong[1], Anderson Yang[1], Matthew Xu[2], Kay Cao[3], Joel Bertinshaw[4], Kee Zhou[4]

[1]Modelling Analysis and Visualisation Section, Methodology and Data Science Division, Australian Bureau of Statistics, Melbourne, Australia. [2]Modelling Analysis and Visualisation Section, Methodology and Data Science Division, Australian Bureau of Statistics, Sydney, Australia. [3]Modelling Analysis and Visualisation Section, Methodology and Data Science Division, Australian Bureau of Statistics, Brisbane, Australia. [4]Australian Climate Service, People and Place Division, Australian Bureau of Statistics, Melbourne, Australia

Abstract

The National Insurance Dataset (NID) provides a valuable snapshot of insurance coverage in residential properties across Australia. However, it does not provide the insurance status of the entire Australian population. To address this, we develop a machine learning model to predict the insurance status of the remaining residential dwellings using geographic and socio-economic data.

Guided by the existing insurance modelling literature, available auxiliary data from Census 2021 and exploratory data analysis, we select variables such as household income, tenure type, dwelling structure, family composition, geographic location, and characteristics of the household reference person (e.g. age and place of birth).

We link the NID to Census 2021 data using an up-to-date Address Register Identifier (ARID). The data are used to train two Random Forest models: a Total population model and an Owner-occupier model. The models produce estimates of insurance coverage at the dwelling level, which are then aggregated to produce small area (SA2) and state level estimates. We also compare our estimates to the results from a logistic demand model.

114 Fortified Assured Streamlined Trusted (FAST) Output Protection

Joseph Chien ORCID iD

Australian Bureau of Statistics, Canberra, Australia

Abstract

ABS DataLab has seen a substantial increase in usage, rendering manual output clearance inefficient, costly, and at risk of human error. This research expands the 5 Safes framework by developing automated output protection tools, addressing these challenges and the disclosure risks from differencing outputs across projects. To improve efficiency and minimise disclosure risks, the Data Access and Confidentiality Methodology Unit is developing Fortified Assured Streamlined Trusted (FAST) tools, which automate output clearance while ensuring safe, quick, and reliable data releases.

This presentation will:

1. Describe improvements to the ABS perturbation methodology in FAST, including the expanded cell key method and sequential descent algorithm.

2. Demonstrate the new tools for researchers, highlighting their benefits for streamlining the output clearance process.

Our primary contribution is a sequential descent algorithm accommodating both symmetric and asymmetric supports for generating perturbation distributions. This approach builds upon existing ABS perturbation methodology by incorporating (ε, δ)-differential privacy parameters in the design of noise distributions. The implementation in Rshiny makes these tools accessible to both users preferring R programming and those who prefer graphical interfaces. The FAST tools will enable more timely and cost-effective output clearance while minimising differencing risks through consistent protection across research projects. Future research directions include developing safeguards against researcher errors, exploring alternative noise generation assumptions, enhancing visualisation interfaces, and implementing web-based services.

123 Seasonal Adjustment Software Redevelopment in the Australian Bureau of Statistics

Michelle Hedger

Australian Bureau of Statsitcs, Adelaide, Australia

Abstract

The Australian Bureau of Statistics (ABS) is undergoing a transformation program to modernise its statistical processes and legacy systems. This includes the redevelopment of its seasonal adjustment software. The ABS developed a suite of time series applications called SEASABS that has been used to produce seasonally adjusted and trend estimates for numerous monthly and quarterly publications. As part of the transformation program, the ABS is replacing SEASABS to better position the organisation for future time series need.

This presentation will describe the journey the ABS is on to replace SEASABS with a secure cloud-based system, with a particular focus on the statistical methods to be introduced in the new system, the methods to be left behind in SEASABS, and the methods that will need to be carried across from SEASABS into the new system.

124 Quasi-Experimental Evaluation Using Linked Administrative Data: Insights and Lessons Learned

Callum Smith, Jennifer Marley, Susan Fletcher

Australian Bureau of Statistics, Canberra, Australia

Abstract

Evaluation is the systematic and objective assessment of the design, implementation or results of a government program for the purposes of continuous improvement, accountability and decision-making. Our talk will focus on the application of robust causal inference techniques to evaluate policy impacts using linked administrative data.

Since December 2021, the Commonwealth Evaluation Policy has encouraged more systematic assessment of programs to support evidence-based policy and service delivery. As Australia's national statistical agency, the ABS supports this shift by enabling the use of rigorous methods and integrated data to inform evaluation practice across government. In particular, the ABS is working to build capability and provide guidance as to how to effectively apply quasi-experimental methods, such as regression discontinuity design (RDD) and difference-in differences (DiD), using the vast volume of data available in ABS integrated data assets, including PLIDA and BLADE.

In our talk, we will share our experiences in applying RDD and DiD to evaluate the effects of government programs, including the lessons learned in identifying meaningful variation in when, where, and to whom a policy or program applies. We will touch on the complexities of constructing valid comparison groups, ensuring sufficient temporal coverage, and dealing with administrative data limitations such as irregular policy rollouts and missing baseline characteristics. These are non-trivial tasks that demand both methodological rigour and contextual understanding.

Our work provides useful guidance for practitioners seeking to apply robust causal inference techniques using linked data assets. By sharing the practical strategies we used to address these challenges, we aim to contribute to the growing body of knowledge on how to extract policy-relevant insights from complex administrative datasets.

Statistical Computing and Tools

10:20 - 11:50am Thursday, 4th December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Jayani Lakshika

15 Rquest: An R Package for Hypothesis Tests and Confidence Intervals for Quantiles and Summary Measures Based on Quantiles

Luke Prendergast ORCID iD[1], Shenal Dedduwakumara ORCID iD[2], Robert Staudte ORCID iD[1]

[1]La Trobe University, Melbourne, Australia. [2]University of Adelaide, Adelaide, Australia

Abstract

Sample quantiles, such as the median, are often better suited than the sample mean for summarising location characteristics of a data set. Similarly, linear combinations of sample quantiles and ratios of such linear combinations, e.g. the interquartile range and quantile-based skewness measures, are often used to quantify characteristics such as spread and skew. While often reported, it is uncommon to accompany quantile estimates with confidence intervals or standard errors. The rquest package provides a simple way to conduct hypothesis tests and derive confidence intervals for quantiles from continuous data, linear combinations of quantiles, ratios of dependent linear combinations (e.g., Bowley's measure of skewness) and differences and ratios of all of the above for comparisons between independent samples. Many commonly used measures based on quantiles are included, although it is also very simple for users to define their own. Additionally, quantile-based measures of inequality are also considered. The methods are based on recent research showing that reliable distribution-free confidence intervals can be obtained, even for moderate sample sizes.

42 metaextractoR: An R Package for Streamlined Data Extraction in Systematic Reviews and Meta-Analyses

Danyang Dai ORCID iD[1], Jason Pole ORCID iD[1], Emi Tanaka ORCID iD[2]

[1]The University of Queensland, Brisbane, Australia. [2]Australian National University, Canberra, Australia

Abstract

Systematic reviews and meta-analyses are pillars of evidence-based medicine, yet data extraction remains labor-intensive and prone to error. Although recent advances have demonstrated the potential of Large Language Models (LLMs) to assist in data extraction, their implementation has largely been limited to proof-of-concept studies, with barriers to broader adoption among researchers. To address these challenges, we present metaextractoR, an open-source R package designed to facilitate and streamline data extraction in systematic reviews and meta-analyses by integrating open-source LLMs via the Ollama framework. metaextractoR enables automated, structured, and reproducible extraction workflows within interactive Shiny applications, in alignment with Cochrane guidelines and veridical data science principles.

Our package incorporates three modular Shiny apps to implement a human-in-the-loop framework: (1) a manual extraction interface for abstracts, (2) refining prompt engineering and model selection, and (3) validation of LLM-generated outputs. These apps enable researchers to iteratively collaborate with LLMs, with each abstract subjected to double extraction – once by a human and once by an LLM – to emulate double extraction recommended by the Cochrane Handbook. Notably, the package runs fully on local machines, with no need for API setup or external data transfer, maximising data privacy and accessibility. Robust logging features further enhance transparency and reproducibility by recording all prompt iterations and outputs. By providing a principled and user-friendly framework, metaextractoR lowers technical barriers and empowers the evidence synthesis community to conduct more efficient and transparent reviews.

134 Uncertainty Visualisation for Holistic Insights

Harriet Mason ORCID iD[1], Dianne Cook ORCID iD[1], Sarah Goodwin ORCID iD[1], Susan VanderPlas ORCID iD[2]

[1]Monash University, Melbourne, Australia. [2]University of Nebraska–Lincoln, Lincoln, USA

Abstract

Incorporating uncertainty into our data visualisations can provide a more comprehensive understanding of the underlying statistics and aid in decision-making. Existing uncertainty visualisations communicate uncertainty as a probability or a distribution, and focus on showing the uncertainty as an isolated variable disconnected from its context. While these visualisations are excellent for understanding the uncertainty itself, they do little to prevent viewers from drawing false conclusions about their data. Successfully communicating uncertainty requires us to convey an estimate and its uncertainty together as a single “validity of signal” variable, something which has proven to be difficult with current graphical tools. In this talk, we will introduce ggdibbler, a ggplot extension that allows us to visualise uncertainty holistically and prevent false signals. We illustrate how ggdibbler can be seamlessly integrated into existing visualisation workflows and demonstrate its usefulness by looking at examples that use ggdibbler to incorporate uncertainty into a choropleth map. In working through these examples, we will discuss which uncertainty visualisation approaches work (or don’t work) for communicating uncertainty, and why.

198 Vectorised Statistics with Composite Data Types

Mitchell O'Hara-Wild ORCID iD, Cynthia Huang ORCID iD

Monash University, Clayton, Australia

Abstract

Statistical analysis on temporal, spatial, graph, and probabilistic data requires special care to appropriately handle their intrinsic structure. Outputs from models typically return these composite data types separately, requiring the user to assemble and apply the results correctly. This reduces the accessibility of statistics and results in error-prone analysis. Representing these data types using composite vector types makes statistical operations and data analysis easier.

In this talk, I will introduce the application of vectorised statistical operations across common dimensions not otherwise handled in traditional data structures. The vectors of vectors data type implemented in the vecvec R package is foundational for creating efficient mixed data types. The distributional and mixtime R packages leverage vecvec in order to create vectors that mix different distributions and temporal granularities together. Storing distributions with different shapes and parameterisations together in the same vector abstracts away the data handling complexity while providing a user-friendly interface for calculating distributional statistics such as the mean, quantiles, and densities. Similarly, storing time at different granularities allows combining data from different sources and facilitates forecasts across multiple levels of temporal aggregation.

Combinations of these data types in a rectangular 'tidy' data structure allow for the flexible application of probabilistic-spatio-temporal-graph data analysis. For instance, the combination of temporal and distributional vectors provides a dataset suitable for working with time series forecasts.

Clinical Trials and Design

10:20 - 11:50am Thursday, 4th December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Anneke Grobler

186 Finding the Balance Between Statistical Significance and a Feasible Sample Size in Clinical Trials – An Example in Paediatric Anaesthesia

Anwen Taplin[1,2,3,4], Daisy Evans[1,2,3,4], Neil Hauser[1,2,3,5], David Sommerfield[1,2,3,5], Britta S von Ungern-Sternberg[1,2,3,5], R Nazim Khan[2,3,4]

[1]Department of Anaesthesia and Pain Medicine, Perth Children's Hospital, Perth, Australia. [2]Perioperative Medicine Team, The Kids Research Institute Australia, Perth, Australia. [3]Institute for Paediatric Perioperative Excellence, The University of Western Australia, Perth, Australia. [4]School of Physics, Mathematics and Computing, The University of Western Australia, Perth, Australia. [5]Division of Emergency Medicine, Anaesthesia and Pain Medicine, Medical School, The University of Western Australia, Perth, Australia

Abstract

Approximately 15% of children undergoing general anaesthesia experience a perioperative respiratory adverse event (PRAE) and the highest risk is associated with emergence and extubation (airway removal). During anaesthesia, endotracheal tubes (ETT) are frequently used in children, and these are either removed awake or deep (during anaesthesia). Deep extubation may be related to a decreased risk of overall complications, while awake extubation is very unpleasant for patients. Another technique is to exchange the ETT for a softer laryngeal mask airway (LMA), followed by awake removal of the LMA, with the aim of reducing the risk of airway obstruction associated with deep removal. The clinicians in our team are interested in determining whether the risk of PRAE is lower when an ETT is exchanged to an LMA, and we needed to choose a sample size for a potential trial. A previous observational cohort of 1,533 patients estimated that 11.8% of patients with awake ETT removal experienced PRAE, compared to 9.5% of patients with their ETT exchanged for an LMA, however there was a large uncertainty in these estimates (OR 0.785, 95%CI 0.44-1.402). A simple calculation using these rates for power 0.8 and 95% significance, results in a sample size of 2823 patients for each group.

Both extubation techniques are routinely used at hospitals and adult studies suggest a much larger difference in PRAE rates, meaning that the intervention is worth pursuing from a clinical perspective. However, the calculated sample size above would take many years to collect, by which time the question may no longer be relevant. This example highlights that sample sizes for clinical trials should consider factors such as the low-risk associated with an intervention, or resource and time restrictions, instead of only statistical sample size calculations which often rely on assumptions with large uncertainty.

189 Current Practices in the Design of Master Protocol Trials: A Scoping Review

Melissa Middleton[1,2], Chris Selman[1,2], Alistair McLean[3,4], Leon Di Stefano[1], Julie Simpson[3,4], Robert Mahar[1,3,4], Katherine Lee[1,2]

[1]Clinical Epidemiology and Biostatistics Unit, Murdoch Children's Research Institute, Parkville, Australia. [2]Department of Paediatrics, University of Melbourne, Parkville, Australia. [3]Methods and Implementation Support for Clinical and Health (MISCH) research Hub, Faculty of Medicine, Dentistry and Health Sciences, University of Melbourne, Parkville, Australia. [4]Biostatistics Unit, Centre for Epidemiology and Biostatistics, Melbourne School of Population and Global Health, University of Melbourne, Parkville, Australia

Abstract

Master protocol designs are clinical trial designs that allow for the simultaneous evaluation of multiple drugs and/or subpopulations within a single overarching protocol. Broadly, there are three main master protocol designs: umbrella (multiple interventions within a single population), basket (single intervention within multiple subpopulations) and platform trials (multiple interventions within multiple subpopulations).

While there has been a rise in trials adopting master protocol designs, it is unclear whether these trials are adhering to best statistical practice given their complex nature.

The aim of this scoping review is to describe current practice in the design and planned analysis of master protocol trials, and identify areas for improvement.

We conducted a scoping review of available documentation for master protocol trials who registered one or more components of their randomised controlled trial (i.e. master protocol, intervention/domain, and/or subpopulation registrations) on the ClinicalTrials.gov platform between January 2022 and December 2024. Information regarding statistical best practice in the design (e.g. use of estimands framework, details of sample size calculation) and analysis (e.g. use of non-concurrent controls, approaches to handle missing data) of master protocol trials will be extracted from the publicly available documentation of the included trials. We identified and screened 495 registrations, with 159 trials, from 213 registrations, satisfying our inclusion criteria and being included for data extraction, which is anticipated to be completed in October 2025. Based on our initial screening of the included trials, platform trials are the most prevalent design (43.4%), and availability of trial documentation was limited with only 16.3% of trials having protocol documents available through either the ClinicalTrials.gov registration or published on PubMed (linked to the registration number).

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85 Rethinking Clinical Trial Design: Should We Consider the Value of Information?

Michael Dymock ORCID iD[1,2], Julie Marsh[2,3], Mark Jones[4], Anna Heath[5,6,7], Kevin Murray[1], Tom Snelling[4]

[1]School of Population and Global Health, The University of Western Australia, Perth, Australia. [2]Wesfarmers Centre of Vaccines and Infectious Diseases, The Kids Research Institute Australia, Perth, Australia. [3]Centre for Child Health Research, School of Medicine, The University of Western Australia, Perth, Australia. [4]Sydney School of Public Health, University of Sydney, Sydney, Australia. [5]Child Health Evaluative Services, The Hospital for Sick Children, Toronto, Canada. [6]Division of Biostatistics, Dalla Lana School of Public Health, University of Toronto, Toronto, Canada. [7]Department of Statistical Science, University College London, London, United Kingdom

Abstract

Clinical trial designs, including adaptive (sequential) designs, are typically narrowly focused on null hypothesis significance testing, but this approach is inadequate in many contexts, particularly when a decision-maker intends to consider multiple relevant clinical and health economic outcomes under uncertainty. Although value of information (VoI) metrics can be used to estimate the monetary value of prospective data collection to the decision-maker, and therefore the value of conducting the proposed research, VoI considerations have rarely been integrated into adaptive designs, partly due to the computational burden.

We propose the value-driven adaptive design, which refocuses trial design on VoI as a metric to direct trial adaptations. Specifically, a VoI analysis is performed at each interim analysis to determine whether or not the trial should proceed to the next analysis (i.e., determine whether further data collection is sufficiently valuable). We provide methods to compute the expected net benefit of perfect information, expected net benefit of sampling (ENBS) for a single analysis and the ENBS for sequential analyses. The approach is flexible to any statistical model, decision model and research cost function, and does not require distributional assumptions about the net benefit. The value-driven adaptive design aligns pragmatic clinical trial design with the requirements of decision-makers. Designs with VoI based adaptations can improve the efficiency of clinical trials.

In this talk, we will motivate the philosophy behind the value-driven adaptive design, highlight its utility for decision-makers and demonstrate its implementation on a case study investigating interventions to prevent respiratory syncytial virus. We will investigate scenarios where a traditionally designed trial stops too early (i.e., collects insufficient data) and stops too late (i.e., wastes resources collecting unnecessary data) and show how a value-driven adaptive design would have outperformed its traditional counterpart.

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106 A Classification of Incomplete Stepped Wedge Cluster Randomised Trial Designs

Ehsan Rezaei-Darzi ORCID iD, Andrew B Forbes ORCID iD, Kelsey L Grantham ORCID iD, Jessica Kasza ORCID iD

School of Public Health and Preventive Medicine, Monash University, Melbourne, Australia

Abstract

Stepped wedge designs are useful cluster randomised trial designs for testing new interventions through a phased rollout of the intervention across clusters. In many trial settings, a stepped wedge design can place a substantial burden on clusters by requiring data collection over the entire trial duration. Incomplete stepped wedge designs, such as the staircase design where clusters collect data for only certain measurement periods, are frequently used to overcome this issue. However, the term “incomplete stepped wedge design” is also used when transition or implementation periods are incorporated in a stepped wedge design. This is thus an imprecise term that does not convey any specific design characteristics. We introduce terminology and notation to classify different forms of incomplete stepped wedge designs, with a particular focus on different variants of staircase designs. We provide definitions of 5 sub-variants of staircase designs and of “irregular” incomplete stepped wedge designs. We also describe design modifications that can be applied to any complete or incomplete design: the inclusion of implementation periods; truncation of the design; and batched variants of these designs. This classification offers standardised terminology that researchers can use to describe their choice of incomplete stepped wedge design. This classification allows for a more nuanced classification of incomplete variants of stepped wedge designs than is currently available, allowing for greater clarity in the communication of design choices.

Lunch

11:50am - 12:00pm Thursday, 4th December, 2025

Grab a packed lunch box before heading to the public lecture...

Public Lecture: Dennis Trewin

12:00 - 1:00pm Thursday, 4th December, 2025
405.201 Haydn Williams LT
Presentation type Keynote
Chairs John Henstridge

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218 Trust in Statistics Is Crucial for the Wellbeing of Societies – But It Must Be Earned

Dennis Trewin

Former Australian Statistician at the Australian Bureau of Statistics, Canberra, Australia

Abstract

We know that trustworthy statistics are a necessary ingredient for the wellbeing of societies. But the increasing use of fake data, data scientists with inadequate statistical skills, the non-use of statisticians where statistical skills are a vital input, and other challenges are having a negative impact on the perceptions of trust in statistics. How do we address this? There are three key elements – Integrity, Quality and Relationships. Often the emphasis is on integrity but it is not the only important element.

The talk will provide examples of the challenges being faced in different fields of statistics. It will then address what needs to be done to increase trust for each of the three elements of integrity, quality and relationships.

SES: Curriculum

1:10 - 2:40pm Thursday, 4th December, 2025
405.201 Haydn Williams LT
Presentation type Contributed Session
Chairs Peter Dunn

8 Hands-On Employability Skills in a First-Year Business Statistics and Insights Unit

Ayse Bilgin ORCID iD[1], Jaslene Huan Lin ORCID iD[2]

[1]Macquarie University, Sydney, Australia. [2]University of Sydney, Sydney, Australia

Abstract

Equipping first-year business students with both statistical knowledge and employability skills is critical for preparing them for the workforce. In a first-year Business Statistics and Insights unit, hands-on learning activities are integrated to bridge the gap between theoretical concepts and real-world business applications. Students engage in practical tasks such as analysing business data, solving real problems, and communicating statistical findings to non-technical audiences. These activities foster key employability skills, including data analysis, critical thinking, and professional communication. By working on authentic business challenges, students not only strengthen their statistical understanding but also develop the competencies needed to navigate data-driven decision-making in the workplace using Excel. The approach has shown to increase student engagement, improve learning outcomes, and boost confidence in applying statistical techniques in professional contexts. The paper discusses the design and impact of these activities, offering strategies for incorporating employability-focused learning in introductory statistics courses, as well as the challenges of embedding employability skills in an introductory-level large statistics unit, offering insights for educators seeking to enhance the practical relevance of their teaching.

57 Reprogramming a First-Year Statistics Unit with a Regression-First Approach

Alethea Rea

Murdoch University, Perth, Australia

Abstract

At Murdoch University, the first-year statistics course was taught with a “foundations first” approach. Students were introduced to descriptive statistics, probability, distributions, sampling strategies and sampling variation before progressing to statistical inference and hypothesis testing. The latter half of the unit covered topics such as analysis of one proportion, one mean, two proportions, two means, multiple proportions, three or more means and the relationship between two numeric variables at a rate of one or two topics a week.

Anecdotally two disadvantages of this approach were observed: firstly, some students struggled to see the relevance of foundational material early in the course; secondly, linear regression-arguably one of the most widely used statistical techniques in the life sciences in its more complex forms-was covered in a single week, 10 weeks into the unit, making it hard emphasis it’s importance.

To overcome these challenges, the unit was reprogrammed to a “regression first” approach. The unit now focusses on the statistical process in the context of this one methodology and introducing other topics (like distributions and probability) just in time to allow an understanding of the five core steps in analysing the relationship between two numerical variables (visualisation, null and alternative hypothesis, assumptions, hypothesis testing, confidence and prediction intervals). In this framework, a two independent samples t-test can then be taught as regression with a

categorical explanatory variable, laying a foundation for students to engage with more advanced forms of regression in later studies.

102 Improving Accessibility in Tertiary Statistics Teaching: Lessons From a New Subject Development

Paul Fijn ORCID iD, Adriana Zanca ORCID iD

University of Melbourne, Melbourne, Australia

Abstract

Statistics education has benefited greatly from advances in technology: advancing the techniques which can be readily applied (and taught) and through accompanying innovations within teaching environments. These advances have not always been equally enjoyed by all students as, for example, statistical software is often chosen for pedagogical reasons rather than accessibility. *Excel*/*R* are ubiquitous in some commercial contexts (providing utility for students' future work-readiness), and *Minitab*/*SPSS* provide simple menu-driven approaches (allowing a focus on teaching the underlying statistical concepts and methods). The use of LaTeX and the consistency of PDF documents are historically valued: for ease of producing documents containing mathematical formulae, and as a professional standard. All of these can create challenges for students with a range of educational needs: for example, screen readers (for students with low/no vision) do not function correctly, and changing formatting is difficult (changing background colours and fonts can be helpful for students with dyslexia and colour-blindness). Additionally, some formats vary in functionality for different operating systems, which can be an impediment for students with limited financial resources.

Designing a new undergraduate statistics course, taught from July to November in 2025, provided a unique opportunity to structure the content, resources, and delivery around best practice for accessibility. We adopted a framework based on Universal Design for Learning (CAST, 2024) incorporating features seen in statistics education more generally: student agency in assessment, authentic learning, and providing opportunities for active learning. Additionally, all resources and subject materials were designed and developed using accessible formats with flexible display options, primarily using *PowerPoint* and *HTML* in conjunction with *MathJax*. We will discuss which aspects were most successful, which would be simplest to implement in existing subjects with established content and resources, and the challenges we faced in implementing universal design for learning in this subject.

196 Pre-Collegiate Statistics Education in the Philippines: A Regional Comparative Study

George Manapat

University of the Philippines Diliman, Quezon City, Philippines

Abstract

In our increasingly data-driven world, statistical literacy has become a fundamental skill. However, pre-collegiate instruction in this subject varies widely. This study compares the structure and content of secondary-level statistics education in the Philippines with those of selected regional peers in the Asia-Pacific. Using curriculum documents and national frameworks, we analyze curricular scope, depth of statistical reasoning, and assessment practices. Preliminary findings reveal that, although the Philippine K-12 mathematics curriculum incorporates statistics, its pedagogical strategies and resource support lag behind neighboring systems. We discuss implications for curriculum enhancement, teacher professional development, and regional collaboration to harmonize educational standards. Our evidence-based recommendations aim to bolster statistical literacy and guide policy dialogue across the Asia-Pacific.

179 Designing for Impact: How Teaching Study Design Shapes Causal Inference and Real-World Relevance

Alexandra Green ORCID iD, Alex Shaw, Kathrin Schemann, Omar Arnaiz, Jim Matthews, Christopher Howden

The University of Sydney, Sydney, Australia

Abstract

As statistical consultants, we're often approached *after* the data has been collected, with researchers hoping we can work our magic to strengthen their results and causal claims. But even the most advanced statistical analyses cannot salvage a poorly designed study or boost the signal to the noise. This talk reflects on the 90-minute study design workshop delivered by the Statistical Consulting Unit at the University of Sydney. While study design varies by discipline, our workshop focuses on the core principles of internal validity, external validity, replication and sample size, domain practice as well as feasibility and impact. It encourages researchers to consider the methods and procedures used to plan, collect and analyse their data ensuring it is reliable and valid for drawing conclusions. Key topics include aligning study type with research questions, sampling and grouping strategies, randomisation and blinding, and visualising relationships between variables. Our workshop and accompanying resources have been instrumental in building core study design literacy across the university. In 2024, our workshop was delivered to 94 researchers over five sessions, at four different locations and online. It has prompted researchers to engage our statistical consulting services before they perform their study or submit their grant application, with the focus of consults shifted to higher level topics such as causal inference and generalisability. Our workshop is evolving with participant feedback and the latest statistical research, equipping researchers to meet reproducibility standards expected by high-impact journals, such as trial pre-registration, statistical analysis plans, EQUATOR guidelines and the use of directed acyclic graphs to strengthen causal inference. Ultimately, the goal of our workshop is to support researchers in undertaking higher-quality studies that drive real-world impact.

IPS: Official Statistics

1:10 - 2:40pm Thursday, 4th December, 2025
405.205
Presentation type Invited Session
Chairs Andrew McMahon

63 Unit-Level Versus Area-Level Approaches to Small Area Estimation with Measurement Errors and Their Application to Indonesian Household Surveys

Ika Yuni Wulansari ORCID iD[1,2,3], Stephen Woodcock ORCID iD[1], James Brown ORCID iD[1]

[1]School of Mathematical and Physical Sciences, University of Technology Sydney, Sydney, Australia. [2]Politeknik Statistika STIS, Jakarta, Indonesia. [3]Statistics Indonesia, Jakarta, Indonesia

Abstract

National Statistical Institutes (NSIs) face growing demands to produce highly disaggregated statistics, particularly to monitor progress toward the Sustainable Development Goals (SDGs). These demands often exceed the capacity of household surveys to generate reliable direct estimates at fine geographic levels. To address this, NSIs increasingly adopt Small Area Estimation (SAE) techniques that combine survey data with auxiliary information through statistical modelling. However, many auxiliary variables used in SAE are derived from surveys and are subject to measurement errors, which can bias model-based estimates if not properly accounted for. This study applies two modelling frameworks: the Fay-Herriot area-level model with measurement error correction following Ybarra and Lohr (2008), and the unit-level model proposed by Datta, Torabi, Rao, and Liu (2018) extended to accommodate functional measurement error under a finite population framework. A simulation study evaluates the performance of these models under scenarios involving a single auxiliary variable and a mixed setting with two auxiliary variables (one error-prone and one error-free). We then apply both models to a real-world case study using Indonesian data. The variable of interest is per capita household expenditure from the National Socio-Economic Survey (SUSENAS). Auxiliary variables are sourced from both the Village Potential Data (PODES), treated as error-free, and additional survey-based variables that may contain measurement error. Estimates are produced at the district level, though the underlying variation reflects sub-district-level patterns due to subsample variability in the design. To support policy use, we present spatial visualisations of the estimates, highlighting any areas where accounting for measurement error notably affects the estimates or their associated confidence intervals. The application demonstrates how utilising auxiliary information from other independent surveys within SAE methods with appropriate adjustments can improve the reliability of disaggregated indicators essential for local development planning.

121 Estimating Nonlinear Population and Superpopulation Parameters with Probability and Nonprobability Samples

Ryan Covey ORCID iD[1], Lucca Buonamano ORCID iD[2]

[1]Australian Bureau of Statistics, Melbourne, Australia. [2]Australian Bureau of Statistics, Sydney, Australia

Abstract

Nonprobability samples, like big data and administrative data, are increasingly used in official statistics and the applied sciences, but statistics computed using only nonprobability samples often suffer from substantial selection bias. This leads to inaccurate estimation and invalid statistical inference. We rectify the issue for a broad class of linear and nonlinear statistics by producing estimating equations that combine probability and nonprobability samples in a way that also accommodates corrections for other sources of error such as nonresponse in the probability sample and measurement error in either sample. Consistency and asymptotic normality are established by building on existing design-based results in the official statistics literature. We construct variance estimators that account for sampling variability introduced by corrections to various sources of error. These variance estimators can be produced with respect to either the sampling design alone or jointly with the superpopulation; in the limit, the joint variance of a sample statistic is equal to its design variance plus the superpopulation variance of the corresponding population parameter. Results are also applicable when using only a probability sample. We illustrate our method for quantiles, the Gini index, linear regression coefficients and maximum likelihood estimators. Our results are also illustrated in simulation and with real data about individual Australian incomes.

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130 Modernising Household Survey Designs

Soraya McPhail, Michael Fairbrother

Australian Bureau of Statistics, Sydney, Australia

Abstract

Declining response rates, alongside increasing collection costs, are an increasing challenge for household surveys introducing potential non-response bias. The ABS are exploring alternative cost-effective sample designs to help reduce non-response bias and improve the representation of the responding sample. The ABS has explored selecting households to respond via an online survey, then only following up a subset of the non-respondents with a more expensive personal interview. The follow-up component can be used to adjust for non-response bias that may be present from the online-only respondents. We have also explored modelling a dwellings propensity to respond and using this information to improve selections. Dwellings with a lower propensity to respond are given a higher chance of selection. While this may decrease the overall achieved response rate this approach seeks to achieve a more representative responding sample. We will discuss these approaches and other avenues the ABS is exploring to improve household surveys.

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145 Technical and Ethical Considerations in Calculating Post-Stratification Survey Weights for the Mayi Kuwayu Study

Benjamin Harrap ORCID iD, Christopher McKay, Masud Hasan, Joanne Thandrayen, Katherine Thurber, Bernard Baffour, Bronwyn Wilkes, Bobby Maher, Siena Montgomery, Raymond Lovett

Australian National University, Canberra, Australia

Abstract

Background

Mayi Kuwayu: The National Study of Aboriginal and Torres Strait Islander Wellbeing (MK) is a large-scale cohort study that is Indigenous-led, with Indigenous Data Governance, that measures a range of unique community-developed cultural indicators, along with health and wellbeing measures. Participants in the first wave of MK came from both a probability sample of Australia's Medicare database and through community-based recruitment and word-of-mouth, meaning the cohort is a mix of probability and non-probability sampled individuals.

One goal of the study is to generate prevalence estimates that were representative of the total Aboriginal and Torres Strait Islander population. To do so, appropriate weights are required.

Methods

Our weighting approach uses post-stratification survey weights. Imputation is used for missing data, and we consider the ethics of imputing a gender binary – given population data informing the weights is provided as a binary.

We also consider the relevance and feasibility of which demographic variables to use in weighting and how to manage issues such as extreme weights.

To examine how our weighting approach performed, we compare weighted prevalence estimates against those reported in the National Aboriginal and Torres Strait Islander Health Survey (NATSIHS).

Results

We present survey weights calculated under different assumptions, the assumptions' impacts on population estimates, and the final weighting method selected for usage with the MK dataset. There were several differences when comparing against NATSIHS prevalence estimates.

Conclusions

We demonstrate how post-stratification weighting can be used to generate more representative population estimates from the MK data, with some important limitations to be considered by other non-probability surveys of Aboriginal and Torres Strait Islander peoples. We discuss future directions of our weighting approach and implications for providers of population data.

Epidemiology 2

1:10 - 2:40pm Thursday, 4th December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Sourav Das

167 Beyond Averages: Generalising Models of Patient Transfer Dynamics in Healthcare Networks

David Wu[1], Michael Lydeamore[1], Tjibbe Donke[2], Nicolas Geard[3], Claire Gorrie[4], Andrew Stewardson[5]

[1]Department of Econometrics and Business Statistics, Monash University, Melbourne, Australia. [2]Institute for Infection Prevention and Hospital Epidemiology, University Medical Center, Freiburg, Germany. [3]School of Computing and Information Systems, University of Melbourne, Melbourne, Australia. [4]Microbiological Diagnostic Unit Public Health Laboratory, Department of Microbiology & Immunology, University of Melbourne, Melbourne, Australia. [5]Department of Infectious Diseases, The Alfred and School of Translational Medicine, Monash University, Melbourne, Australia

Abstract

The transmission of antimicrobial-resistant infections within and between hospitals remains a significant global health challenge. To effectively model and control these infections, it is critical to understand how patients—and pathogens—move through healthcare networks. However, existing models of inter-hospital transmission often oversimplify patient flow, either by collapsing temporal patterns into static mean flows or by relying on narrowly tailored agent-based simulations. There remains a clear need for generalisable, data-driven models that accurately reflect patient movement across time and settings.

By using a line-listed patient admission dataset spanning 10 years in Victoria, we build a network of hospital transfers. In this weighted, directed, temporal network, each node represents a healthcare facility, and edges between nodes represent the number of patients transferred between the two facilities.

By constructing three models at different temporal resolutions, we demonstrate that the method used to collapse the temporal structure of patient transfers can substantially influence estimates of transmission potential. We observe several temporal factors—such as weekday/weekend variation and public holidays—that significantly affect transfer rates. We also show that incorporating the time patients spend at home between hospital admissions is essential for accurately reconstructing the transmission network. These algorithmic developments help to set up robust models of surveillance and outbreak control, offering practical guidance for future interventions in healthcare settings.

200 The Moderating Role of Personality Traits in Internet Addiction and Mental Health: A Multilevel Modelling Study

Ashish Yadav ORCID iD, Ruchi Yadav

Institute of Medical Sciences, Banaras Hindu University, varanasi, India

Abstract

The present study investigates the moderating effect of personality traits on the relationship between internet addiction, general mental health, and well-being among university students. The study wishes to explore the relationship between mental health and internet addiction while considering personality traits as an exposure variable. To confirm the theoretical structure obtained through a systematic review a confirmatory factor analysis will be performed. A multilevel modelling approach will be used to explore the relationship between the outcome

and the independent proximate variables. Once the models are developed, model diagnostics will be performed accordingly. The research begins with a comprehensive systematic review and meta-analysis, providing an unbiased estimate of depression, anxiety, stress, and internet addiction prevalence among university students. This forms the foundation for understanding the mental health challenges faced by the youth. The development and validation of Multilevel Hierarchical Models enhances the study's methodological rigour, considering the nested structure of individuals within the university population. Model diagnostics ensure the reliability and credibility of the findings. This study is expected to bring about several benefits. It will provide a comprehensive understanding of the prevalence of internet addiction and mental health issues among young people. Additionally, it will advance our knowledge of how personality traits affect these relationships. Moreover, it will offer universities and other institutions evidence-based recommendations to enhance the well-being of their students. This research endeavours to inform policies, treatments, and decision-making processes specific to universities by taking a holistic approach to understanding the complexities of student mental health and well-being. Its implications extend beyond individual universities and can be applied to large young populations.

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78 Modelling Tree Population Dynamics Using Spatial Birth–Death Point Processes

Shamali Sujeewa Kumari Pradana Mudiyanselage ORCID iD[1], Yan Wang ORCID iD[1], Ian Flint[2], Laleh Tafakori ORCID iD[1]

[1]Department of Mathematical and Geospatial Sciences, RMIT University, Melbourne VIC, Australia. [2]School of Agriculture, Food and Ecosystem Sciences, the University of Melbourne, Parkville VIC, Australia

Abstract

Accurate modelling of tree population dynamics in forests depends on understanding how birth and death events interact with the spatial layout of the ecosystem. Traditional models often neglect local interactions, spatial heterogeneity, and the irregular timing of ecological observations. Addressing these limitations, this study introduces an advanced modelling framework based on spatial birth–death point processes (SBDPPs) to analyse the spatiotemporal dynamics of tree populations with fine ecological and statistical resolution.

The model advances the field by proposing both continuous-time and discretely observed time formulations of SBDPPs, enabling it to reflect the structure of real-world ecological monitoring data. Birth and death events are modelled as spatially dependent stochastic processes, where event intensities are functions of local configuration and interaction distances. We derived parametric maximum likelihood estimators for both continuous-time and discretely observed time event and introduce a novel likelihood formulation that facilitates tractable inference from discretely observed data. Interaction distances are estimated using profile likelihood, enhancing the model’s ecological interpretability. Comprehensive simulation studies demonstrated the model’s ability to estimate underlying parameters accurately, with improved precision at higher sampling frequency and longer observation windows. The results indicated that estimates of birth and death rates are sensitive to the assumed interaction range between trees, emphasizing the need to select an appropriate spatial scale in ecological analysis. The model was applied to over 40 years of tree census data from CSIRO’s North Queensland rainforest plots, uncovering unique population trends and spatial arrangements for different species. It revealed patterns of stability, decline, and growth that closely reflect actual birth and death events recorded in the data.

Regression Methods

1:10 - 2:40pm Thursday, 4th December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Soudabeh (Soudi) Shemehsavar

58 Robust Spherical Regression in an Extrinsic Semiparametric Framework

Houren Hong, Janice Scealy, Andrew Wood, Yanrong Yang

Research School of Finance, Actuarial Studies and Statistics, Australian National University, Canberra, Australia

Abstract

Robust regression for spherical data with Euclidean covariates is an important yet under-explored problem relevant to many fields. This talk introduces a flexible and efficient single-index framework for semiparametric regression in an extrinsic setting, enabling robust estimation through a suitable choice of loss function. Large-sample properties of the resulting estimators are established in a general setting. Moreover, analyzing robustness is complicated by the spherical geometry and semiparametric structure, where traditional tools such as the influence function become inadequate. We address this by using the standardized influence function. Special attention is given to the exponential squared loss (ESL), whose application to spherical regression appears to be new. We demonstrate that ESL achieves comparable efficiency and superior robustness to least squares under high concentration. Theoretical analysis of its tuning parameter reveals a trade-off between efficiency and robustness, guiding its optimal selection. Simulations and applications to geochemical data validate the computational efficiency and robustness of our approach.

70 Meta-Analysis of Medians, Other Quantiles and Functions of Quantiles With Applications to Diabetes Outcomes

Udara Kumaranathunga ORCID iD[1], Luke Prendergast ORCID iD[1], Jonathan Shaw ORCID iD[2], Alysha De Livera ORCID iD[1]

[1]La Trobe University, Melbourne, Australia. [2]Baker Heart and Diabetes Institute, Melbourne, Australia

Abstract

Meta-analysis of continuous outcomes has traditionally relied on methods that use moment-based statistical measures such as means and standard deviations. However, it is found that in practical scenarios, outcome measures are often skewed. Diabetes-related biomarkers, such as creatinine, triglycerides, and microalbumin are several examples among many that commonly display skewed underlying distribution. For such skewed outcome measures, we introduce meta-analyses of medians, other quantiles, and functions of quantiles to comprehensively synthesise information. We consider commonly reported scenarios of quantiles, namely the five-number summary (minimum, first quartile, median, third quartile, maximum) and the three-number summaries ({first quartile, median, third quartile} or {minimum, median, maximum}), each with their sample sizes. Within a novel density-based framework, without making any prior assumptions about the underlying distributions, we use flexible quantile-based distributions with percentile matching to estimate the unknown parameters. We conduct simulation studies to evaluate the approaches and present results for varying number of studies, study sizes, distributions of the outcome measures, and heterogeneity. Together with the simulation results, we report results from an empirical study of meta-analysis of quantiles and interquartile range widths as an example of meta-analysis of functions of quantiles, using real-world data from the Australian Diabetes, Obesity, and Lifestyle (AusDiab) study. We further present our CRAN-based R package (metaquant), which contains the necessary code for straightforward implementation of the introduced approaches.

141 A Practical use of the Moments of the Logit-Normal Distribution: How to Avoid Numerical Integration in the Expectation Propagation Algorithm for Logistic Regression.

John Holmes

University of Canterbury, Christchurch, New Zealand

Abstract

The logit normal distribution is a distribution supported on $0 < x < 1$, such as $logit(x) = log(x/(1-x))$ is distributed normal. It is generally believed that the moments of the logit normal distribution are not known analytically and must be solved using numerical integration techniques. This is not quite true, as the expected value of a logit-normal random variable is an example of a Mordell integral, which can be written as an infinite sum, the terms of which decay rapidly. Furthermore, the properties of the logit-normal distribution allow the expected value to be simplified further from the standard Mordell integral result as shown in Holmes and Schofield (2022). However, extending the Mordell integral based result to finding higher moments has proven difficult/impossible.

Knowing the moments of the logit-normal distribution has the potential to expand the range of inferences that can be made in logistic generalised linear models. In this talk though, we will focus on the need to determine functions of moments of the logit normal distribution in approximate Bayesian inference approaches for logistic regression, specifically Expectation Propagation. In order to determine these moments, we will introduce a new highly accurate approximation to the softplus and logistic functions over all values of x such that the approximation of the expectations needed in the Expectation Propagation moment matching algorithm used to estimate the approximate posterior parameters can be readily found analytically, thus removing the need for numerical integration.

150 A Set of Precise Asymptotics for Gaussian Variational Approximation in Generalised Linear Mixed Models

Nelson Chua[1,2], Francis Hui[1], Luca Maestrini[1], Alan Welsh[1]

[1]Australian National University, Canberra, Australia. [2]Australian Bureau of Statistics, Canberra, Australia

Abstract

Generalised linear mixed models (GLMMs) are used to model data with a clustered or hierarchical structure, capturing intra-cluster dependence through the use of (latent) random effects. However, this dependence structure results in a likelihood function that is typically computationally intensive to evaluate, and so approximate likelihood approaches are often used instead for model fitting and inference. One such approach which has grown in popularity over the past decade is Gaussian variational approximation (GVA). In addition to estimates of the model parameters, GVA concurrently produces random effects predictions and associated uncertainty measures.

In this talk, we formulate a set of precise asymptotic properties for both the parameter estimates and random effects predictions obtained from GVA, focusing on independent cluster GLMMs. We find that these properties change substantially depending on whether or not the random effects are conditioned on. By comparing GVA's asymptotic properties and empirical finite-sample performance with that of other commonly used approximate likelihood approaches, we highlight situations in which the use of GVA would be advantageous in practice.

188 A Numerically Stable Rescaling Method for Penalized Likelihood Estimation of Semiparametric AFT Models

Aishwarya Bhaskaran ORCID iD[1], Ding Ma ORCID iD[2], Jun Ma[1]

[1]Macquarie University, Sydney, Australia. [2]The University of Queensland, Brisbane, Australia

Abstract

Cox proportional hazards and accelerated failure time (AFT) models are two principal frameworks for assessing covariate effects in survival analysis. The use of an AFT model is particularly appealing as it offers an intuitive interpretation by directly linking covariates to time-to-event outcomes through a multiplicative acceleration factor. While semiparametric AFT models allow for flexible modelling without specifying the error distribution or baseline hazard, they pose significant computational challenges, especially with interval-censored data. Unlike Cox models, estimation in semiparametric AFT models is markedly more demanding, as the transformation of failure times used to estimate the baseline hazard is dependent on the regression coefficients and must be updated iteratively, often leading to instability or convergence issues. Moreover, incorporating interval censoring complicates the likelihood expression, introducing algebraic challenges in deriving gradients and Hessians, particularly when the intervals are narrow.

To address these challenges, we propose a maximum penalized likelihood approach using rescaling techniques for fitting semiparametric AFT models to data with partly interval-censored failure times. Our method employs M-splines to approximate the nonparametric baseline hazard, a constrained optimization framework for stable coefficient estimation and an automatic smoothing selection criterion that accounts for active constraints. Simulation studies demonstrate the robustness and accuracy of our approach across a range of censoring scenarios.

Afternoon tea

2:40 - 3:00pm Thursday, 4th December, 2025

SES: Improving students' learning experiences and support

3:00 - 4:30pm Thursday, 4th December, 2025
405.201 Haydn Williams LT
Presentation type Contributed Session
Chairs Amanda Shaker

142 Bridging the Gap in Statistical Learning through the Maths Hub Model in Higher Education

Ravindi Nanayakkara ORCID iD

La Trobe University, Melbourne, Australia

Abstract

In today's data-driven world, statistical literacy is vital across diverse academic disciplines. Mathematics and statistics underpin many fields, making strong competency essential for successful undergraduate and postgraduate studies. However, many students enter university with limited preparation in these areas, often resulting in statistics anxiety and academic difficulties. It is challenging for universities to meet diverse student needs due to limited time and resources for personalized support. To address this gap, the Maths Hub was established as a multi-campus, multidisciplinary support centre focused on improving learning outcomes in mathematics, statistics, and related fields through tailored, discipline-specific assistance.

The Maths Hub offers tutoring, workshops, and online resources, delivering support through a blended learning model combining face-to-face and Zoom sessions. This flexible approach accommodates students from different campuses and varied learning paces, fostering active engagement with complex quantitative concepts. The Hub's services include assistance with lecture notes, lab classes, problem-solving, quizzes, assignment guidance, exam preparation, statistical data analysis using Jamovi and SPSS software, and R programming. Additionally, support is provided for LANTITE numeracy and specific Maths Hub modules.

This study explores how the Maths Hub model supports statistical learning and bridges the gap in students' understanding of quantitative concepts in higher education. Using a mixed-methods approach, the research integrates student performance metrics with qualitative feedback from discipline-specific surveys to evaluate the Hub's impact. The findings indicate that the Maths Hub, particularly in its blended learning format, plays a key role in enhancing students' statistical skills by increasing access to tailored support, fostering deeper engagement, and building confidence in applying statistical knowledge. The model not only supports academic performance but also reduces barriers in quantitative learning, demonstrating its value as an inclusive and flexible support structure within the university context.

120 Exploring the Role of Mathematics Background and Support in Student Success Across Introductory Statistics Subjects

Gizem Intepe ORCID iD, Merryn Horrocks

Western Sydney University, Sydney, Australia

Abstract

This study investigates how mathematics background and engagement with academic support influence performance in four first-year statistics subjects tailored for students in Business, Science, Engineering, and IT/Data Science programs. These service-taught subjects incorporate discipline-specific applications while covering core statistical concepts. Institutional data from 2019–2023 was analysed, including final grades, secondary school mathematics background, and participation in support workshops and drop-in sessions.

Results from ANCOVA and multiple linear regression analyses indicate that secondary school mathematics background is a significant predictor of final grades across all subjects. Both workshop attendance and engagement with mathematics support services were found to be significant positive predictors of student performance, regardless of mathematics background. Students with stronger mathematics backgrounds consistently achieved higher grades; however, the benefit of support attendance was uniform across all levels. No significant interaction effects were observed.

These findings highlight not only the value of embedded academic support but also raise important questions about the role of mathematics preparation in student success. As universities increasingly enrol students with diverse educational backgrounds, understanding and addressing the persistent effects of prior mathematics education becomes critical in designing equitable and effective statistics curricula.

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21 Addressing Barriers and Advancing Practice: Innovations in Student Experience Feedback

Sumaira Qureshi ORCID iD[1], Luke Nguyen-Hoan[1], M Ejaz Qureshi[2,3]

[1]University of Canberra, Canberra, Australia. [2]The Australian National University, Canberra, Australia. [3]The University of Agriculture, Faisalabad, Pakistan

Abstract

Effective feedback practices – including constructive comments that justify grades and identify strengths and weaknesses - play a critical role in higher education. This study investigates barriers to student engagement with feedback at an Australian university, focusing on differences between postgraduate (PG) international and undergraduate (UG) domestic students. Employing a mixed methods research design, it combines quantitative survey analysis with qualitative thematic coding to gain comprehensive insights into students' perceptions of the Student Experience Questionnaire (SEQ) process.

Data were collected through an anonymous online survey comprising 25 questions, including scaled-response items designed to capture quantitative insights and open-ended prompts to gather qualitative feedback. Thematic analysis revealed key emotional and structural barriers, such as anxiety when receiving feedback, perceived staff defensiveness, and a lack of visible outcomes from previous SEQ responses. Statistical analysis identified notable distinctions between PG international and UG domestic students: while both groups acknowledged communication challenges, they differed in how these were perceived and prioritised. PG international students expressed stronger support for strategies to enhance engagement with the SEQ (e.g., in-class time, reminders), whereas UG domestic students were more sceptical about its value and effectiveness.

Findings underscore the need to address both emotional and practical dimensions of feedback. Students advocated for more transparent communication, independent feedback channels, and greater visibility of how feedback is acted upon. These recommendations align with broader literature on student voice and engagement, highlighting the importance of co-designed solutions that foster mutual trust and accountability between students and educators.

By centring student perspectives and offering actionable insights, this study contributes to advancing feedback practices and student engagement. The results have implications for institutional policy, particularly in designing

more inclusive, transparent, and responsive feedback systems that reflect the diverse needs of contemporary university cohorts.

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20 Empowering Learning Through Live Data: Exploring Autonomy, Engagement, and Student Voice in a Hybrid Statistics Unit

Sumaira Qureshi ORCID iD[1], Zeena Alsamarra'I[1], Qishang Chua[1], Luke Nguyen-Hoan[1], M Ejaz Qureshi[2,3]

[1]University of Canberra, Canberra, Australia. [2]The Australian National University, Canberra, Australia. [3]The University of Agriculture, Faisalabad, Pakistan

Abstract

This study investigates how autonomy-supportive practices, real-time data collection, and hands-on analysis can enhance student engagement and deepen conceptual understanding in a hybrid statistics unit co-taught to undergraduate and postgraduate students. Conducted over two semesters, the project engaged on-campus, online, and self-paced learners, incorporating student-generated data into weekly learning activities. It also examines whether students' chosen mode of participation correlates with academic performance, offering insight into how autonomy influences learning outcomes.

Each week during the optional drop-in sessions for the on-campus and online cohorts, students completed an anonymous online questionnaire aligned with the statistical concept being taught. The collected data were cleaned collaboratively at the start of each session and analysed in real time using the week's topic. This hands-on process allowed students to engage with the full data cycle – from collection to cleaning, analysis, and interpretation. Self-paced students accessed the same materials through recordings and step-by-step resources.

This approach was designed to foster student ownership, motivation, and real-world application of statistical knowledge. In Week 13, 107 out of 210 enrolled students (49% response rate) reflected on the experience. Preliminary analysis of this feedback indicates that students perceived the live data collection and analysis as "interactive", "transparent", and "practical", with many reporting increased confidence in using statistical software and applying common tests. Several students highlighted that seeing their own data analysed helped demystify complex concepts and reinforced their learning through relevance. Suggestions for improvement included diversifying datasets, enhancing prompts to encourage greater participation, and providing clearer access to data.

By integrating autonomy, interactivity, and student voice, this project presents a scalable and inclusive model for teaching quantitative methods. It illustrates how live, student-generated data can serve not only as a pedagogical tool but as a catalyst for deeper engagement, conceptual understanding, and practical skill development across varied learning contexts.

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62 Exploring Educational Performance in Saudi Arabia by Gender, Early Literacy, and SES Using TIMSS 2015, 2019, and 2023 and Grade 4 Numeracy

Asmaa Khalid S Binkhamis ORCID iD, Stephen Woodcock ORCID iD, James Brown ORCID iD

University of Technology Sydney, Syndey, Australia

Abstract

Despite substantial investment in education, Saudi students continue to perform below international benchmarks, emphasizing the importance of data-driven reform efforts. This research investigates Saudi Arabia's educational performance in international assessments in support the goals of Vision 2030, particularly focusing on Trends in International Mathematics and Science Study (TIMSS). The primary objective is to identify the key factors influencing mathematics achievement using TIMSS data across multiple assessment years.

In alignment with Saudi Arabia's Vision 2030 goals and its emphasis on evidence based educational transformation, this research analyses student performance patterns in mathematics for Grades 4 and 8 using aggregated TIMSS data from 2015, 2019, and 2023. The analysis focuses on structural trends related to gender, early literacy acquisition and school-level socioeconomic composition.

A general linear modelling framework is employed to assess these predictors on average student performance, incorporating interaction terms to explore the joint effects of variables, allowing for a more realistic modelling of complex education systems.

Preliminary findings suggest the gender gap has narrowed over time, and early literacy remains a consistent predictor of mathematics performance. The influence of socioeconomic background across assessment years consistently favours students attending more advantaged schools.

By combining three TIMSS cycles, the study enhances statistical power and enables the identification of persistent pattern over time. This multi-year perspective offers deeper insight into these variables, which are pillars of the Saudi national education strategy. The findings aim to inform targeted interventions that enhance early literacy skills and resource allocation across schools, to improve learning outcomes and achieve long-term educational equity.

Official Statistics 2

3:00 - 4:30pm Thursday, 4th December, 2025
405.205
Presentation type Contributed Session
Chairs Joseph Chien

47 A Survival Analysis of Survey Responses and the Impact of Follow-up

Michael Fairbrother

The Australian Bureau of Statistics, Sydney, Australia

Abstract

Surveys at the ABS are typically collected over an enumeration period. During this period, phone calls and reminder emails will be sent to businesses to boost response rates. Strategies are put in place to outline when reminders should be sent, and who they should target. Monitoring response rates as you go can inform how to best use available staff for the remainder of the enumeration period. In this study, we use survival analysis techniques to forecast final response rates for the Wage Price Index Survey. We design a model that can be used at any stage of the enumeration period and can include planned strategies under different scenarios in its forecasts. This can be used to inform decision making min-enumeration and can be used to assess the impact of follow-up strategies post-enumeration.

112 Small Area Estimation of Australian Labour Force Statistics

Mark Ioppolo[1], Summer Wang[2]

[1]Australian Bureau of Statistics, Perth, Australia. [2]Australian Bureau of Statistics, Brisbane, Australia

Abstract

The Australian Bureau of Statistics (ABS) Labour Force Survey provides the official estimates of employment, unemployment, and labour force participation across Australia. It serves a diverse range of data users from government agencies, businesses, and academic institutions. While there is strong demand for timely and disaggregated labour market data, traditional design-based estimates become unreliable at fine granularities due to small sample sizes. To address this, the ABS has begun using administrative tax and Jobseeker data in conjunction with a Rao-Yu time series model to produce and publish smoothed monthly labour statistics at the Statistical Area 4 (SA4) level. This presentation will outline the ABS’s ongoing efforts to enhance small area labour force statistics to address long-standing data needs, including developments in producing estimates disaggregated by age and sex at the SA4 level, and extending coverage down to finer geographic levels. We will outline key achievements, practical applications, and the challenges that remain.

125 Development of a Longitudinal Weighting Methodology for the ABS Longitudinal Labour Force Microdata

Aparjit Singh, Jennifer Marley

Australian Bureau of Statistics, Canberra, Australia

Abstract

A relatively recent development by the ABS has been the creation of the Longitudinal Labour Force microdata asset (LLFS). It comprises the monthly data collected in the ABS Labour Force Survey (LFS) longitudinally linked across time. Currently, the LLFS contains over 42 years of LFS data from October 1982 to the latest LFS publication month and is available in the ABS DataLab for researchers to use. However, a crucial piece is missing – longitudinal weights.

Calculating longitudinal weights for the LLFS presents methodological challenges as the LFS is not designed as a longitudinal study. The LFS is a large cross-sectional household survey that collects information monthly from the Australian population aged 15 years and over about their labour market characteristics from which headline figures such as the monthly unemployment rate estimates are calculated. The monthly LFS sample consists of eight sub-samples, referred to as rotation groups, where each sub-sample remains in the LFS for eight months. A new rotation group is introduced each month to replace an out-going rotation group.

In our presentation, we will describe the overall approach used to arrive at monthly and multi-monthly longitudinal weights. We will also compare and contrast our longitudinal weighting approach to other Australian longitudinal studies, with our deviation from common practice particularly noteworthy in accounting for the LFS's short, rolling panels of survey respondents.

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127 Analysis of ABS Labour Market Microdata: Employment Outcomes for Potential Workers

Susan Fletcher, Jennifer Marley

Australian Bureau of Statistics, Canberra, Australia

Abstract

In recent years, the Australian Government has boosted its focus on policies and programs that aim to increase the participation of people aged 15 years and over in the Australian labour market. Potential workers, those who are not currently employed but who want to work, are of particular interest with the goal being to match them with meaningful and enduring job placements.

As part of our initiatives at the ABS to address critical labour market data gaps, we have applied a multivariate modelling approach to conduct a novel analysis of two ABS data sources - the Longitudinal Labour Force Survey microdata and the Participation, Job Search and Mobility survey data from 2015 onwards – to answer key research questions about potential workers transitioning into employment after one month and three months. Our work is a leading example of how available ABS labour market microdata can be analysed to discover new, policy-relevant and actionable insights.

Our modelling process follows a two-step approach: a bootstrapping-type selection process to identify significant variables from a set of candidate variables, followed by a cross-validation technique to assess accuracy and model fit. Our analysis identified three distinct groups of potential workers, namely those with distinctly higher, moderate and distinctly lower transition rates to employment. We fit logistic regression models with main effects interacted with a time effect to (i) estimate the probabilities of potential workers in these three groups transitioning to employment after one month and three months, (ii) identify characteristics and circumstances associated with higher probabilities of gaining employment and also potential barriers to employment, and (iii) understand how these probabilities, characteristics and barriers have changed over time.

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159 Modernising Agricultural Statistics

Rachael Russo

Australian Bureau of Statistics, Brisbane, Australia

Abstract

The Australian Bureau of Statistics has produced agricultural statistics for over a century by collecting data directly from farmers in Agricultural Censuses and large-scale sample surveys. In recent years this has been replaced with an administrative data first approach with limited input from ABS surveys. The increasing availability of quality external data, together with a commitment to reduce reporting burden and declining survey response rates, have triggered this change in the way Australia’s official agricultural statistics are produced. This talk focuses on Broadacre Cropping statistics. We explain the new data that is available and how this is being used to produce estimates. Particular attention is given to the role and demands placed on survey methodologists and how this has changed now that there is no survey.

Law, Policy, and Society

3:00 - 4:30pm Thursday, 4th December, 2025
402.220 LT
Presentation type Contributed Session
Chairs Katharina Schueller

105 Have You Met Your Digital Doppelgänger?

Alex Jackson

Symbolix, Melbourne, Australia

Abstract

Twin Cities is a statistically representative, yet completely synthetic, dataset of households, families, and persons in Australia. In this talk, I will discuss the motivations, methods, differences between the digital twin and ABS microdata, and applications.

This product was born out of the desire to be able to undertake rigorous yet flexible agent-based modelling. Using publically available ABS Census data as feedstock, we sequentially augmented the Geoscape G-NAF address list with household attributes. This flowed on to create families of persons within the household, and the persons themselves - all with statistically representative attributes derived from the Census and other reliable data sources.

Twin Cities is not seeded/based upon a resampled microdata. This is a unique approach, which while computationally complex, gives control and flexibility over forecasting scenarios.

Like ABS microdata, Twin Cities gives the user unit level responses. However, as it is completely synthetic, there are no privacy issues. The unit level data can be used for further applications, among them microsimulations for traffic models, behaviour change scenario modelling, and forecasting.

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113 Where’s the Hype for Causality? Unleashing the Power of Causation Over Correlation.

Christopher Howden ORCID iD

The University of Sydney, Sydney, Australia

Abstract

“Correlation isn’t Causation” - is a well know adage amongst researchers, but do they really understand the power of causation over correlation? The power to accelerate their careers using stronger causal research to identify which levers drive real-world change, make more meaningful research contributions, publish in higher-impact journals, and secure a seat at the table when society discusses its most pressing challenges and their field of research.

Shouldn’t we be saying *Causation is **better** than Correlation*? And that we statisticians have the tools and tricks to differentiate between the two and help researchers improve the causal strength of their research using methods such as: study designs with strong causal power e.g. Before After Control Impact (BACI) designs, causal pathway analysis, Directed Acyclic Graphs (DAG’s), and the Bradford Hill criteria (a list of analyses and other tricks that strengthen causal inference).

In this talk, I will discuss the importance of building interpretable models with strong causal power, the opportunity to use them in elevating statistics reputation in a world distracted by the hype of AI and ChatGPT, and some

methods that improve causal inference. I'll also explore how the statistics profession can reclaim leadership in this crucial area - by guiding researchers, shaping discourse, and reinforcing our discipline's relevance in solving society's most pressing challenges.

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185 Statistical Challenges in Class Actions in Australia

Ian Gordon ORCID iD

Statistical Consulting Centre, The University of Melbourne, Melbourne, Australia

Abstract

The estimation sometimes required in class actions in Australia entails a mix of practical and more classically statistical challenges. These include varying legal manoeuvres, definition of the class and its relation to a feasible sample frame, estimation of a total when there is uncertainty about the population size, the feasibility and merit of using stratified sampling, and the contribution that statistical estimates make to the overall process.

In this paper I draw on experiences of engaging in this activity in a number of large Australian class actions, to demonstrate these challenges and suggest approaches that balance statistical rigour with practical implementation.

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71 Bayesian Spatiotemporal Modelling of Political Violence and Conflict Events Using Discrete-Time Hawkes Processes

Raiha Browning[1], Hamish Patten[2], Judith Rousseau[3], Kerrie Mengersen[4]

[1]Walter and Eliza Hall Institute of Medical Research, Melbourne, Australia. [2]Freelance consultant, Lausanne, Switzerland. [3]Paris Dauphine University, Paris, France. [4]Queensland University of Technology, Brisbane, Australia

Abstract

Monitoring of conflict risk in the humanitarian sector is largely based on simple historic averages. To advance our understanding, we propose Hawkes processes, a self-exciting stochastic process used to describe phenomena whereby past events increase the probability of future events occurring. The overarching goal of this work is to assess the potential for using a more statistically rigorous approach to monitor the risk of political violence and conflict events in practice and characterise their temporal and spatial patterns.

The region of South Asia was selected as an exemplar of how our model can be applied globally. We individually analyse the various types of conflict events for the countries in this region and compare the results. A Bayesian, spatiotemporal variant of the Hawkes process is fitted to data gathered by the Armed Conflict Location and Event Data (ACLED) project to obtain sub-national estimates of conflict risk over time and space. Our model can effectively estimate the risk level of these events within a statistically sound framework, with a more precise understanding of the uncertainty around these estimates than was previously possible. This work enables a better understanding of conflict events which can inform preventative measures.

We demonstrate the advantages of the Bayesian framework by comparing our results to maximum likelihood estimation. While maximum likelihood gives reasonable point estimates, the Bayesian approach is preferred when possible. Practical examples are presented to demonstrate how the proposed model can be used to monitor conflict risk. Comparing to current practices that rely on historical averages, we also show that our model is more stable and robust to outliers. In this work we aim to support actors in the humanitarian sector in making data-informed decisions, such as the allocation of resources in conflict-prone regions.

Consulting

3:00 - 4:30pm Thursday, 4th December, 2025
407.208 LT
Presentation type Contributed Session
Chairs Sue Finch

50 Streamlining Analysis Workflows: Using R to Seamlessly Set up Projects and Generate Consistent, Professional Reports

Robin Cook ORCID iD, Zac Dempsey ORCID iD, Matt Cooper ORCID iD

The Kids Research Institute Australia, Perth, Australia

Abstract

As consultant statisticians, we work across many projects that vary in scope, complexity, and duration. Setting up a new project is a critical time when numerous decisions must be made: What folders will I need for this project? What was the best template I last used for a project like this? Ad hoc decisions at this stage often lead to disorganised analytical workspaces, hinder project handover between statisticians, and compromise reproducibility and archiving efforts.

Researchers and students beginning to undertake their own analyses in R often face many steep learning curves. Beyond the fundamental question: “Which statistical test is appropriate for my data?” they are confronted with many other distracting decisions: How do I standardise the look of tables and figures produced by different packages? As internal consultant statisticians, we are best placed to facilitate and educate on good data practices.

To address these challenges, we have developed an R package that streamlines project setup and promotes consistent, professional report writing. Owing to its low barrier for entry, this package has been widely adopted among analytical teams within our institute. Leveraging RStudio add-ins, graphical user interfaces, and a suite of custom functions, our package automates many routine decisions. This takes the guess work out of best practices for new users while preserving a high level of customisability for experienced users.

Our branded Quarto templates (for HTML and Word), alongside functions for standardising the appearance of tables and figures, significantly reduce the time and effort required to progress from raw data to a polished report. With easily customisable branding parameters, this package offers broad utility for individuals and organisations alike. This workflow meaningfully reduces project management overhead and enhances standardisation across teams, maximising the focus and time spent on meaningful data analysis.

96 Statistical Consulting: Guidelines to Get You Started.

Jono Tuke ORCID iD

University of Adelaide, Adelaide, Australia

Abstract

Having a degree, masters, PhD in statistics is only the start of your journey into statistical consulting. In this talk, I will discuss some of the key skills that I have found useful as a statistical consultant. Please feel free to join me to nod wisely at my sage comments, or even better tell me where I am wrong and can improve.

131 Wait a Minute, is AI Making R and Subsequently Statistics Learning Harder Rather than Easier?

Omar Arnaiz ORCID iD, Alexandra Green

The University of Sydney, Sydney, Australia

Abstract

As a statistical consulting group with over 500 consults per year, we focus on applied statistical advice and workflows rather than spending our time teaching people how to code during our consults and workshops. However, more and more the 'undoing' of AI R coding answers seems necessary during our consulting to get our clients back on the right path. Is the rise of AI in learning making R coding and subsequently statistics harder rather than easier?

AI marketing would certainly have you believe it is essential tool for tasks commonly faced by researchers or the applied statistician. Indeed, Google's AI Overview says, "AI significantly enhances statistical methods by automating tasks, improving model accuracy, and enabling more complex analyses".

This talk explores the typical use of AI in R coding statistical analyses across several research domains. Positive impacts such as fast-tracking learning to code and think in R along with the negative consequences such as overly complicated workflows that are difficult to understand are discussed. The utility of AI in more difficult problems is further explored with an advanced R code example used on real-world data to create multiple complex models and subsequent report output.

Finally, the broader impact on researcher statistical analysis plans is discussed with a focus on exploring how our consulting approach can improve the use of AI as tool for learning to code without adversely impacting critical thinking and the quality of research.

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180 Changing 'Can't Do' to 'How To': Empowering Researchers by Fostering a Statistical Literacy Growth Mindset

Kathrin Schemann ORCID iD, Alex Shaw, Alexandra Green, Omar Arnaiz, Chris Howden

The University of Sydney, Sydney, Australia

Abstract

Researchers need statistical literacy to produce rigorous and reproducible high impact research. In practice this means being able to: i) produce efficient study designs and analyses, ii) critically appraise study methods and interpretations, and iii) communicate their results and research impact to disparate audiences such as industry, policy makers and the public for decision-making. Despite statistical literacy clearly being a key competency for academics, researchers often lack confidence or express a closed statistical literacy mindset of "I can't do statistics".

This presentation will explore the implementation of an ecosystem of interconnected services to upscale statistical skills development and the fostering of a statistical literacy growth mind set. Over the last five years the Unit developed interconnected services comprising of 12 modular training workshops, one-on-one consulting and statistical resources. In 2024 alone the Unit trained over 1,000 researchers in 50 workshops in hybrid-mode (in person and on Zoom) and conducted over 500 consultations. All workshops are delivered using practical "how to" software-agnostic workflows and enhanced with software workflows for the most common software applications. The "how to" workflows allow for applied communication of statistical concepts in a logical and easy to understand way, which is crucial for successfully working with individual researchers from different disciplines. The services acknowledge discipline specific challenges and practices and draw on the wide-ranging examples from the consulting experience across the University and the team's varied experience in government, academia and the commercial sector. In addition to working with individual researchers, the Unit trains champions within research groups to leverage the services and fostering of a growth mindset across the University. Ultimately, researchers are empowered to tackle their research with confidence.

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146 What The Statistical Consulting Network Can Do For You

Sue Finch

University of Melbourne, Parkville, Australia

Abstract

Statistical consultants often work in isolation or very small work groups, resulting in limited opportunities for mentoring or problem solving and brain-storming. The Statistical Consulting Network of the SSA was established in 2020 as a way of providing support to and advocating for statistical consultants. It also aims to provide training and professional support opportunities. In this talk, we will review the activities of the Statistical Consulting Network, including the monthly online meet-ups which commenced in April 2022. We will provide examples of the wide-ranging issues and problems raised in the meet-ups, and the contributions and perspectives offered by meeting participants.

Pitman Medal and Service Awards

4:30 - 5:00pm Thursday, 4th December, 2025
405.201 Haydn Williams LT
Chairs Lynne Giles

ASC Closing Ceremony

5:00 - 5:30pm Thursday, 4th December, 2025
405.201 Haydn Williams LT
Chairs Nicola Armstrong

SES Workshop: Teaching Students Who Feel Apprehensive Towards Statistics - What Works?

9:30 - 10:30am Friday, 5th December, 2025
405.201 Haydn Williams LT
Chairs Karol Binkowski, Amanda Shaker

Various strategies are suggested in the literature in order to help address statistics anxiety. Additionally, our own research has provided us with insights into what can help to alleviate statistics anxiety, including help-seeking, weekly quizzes, and developing an interest in statistics. In a multi-institutional study, we asked students to rate various strategies in terms of how they affected students' feelings towards statistics (e.g. teacher attitudes, using software, etc.). The results indicated that strategies related to instructors' behaviours and attitudes had the most impact and, given the wide range of possible strategies available, and educators' limited time, we believe that focusing on these elements will realise the largest benefits in terms of addressing student perceptions of statistics while teaching.

In this interactive and collaborative workshop, we will present some key findings, and facilitate discussion regarding what works to help address statistics anxiety and foster positive attitudes towards statistics, especially when teaching students who feel apprehensive towards statistics. This workshop will be of interest to anyone who teaches statistics, including both early career and more experienced educators, as well as those who teach at any scale from one-to-one teaching / support, to large cohort teaching.

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103 Teaching Students Who Feel Apprehensive Towards Statistics – What Works?

Anna Riach[1], Amanda Shaker[2], Ellen Marshall[3], William Kay[4], Karol Binkowski[5]

[1]University of Leeds, Leeds, United Kingdom. [2]La Trobe University, Melbourne, Australia. [3]Coventry University, Coventry, United Kingdom. [4]Cardiff University, Cardiff, United Kingdom. [5]Macquarie University, Sydney, Australia

Abstract

This is a workshop session.

Various strategies are suggested in the literature in order to help address statistics anxiety. Additionally, our own research has provided us with insights into what can help to alleviate statistics anxiety, including help-seeking, weekly quizzes, and developing an interest in statistics. In a multi-institutional study, we asked students to rate various strategies in terms of how they affected students' feelings towards statistics (e.g. teacher attitudes, using software, etc.). The results indicated that strategies related to instructors' behaviours and attitudes had the most impact and, given the wide range of possible strategies available, and educators' limited time, we believe that focusing on these elements will realise the largest benefits in terms of addressing student perceptions of statistics while teaching.

In this interactive and collaborative workshop, we will present some key findings, and facilitate discussion regarding what works to help address statistics anxiety and foster positive attitudes towards statistics, especially when teaching students who feel apprehensive towards statistics. This workshop will be of interest to anyone who teaches statistics, including both early career and more experienced educators, as well as those who teach at any scale from one-to-one teaching / support, to large cohort teaching.

Morning tea

10:30 - 11:00am Friday, 5th December, 2025

SES: Innovation in Teaching and Assessment

11:00am - 12:00pm Friday, 5th December, 2025
405.201 Haydn Williams LT
Presentation type Contributed Session
Chairs Ayse Bilgin

174 Design and Implementation of (Nearly Automated) Deep Question Banks for Formative Assessment

Wesley Burr, David Riegert

Trent University, Peterborough, Canada

Abstract

The WeBWorK2 project from the Mathematics Association of America is a long-running open source project designed as a platform for auto-graded student assessment items (e.g., assignments, quizzes, and tests). All problems are written in a custom form of Perl, based on extensive templates. The R programming language has been integrated into this platform, allowing for WeBWorK questions to query R and ask questions based on the inputs and outputs of that interface. One of the traditional strengths of WeBWorK is the ability to randomise assignment questions in multiple ways: at the level of the question (i.e., different distinct question from a bank); at the level of the parameters (i.e., the same question, but with different numbers); and at the level of the context (i.e., some portion of the prompt varies per student).

In this presentation, we will discuss recent efforts to develop a (nearly automated) pipeline for creation of deep (that is, extremely large) question banks for use in lower-level statistics courses. The goal is for the question bank to be sufficiently large as to allow every student to receive a unique problem for every question on a formative assessment, encouraging individual effort and reducing the most blatant of copying. The questions, formed via a pipeline written mostly in R, are stored in a SQL database, and then accessed by WeBWorK via a custom-designed R package, allowing for pre-computation of all question/answer pairs, and auditing possibilities for the questions.

We will conclude by sharing some lived experience and student feedback, having used these very large question banks in two introductory statistics courses taught to science majors at Trent University in Ontario, Canada for the last 4 terms.

176 The Crucial Role Of Transparent Visual Encoding

Sue Finch, Ian Gordon

University of Melbourne, Parkville, Australia

Abstract

Cleveland and McGill (1985, 1987) framed the task of graphical perception as the visual decoding of encoding information. Finch and Gordon (2014) articulated five principles of good graphs, including the principle of keeping the visual encoding transparent. This entails the construction of a graph that makes the viewer's task of visual decoding as simple as possible. Statistical educators should be aware of this principle, and the implications for teaching statistics to students at the tertiary level. We illustrate examples of graphics that either observe or ignore this principle, from key graphical contexts of the media, journal articles and financial reporting, including company annual reports. We consider the nature and transparency of visual encoding in introductory graphical forms, and the role of software in supporting or impeding the development of statistical thinking in the interpretation of graphical forms.

192 Building Capability Through Practice: A Model for Statistical Training and Ongoing Peer Learning in the Grains Industry

Sharon Nielsen, Sam Rogers, Annie Conway

Adelaide University, Adelaide, Australia

Abstract

Upskilling practitioners in applied statistics requires more than short-term training. To support meaningful and lasting capability development, we have developed and delivered a suite of applied statistical workshops tailored to the needs of professionals working with agronomic and on-farm trial data. These workshops focus on core concepts of experimental design and analysis, including model-based approaches and the use of R for implementation.

Crucially, each workshop is embedded within a broader model of peer-supported learning. Participants are invited to join monthly online Community of Practice (CoP) meetings where they share real examples, discuss challenges, revisit workshop content, and deepen their understanding through group problem solving. These sessions foster collaboration across organisations, regions, and roles—supporting the adoption of sound statistical practices in diverse applied contexts.

This abstract outlines our training approach, shares examples of participant engagement and feedback, and reflects on the value of CoP meetings as a mechanism for continued professional development. The model has proven effective in building confidence, improving practice, and sustaining learning beyond the workshop setting. We propose it as a scalable and transferable approach for supporting statistical capacity in applied sectors.

209 Keeping it Real: Engaging kids in statistics with authentic problems

Linda McIver

Australian Data Science Education Institute, Melbourne, Australia

Abstract

By the time students reach university, their school experience has often taught them that they are no good at maths or computation, that statistics is scary, and that it’s a meaningless jumble of coloured balls in urns, fictional sports statistics, or made up sales figures. Their confidence and motivation have been shredded.
The Australian Data Science Education Institute is a charity working to build teacher confidence using a Problem Based learning approach. We use authentic problems relevant to the students and the school community, and real, messy, complicated datasets to teach Statistics, Data Science, and other STEM skills from primary school through to university. By giving students meaningful problems to solve, we build their confidence and show them that these skills can help them make real change in their own communities. We focus on authenticity, not technical skills. Once students have the motivation, they are more than capable of learning the skills. This talk will outline our approach, and provide resources to support your teaching.

SES Discussion

12:00 - 12:30pm Friday, 5th December, 2025
405.201 Haydn Williams LT

Discussion: What do you want Statistical Education Section to do?

SES Closing Ceremony

12:30 - 1:00pm Friday, 5th December, 2025
405.201 Haydn Williams LT

Discussion: What do you want Statistical Education Section to do?